# LUMA

## A little light. A lot of personality.

Printable robotic desk companion

## Illustrated assembly manual + engineering drawings

Raspberry Pi 4 · four DS3218MG/PCA9685 joints · round animated LCD (1.85-inch, or optional 4-inch DSI head) · five pedestal sensors · white light + RGB halo

Revision C / 07 September 2026 / Digital prototype release

# Inside the build

# Meet LUMA

LUMA is a printable Raspberry Pi 4 desk companion with an articulated arm, a circular animated face and two independently controlled light rings. It is an original prototype inspired by the posture and personality of the Autonomous Lamp. It is not a conversion kit for the commercial lamp, and its parts do not claim compatibility with that product.

The head contains a Waveshare ESP32-S3-Touch-LCD-1.85 display, a small white-light ring and a larger RGB halo. The Raspberry Pi makes the behavior decisions. The ESP32-S3 built into the display serves as a graphics peripheral over USB. This preserves the requested Pi 4 control while using the actual QSPI display interface supported by the Waveshare board.

The selected 1.85-inch screen is a 360 × 360 IPS TFT LCD, not an OLED. Its visible circular area is approximately 45.68 mm across. Do not substitute a similarly named Waveshare B, C, AMOLED or bare display module without checking both the mechanical envelope and firmware pin mapping.

Revision C uses four matching on-hand Miuzei DS3218MG 270° servos through the user-selected PCA9685. Each wider enclosed link has a driven straight-horn side and an opposite625-bearing side. All five VL53L1X boards move from external head pods to direct mounts inside the fixed pedestal wall. The optional Waveshare4inch DSI LCD(C) head remains available, keeps the RGB halo, and has no separate white ring.

Optional 4-inch DSI head on the same base and enclosed arm.

## What the prototype does

- Wink: one eye closes briefly, the head makes a small gesture and the halo responds.
- Say Hi: a friendly greeting appears on the face, with an audible greeting through the specified audio hardware.
- Be happy: smiling eyes, a bright expression and a gentle arm gesture.
- Sense five horizontal pedestal sectors using Adafruit VL53L1X STEMMA QT boards on separate I2C multiplexer channels.
- Illuminate the face area with the white channel of a 24-pixel RGBW ring and express color with a separate 60-pixel RGB ring.

The provided behavior software runs locally. A conversational language model, camera, microphone, speech recognition and cloud account are not required by this design.

## What is in the project

The printable STL files and editable CAD define the mechanical prototype. The drawings in the back of this manual document those parts. The bill of materials identifies purchased hardware separately from printed parts. Wiring instructions and software source accompany the manual, along with three rendered promotional videos and their editable production scripts.

The project is a digital prototype release. Mesh and software checks can verify files and logic, but they cannot establish physical fit, holding torque, optical performance or print strength. The first assembled unit is the fit-and-function validation build. Record component revision, printer settings and any measured corrections before duplicating it.

## Design choices that matter during assembly

The 55 mm square display board cannot fit through the small light ring's roughly 52.2 mm center. It is installed behind that ring in a separate plane. Follow the head stack order instead of trying to place all boards side by side.

Each enclosed arm link clamps one DS3218 case between its halves. Its driven side uses the servo's straight metal arm, while a625 bearing supports the opposite side. Insert the captive M5 idler nut, servo, lead and harness before closing the three through bolts. The optional4-inch head mounts its display on a carrier plate attached to the display's M4 bosses.

The large RGB ring consists of four purchased 15-pixel quarter boards. Its printed support carries the mechanical load; the solder bridges only connect the electronics. All four arcs are needed to create the complete 60-pixel ring.

Frosted material belongs only over the light rings. Keep the LCD and five pedestal apertures open. A cover over a ranging sensor can create reflection and crosstalk.

The two NeoPixel rings use different pixel formats. The large ring is RGB and the small ring is RGBW. They use separate data outputs and separate driver objects. The normal brightness limits are part of the power and thermal design.

## Reading the drawings

All mechanical dimensions are in millimeters. STL files do not carry a unit declaration: select millimeters in the slicer and confirm the imported bounding dimensions before printing. Use the drawing's stated scale; page fitting in a PDF viewer changes paper scale. Numerical dimensions and the editable model control geometry.

Purchased circuit boards, bearings, fasteners and servo horns are not printable substitutes. Match the exact part numbers in the bill of materials. Minor manufacturing differences are handled by the stated clearances and the first fit coupon, not by forcing a board into a printed pocket.

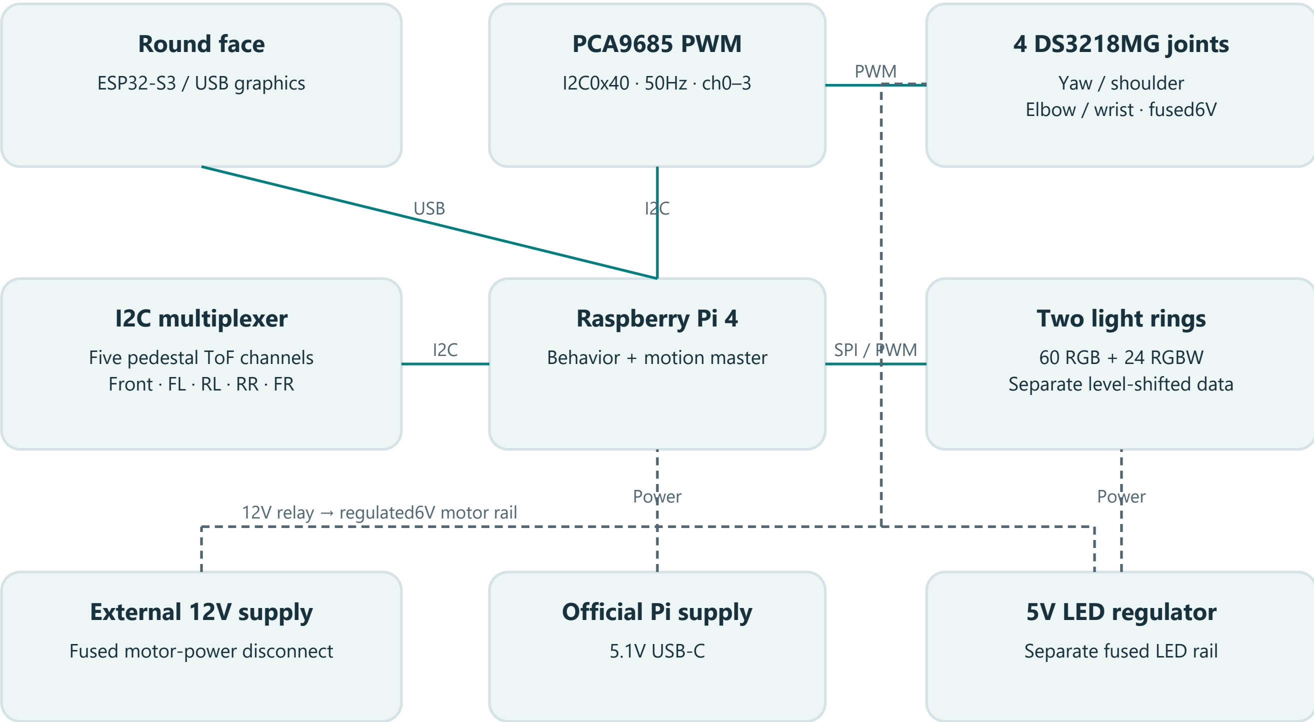
## Build sequence

1. Check the purchased component revisions and inspect the dimensioned interfaces.
2. Print and measure the fit coupon, then slice the structural parts and diffusers.
3. Bench-test each electrical subsystem before fitting it into the arm.
4. Mount the five sensors inside the pedestal and assemble the base, joint supports and links with motors unpowered.
5. Assemble the layered head, display and light rings.
6. Route and restrain the wiring, leaving service loops at every moving joint.
7. Center each unloaded PWM servo, calibrate neutral pulses/signs and enable a restricted first movement.
8. Validate sensing, fault response, balance, temperature and the three expressions.

## Reference and attribution

The appearance reference is the Autonomous Lamp. The LUMA geometry, manual layout and animations are newly authored for this project. Vendor specifications and documentation are identified in the component source register and bill of materials. Promotional files show the digital prototype, not a filmed or physically tested product.

# System at a glance




Solid: data. Dashed: power. Wiring chapter defines grounds, fuses and the motor feed.

The Pi supplies control commands. Dedicated motor and LED power paths keep the actuator load off the Pi's supply. The detailed wiring chapter is authoritative for the power-disconnect circuit and common grounds.

# Parts and purchasing

Planning allowance for the standard build below: **US $745.07** before shipping and tax. The optional 4-inch DSI head adds **US $60.80** of purchased parts and removes the 1.85-inch display, the inner RGBW ring and their fasteners (about US $68.35). Figures are budgeting allowances, not checked-out prices. Filament and any workshop tools are included only where explicitly listed. Reuse of existing supplies can change the total substantially.

Order the exact display and voltage variant specified. Generic fasteners, connector harnesses, relay contacts and wire must meet the listed dimensions and electrical ratings. Buy the servo horns and their screws with the servos; verify their actual thread and engagement before ordering extras.

## Controller, sensing, lights and power

| Ref / qty | Item | Required specification | Budget / each |
|---|---|---|---|
| E01 / 1 | Raspberry Pi Raspberry Pi 4 Model B 4GB | Master controller. 40-pin GPIO and four USB ports | $60.00 |
| E02 / 1 | Raspberry Pi 15W USB-C Power Supply | Separate Pi supply. 5.1V 3A USB-C | $12.00 |
| E03 / 1 | Reputable flash vendor 32GB or larger microSD A1 | Operating system card. High endurance preferred | $12.00 |
| E04 / 1 | Waveshare ESP32-S3-Touch-LCD-1.85 SKU28514 | Central face display. 55x55mm;360x360 ST77916 QSPI;USB-C;no B/C suffix | $35.00 |
| E05 / 5 | Adafruit 3967 VL53L1X STEMMA QT | Five interaction ranging sensors. 25.5x17.5x4.6mm;I2C0x29 | $14.95 |
| E06 / 1 | Adafruit 5626 PCA9548 STEMMA QT | Mux for five same-address sensors. TCA9548A-compatible;40.6x20.2x4.8mm;0x70 | $6.95 |
| E07 / 5 | Adafruit STEMMA QT JST-SH4 cable | One separate cable per sensor. Length selected to assembled routing;100-300mm nominal | $2.00 |
| E08 / 1 | Adafruit STEMMA QT to female header cable | Pi to mux upstream connector. 3.3V logic only | $2.00 |
| E09 / 4 | Adafruit 1768 quarter NeoPixel60 ring | Outer RGB emotion halo. 15RGB per quadrant;four form OD157.5mm ID144.8mm ring | $9.95 |
| E10 / 1 | Adafruit 2862 NeoPixel24 RGBW natural white | Inner white lamp ring. 24RGBW;OD65.5mm;only W channel used;4500K nominal | $24.95 |
| E11 / 1 | Adafruit 1787 74AHCT125 DIP14 | Two data level shifters. 5V VCC;3.3V logic accepted;AHCT required | $1.50 |
| E12 / 2 | Generic 330ohm 0.25W resistor | LED input series resistors. One per independent data line | $0.10 |
| E13 / 2 | Generic 1000uF 10V electrolytic | LED ring supply reservoir. Observe polarity;one each ring | $1.00 |
| E14 / 1 | Generic 100nF ceramic | 74AHCT125 decoupling. Across DIP14 and DIP7 at socket | $0.10 |
| E15 / 1 | Pololu D24V50F5 item2851 | LED5V buck converter. 5V5A typical thermal-limited;17.8x20.3x8.8mm | $32.95 |
| E16 / 4 | Miuzei DS3218MG 20kg 270-degree | Four inventory joint servos. 4.8-6.8V; PWM500-2500us;1500us neutral;50-330Hz;2.2A stall and21.5kg-cm at6.8V;40x20x40.5mm | $0.00 |
| E17 / 1 | Generic PCA9685 16-channel PWM board | Operator-selected four-servo controller. I2C address0x40;50Hz;channels0-3;3.3V logic VCC;servo V+ supplied separately | $0.00 |
| E18 / 1 | Mean Well GST220A12-R7B | External enclosed motor/LED supply. 12V15A model;R7B powerDIN;keep outside printed enclosure | $95.00 |
| E19 / 1 | Generic Matching R7B powerDIN harness | DC supply breakout. Correct polarity;all specified parallel contacts;15A aggregate rated | $15.00 |
| E20 / 1 | Generic IEC C13 mains cord | External supply cord. Local approved cord;earth conductor intact | $7.00 |
| E21 / 1 | Generic Latching mushroom stop two NC contacts | Motor stop plus isolated Pi sense. NC1 interrupts12V relay coil;NC2 connects BCM27 to GND | $15.00 |

| Ref / qty | Item | Required specification | Budget / each |
|---|---|---|---|
| E22 / 1 | Generic 12VDC coil relay with30A DC NO contact and socket | Hardware motor supply disconnect. Coil<=200mA;NO contacts>=15A at12VDC;socket16AWG | $12.00 |
| E23 / 1 | Generic 1N4007 diode | Relay coil flyback. Stripe/cathode to coil positive | $0.10 |
| E24 / 1 | Generic 15A DC fuse and holder | Main12V input branch. Holder and connector rated>=15A | $3.00 |
| E25 / 4 | Generic 3A time-delay DC fuse and holder | Individual6V servo feeds. 18AWG star feeds;one fuse per DS3218 | $3.00 |
| E26 / 1 | Generic 2A DC fuse and holder | Buck12V input branch. LED firmware capped20percent | $3.00 |
| E27 / 1 | Generic 7.5A time-delay DC fuse and holder | Servo regulator12V input branch. Between motor relay output and6V regulator VIN;16AWG | $3.00 |
| E28 / 1 | Generic 0.5A DC fuse and holder | Relay coil branch. NC1 switches coil only | $3.00 |
| E29 / 1 | Generic 3A DC fuse and holder | 5V LED output branch. ProtectsLED18-22AWG harness | $3.00 |
| E30 / 1 | Adafruit 3369 mini USB speaker | Spoken Hi. USB audio;84x43x32mm;73.6g;keep volume modest | $12.50 |
| E31 / 1 | Generic USB-A to right-angle USB-C data cable | Pi to display. Flexible cable;strain relief;15mm rear plug envelope assumed | $7.00 |
| E33 / 1 | Generic 16AWG /18AWG silicone wire and rated distribution terminals | Power harness. 16AWG trunk;18AWG motor branches;22AWGLED branches | $20.00 |
| E34 / 1 | Generic Small perfboard and DIP14 socket | Level shifter assembly. Insulate underside;strain relief all wires | $5.00 |
| E35 / 1 | Generic Pi4 heatsink and ventilation hardware | Thermal management. Check CPU temperature after assembled operation | $7.00 |
| E36 / 1 | Generic 10kohm resistor | External E-stop sense pullup. BCM27 to3.3V;open NC2 means asserted | $0.10 |
| E37 / 1 | Pololu D42V110F6 item5673 | Servo rail step-down regulator. 6V output;11A nominal;6-60V input;31.8x43.2x9mm;four2.2mm mounting holes | $59.95 |
| E38 / 1 | Generic 2200uF 10V low-ESR electrolytic | Servo rail bulk capacitor. Across6V star after regulator;observe polarity;voltage rating at least10V | $3.00 |

## Mechanical hardware and consumables

| Ref / qty | Item | Required specification | Budget / each |
|---|---|---|---|
| M01 / 1 | Local metal supplier Custom ballast disc | Steel ballast plate. Mild steel diameter180 thickness6 mm; 4 holes diameter4.4 at XY(+/-45,+/-35); 4 holes diameter3.4 at XY(+/-26,-36/-8); deburr; approx1.19kg | $25.00 |
| M02 / 1 | Generic 6808-2RS | Yaw support bearing. 40mm bore x52mm OD x7mm width; sealed deep groove | $12.00 |
| M03 / 4 | Generic M4x25 | Base and tray bolts. M4 x25 socket head; steel plate6 + floor3 + spacer5 + tray3 | $0.25 |
| M04 / 4 | Generic M4x5 spacer | Tray standoffs. Metal spacer5mm length; 4.3mm bore; OD8 minimum | $0.40 |
| M05 / 4 | Generic M4 nut | M4 lock nuts. ISO metric nyloc | $0.12 |
| M06 / 8 | Generic M4 washer | M4 load spreading washers. OD12 minimum | $0.06 |
| M07 / 4 | Generic M3x16 | Yaw cradle through bolts. Through printed cradle3 steel6 and base3; nuts below base; tray has7mm head access | $0.16 |
| M08 / 2 | Generic M3x50 | Yaw rear body clamp bolts. Through bottom3 + body40.5 + keeper2.6; do not overtighten case | $0.30 |
| M09 / 2 | Generic M3x60 | Shoulder clamp bolts. Clamp through the52mm tower width and keeper plus nuts | $0.40 |
| M10 / 4 | Generic M3x16 | Shoulder foot bolts. 6mm tower foot plus6mm platform plus nuts | $0.16 |
| M11 / 6 | Generic M3x70 | Arm link through bolts. 5+52+5mm stack plus nut;3 per link at X35 and around each servo block | $0.45 |
| M12 / 8 | Generic M3x12 nominal with matching nuts | DS3218 straight-horn attachment sets. Two per joint through adjustable printed slots; actual supplied horn hole size and thread are not documented | $0.25 |
| M14 / 6 | Generic M3x12 | Base lid screws. Into2.8mm plastic-thread-forming printed pilots | $0.15 |
| M15 / 4 | Generic M3x12 | Head yoke screws. 6mm fork mounting face +3mm shell floor; nuts on inside | $0.15 |
| M16 / 8 | Generic M3x10 | Outer bezel screws. 4mm bezel +1mm shim;2.8mm plastic-thread-forming pilot | $0.15 |
| M17 / 4 | Generic M3x12 | Central face screws. 4mm plate +1mm shim;2.8mm plastic-thread-forming pilot | $0.15 |
| M18 / 4 | Generic M3x10 | White ring carrier screws. 2mm carrier on dedicated radius45 bosses at z32.5 | $0.15 |
| M19 / 4 | Generic M3x25 | LCD cradle and retaining plate screws. 3mm retainer +15mm cradle; thread into head pillar at z14 | $0.20 |
| M20 / 12 | Generic M3x1 shim | Precision axial shims. M3 bore; thickness1mm;12 under face and bezel | $0.12 |

| Ref / qty | Item | Required specification | Budget / each |
|---|---|---|---|
| M21 / 20 | Generic M2.5x6 | Pedestal sensor board screws. Four per Adafruit3967 board into2.1mm blind pilots;20.32x12.70mm pattern | $0.12 |
| M24 / 30 | Generic M3 nut | General M3 nuts. Include spare quantities for clamps horn interfaces and platform | $0.10 |
| M25 / 50 | Generic M3 washer | General M3 flat washers. Nominal0.5mm; not replacement for1mm axial shims | $0.05 |
| M26 / 4 | Generic M2.5x6 | Raspberry Pi fasteners. Pi4 four mounting holes; printed standoffs may be tappedM2.5 | $0.12 |
| M27 / 4 | Generic Rubber feet | Adhesive anti slip feet. 20mm diameter x5mm high minimum; place radius90 | $0.40 |
| M28 / 1 | Generic Translucent PETG | Frosted optical printing filament. Natural/translucent; approximately60g including trials; solid1.2mm optical skin | $20.00 |
| M29 / 1 | Generic PETG | Structural filament. Approx1kg roll; actual slicer mass varies;6 walls on the arm link halves | $22.00 |
| M30 / 1 | Generic Neutral cure silicone | Small adhesive tube. 3 small removable dots for inner diffuser; electronics compatible | $6.00 |
| M31 / 1 | Generic Thin PCB foam | Closed cell foam tape. 0.5 and1mm thickness; LCD edge preload only | $4.00 |
| M32 / 1 | Generic LED retaining tape | Electronics compatible thin adhesive strips. Attach LED ring PCB on support posts without covering LED optics | $3.00 |
| M33 / 1 | Generic Desk clamp | Optional commercial padded desk clamp. Use only if assembly reach/mass exceeds validated stable envelope; not a printed load bearing clamp | $12.00 |
| M34 / 12 | Generic M2.5x12 | Optional harness clip screws. For6 printed cable saddles; use selftapped pilots or cableties | $0.12 |
| M35 / 3 | Generic 625-2RS | Pitch-joint idler bearings. 5mm bore x16mm OD x5mm width; sealed | $3.00 |
| M36 / 3 | Generic M5x16 shoulder bolt or precision-shank bolt | Pitch-joint idler axles. Smooth5mm shank across each625 inner race; one shoulder elbow wrist | $0.80 |
| M37 / 3 | Generic M5 nyloc nut | Pitch-joint captive idler nuts. ISO metric; one in each printed hex trap | $0.20 |
| M38 / 6 | Generic M5 washer | Idler bearing thrust washers. One each side of every625 inner race; thin hardened steel | $0.12 |

## Optional 4-inch DSI head (not in the standard total)

Buy these only for the optional head described in the mechanical chapter. With this head, do not buy E04 (1.85-inch display), E10 (24-pixel RGBW ring), E31 (display USB cable), M18 (white ring carrier screws) or M19 (LCD cradle screws).

| Ref / qty | Item | Required specification | Budget / each |
|---|---|---|---|
| D01 / 1 | Waveshare 4inch DSI LCD (C) | Optional head display. Round720x720 IPS DSI;126mm outline;101.52mm active;6mm case;four M4 rear bosses on75x75;includes two50mm 15-pin FPC | $45.00 |
| D02 / 1 | Generic 15-pin 1.0mm pitch FFC type A 800mm | Pi DSI to optional head. Same-side contacts;0.3mm thick;800mm nominal from Pi DSI connector through both arm cable ports;bench-verify signal integrity before closing the arm | $8.00 |
| D03 / 1 | Generic 2-conductor 22AWG silicone lead with2.0mm pitch 4-pin plug | Optional head5V feed. Pi header5V/GND to LCD HP2.0 4-pin;SDA/SCL contacts left empty;800mm | $4.00 |
| D10 / 4 | Generic M4x8 | Optional LCD carrier to display screws. Through5mm carrier pad into the4mm deep M4 case bosses; longer screws bottom out | $0.15 |
| D11 / 4 | Generic M3x8 | Optional carrier to shell screws. Into2.8mm printed pilots at(+/-55,+/-38) top20.5 | $0.15 |
| D12 / 4 | Generic M3x12 | Optional face ring screws. 4mm ring +1mm shim;2.8mm pilots at radius67.8 | $0.15 |
| D14 / 1 | Generic Thin closed cell foam tape | Optional LCD rim preload. 0.5mm between the display case rim and the face ring; also under the carrier pads if the boss ends are not coplanar | $2.00 |

## Printable part schedule

24 unique STL files; 31 printed pieces for the standard build, including the fit coupon and optional cable clips. Quantities control the print run. The drawing index uses the same part order.

| Drawing / qty | STL identifier | Material / print orientation |
|---|---|---|
| P01 / 1 | `base_tub` | PETG; floor down |
| P02 / 1 | `base_lid` | PETG; flat annulus down |
| P03 / 1 | `electronics_tray` | PETG; plate down |
| P04 / 1 | `yaw_mount` | PETG; base down |
| P05 / 1 | `yaw_cap` | PETG; flat down |
| P06 / 1 | `turntable` | PETG; neck down |
| P07 / 1 | `shoulder_tower` | PETG; foot down |
| P08 / 1 | `shoulder_cap` | PETG; flat down |
| P09 / 1 | `upper_arm_left` | PETG; plate down |
| P10 / 1 | `upper_arm_right` | PETG; plate down |
| P11 / 1 | `forearm_left` | PETG; plate down |
| P12 / 1 | `forearm_right` | PETG; plate down |
| P13 / 3 | `horn_spacer` | PETG; flat down |

| Drawing / qty | STL identifier | Material / print orientation |
|---|---|---|
| P14 / 1 | `head_yoke` | PETG; rear mounting plate on bed; rotate180X |
| P15 / 1 | `head_shell` | PETG; rear disc down |
| P16 / 1 | `outer_bezel` | PETG; front lip down |
| P17 / 1 | `face_center` | black PETG; flat front down |
| P18 / 1 | `white_carrier` | PETG; flat down |
| P19 / 1 | `lcd_cradle` | PETG; flat base down |
| P20 / 1 | `lcd_retainer` | black PETG; flat down |
| P21 / 1 | `outer_diffuser` | natural/translucent PETG; smooth optical face on bed |
| P22 / 1 | `inner_diffuser` | natural/translucent PETG; smooth optical face on bed |
| P23 / 6 | `cable_clip` | PETG; flat down |
| P24 / 1 | `fit_coupon` | PETG; flat down |

The optional 4-inch DSI head prints 3 additional files (3 pieces) and omits `head_shell`, `face_center`, `white_carrier`, `inner_diffuser`, `lcd_cradle`, `lcd_retainer` from the list above.

| Drawing / qty | STL identifier | Material / print orientation |
|---|---|---|
| P25 / 1 | `head_shell_dsi` | PETG; rear disc down |
| P26 / 1 | `lcd4_carrier` | PETG; flat down |
| P27 / 1 | `face_ring_dsi` | black PETG; flat front down |

# Printing and finishing

## Prepare one interface before a full print run

Use a 0.4 mm nozzle as the baseline. Print the fit coupon supplied with the CAD, measure its holes and pockets with calipers and test the actual fasteners and bearings. This is especially useful for horizontal holes, which can print smaller than their modeled diameters. Keep the slicer at 100% scale; compensate holes locally or revise the relevant CAD clearance rather than scaling the complete machine.

The part manifest gives quantities, material and orientation. The STL orientation is the manufacturing orientation when the manifest identifies it as such. The assembled model is for viewing and should not be sliced as a single object. Check that each sliced part has continuous perimeters and that screw holes remain open.

## Structural material

Use PETG for the first functional build. For the four enclosed arm halves and load-carrying brackets, start with0.20mm layers, six perimeters, six top/bottom layers and40% gyroid infill. Print each half with its5mm plate on the bed. The26mm walls, servo block and through-bolt boss then rise vertically. The idler halves include16.2mm 625 pockets and distal captive-nut bosses; inspect their slicer paths and use local solid infill. Print one idler half and one driven half per link.

For non-load-bearing covers, start with four perimeters, 0.20 mm layers and 20% infill. Use the orientation and support notes specific to each part. Remove supports without gouging bearing seats or the surfaces that locate circuit boards.

The optional4-inch DSI head adds three files: `head_shell_dsi`, `lcd4_carrier` and `face_ring_dsi`. Print them only for that head and omit the six standard-head parts listed in the schedule. Revision C has no `brow_bracket`, `sensor_pod` or `sensor_retainer` print.

PLA can be useful for quick dimension checks, but warm motors, LED boards and sustained joint loads make it a poor default for the final load-carrying structure. A material change also changes shrinkage and fit; repeat the coupon.

## Frosted light covers

Print both annular diffusers in natural translucent PETG. The small ring covers the white LEDs and the large ring covers the RGB LEDs. Use 0.12–0.16 mm layers and a continuous solid optical face. Preserve the modeled thin wall; do not replace it with sparse infill that leaves a visible lattice in the illuminated area.

Make a small test of the intended wall thickness before committing both rings. Translucent filaments vary greatly in pigment and light transmission. A nominal 1.2 mm optical face is a starting point, not a measured optical specification.

Wet-sand the outside very lightly with 600–1000 grit abrasive to obtain a uniform matte surface. Wash and dry the part before installation. Do not sand the locating lip so far that the diffuser becomes loose. Check the result at the firmware's normal brightness limit with the head closed: rings that look uniform in room light can reveal hotspots when lit.

Keep the opaque separator between the two light paths. A black or charcoal separator helps prevent RGB light washing across the face. Leave an air gap between LEDs and diffuser as defined by the head stack. Avoid adhesives that touch the LED emitters or the LCD glass.

## Fasteners and tolerance

Use the specified metal nuts, washers,625 bearings and servo horns. Deburr holes before insertion. Test the16.2mm pocket and horn slots on `fit_coupon`. A bolt must enter a clearance hole by hand. The M5 idler bolt must clamp the bearing inner race without rubbing the outer race. Tighten opposing fasteners alternately and stop before printed layers or a PCB deform.

If a heat-set insert is specified, test the insert type and soldering-iron temperature on scrap from the same filament. Support the boss while inserting it, allow it to cool completely, and confirm the insert is perpendicular. Do not add heat-set inserts to a design location intended for a through-bolt and captive nut.

Press bearing outer races squarely into their stated seats. Apply force to the race being fitted. A bearing that becomes rough after installation is a sign of an undersized or distorted seat; correct the seat before assembling the joint.

## Harness preparation

Make a service loop at each joint, and move the unpowered mechanism through its intended travel while watching the loop. The wire must not become taut, rub a sharp printed edge, enter a gear/horn gap or prevent the joint from reaching its limit. Fit the provided cable guides and add soft sleeving where the harness moves against a surface.

Use flexible stranded wire for moving runs. The five short sensor branches remain fixed in the pedestal. Keep their I2C wiring away from the6V servo distribution and strain-relieve each small connector. A STEMMA QT connector is a signal interconnect, not a high-current power connector.

## Record the first print

Record printer, nozzle, material brand, layer height, wall count, infill, measured coupon hole sizes and any CAD clearance change. Weigh the completed head with its display, lights, screws and wiring. Use the measured head mass when reviewing the arm torque calculation; record the five fixed pedestal sensors with the base mass.

Estimated print duration and filament mass depend on the slicer and printer. The CAD validation report includes geometric volume; the slicer's estimate after applying the actual walls and infill is the useful production estimate.

# LUMA mechanical assembly, revision C

The editable geometry is `cad/luma.scad`. `cad/build.py` exports printable parts, two assembly manifests, purchased-part envelopes, and mesh measurements. Open `cad/assembly.scad` for the standard head or `cad/assembly_dsi.scad` for the optional4-inch DSI head. Files under `cad/visual_only` are not printable parts.

Revision C moves all five Adafruit3967 VL53L1X boards from external head pods to direct mounts behind the pedestal wall. It removes the pod, retainer, and DSI brow-bracket parts and closes the old head penetrations. It also replaces four dual-output ST3215 bus servos with four inventory Miuzei DS3218MG 270-degree PWM servos. The links and yoke are wider, each pitch joint has a625 bearing opposite its single driven horn, and the pedestal/yaw stack is6mm taller. Link pitches, indexed angles, displays, and light-ring geometry remain unchanged.

The indexed neutral has the upper link115° and forearm45° counterclockwise from forward+Y. The relative elbow angle is−70°. Pitch axes run alongX and yaw runs alongZ. Dimensions are: base diameter216mm and wall height56mm; shoulder axisZ121mm; upper pitch140mm; forearm pitch120mm; head diameter176mm and optical depth42mm. The rear head plane center is aboutY46.7mm/Z332.7mm, and the housing top is about420.7mm. Rubber feet add5mm.

## Purchased geometry and fit datums

The Miuzei DS3218 datasheet specifies a40 × 20 × 40.5mm body, 4.8-6.8V PWM control, and a300mm lead. The model uses a conservative body envelope fromX−30..10 relative to the shaft, Y±10, and Z−40.5..0 from the output-side case plane. The mounting-tab envelope is54.5mm long at the datasheet's27.7mm height datum. The included straight metal arm and25T spline stay purchased parts. Its attachment-hole positions and thread are not dimensioned, so the printed driven interfaces use a7mm center clearance and two3.4mm radial slots at12-18mm and20-25mm. Measure the actual horn before selecting bolts. Source: https://images-na.ssl-images-amazon.com/images/I/81Lbgu%2BnG6L.pdf .

Each pitch joint is supported on both sides. The driven plate attaches to the straight metal servo arm through a3.3mm printed spacer. The opposite plate carries a625-2RS bearing, 5mm bore ×16mm OD ×5mm wide. An M5 precision-shank or shoulder bolt clamps only the bearing inner race to a captive M5 nut in the parent idler boss. Use thin thrust washers that do not touch the outer race. The yaw platform remains independently supported by one6808-2RS bearing, 40 ×52 ×7mm.

The Adafruit3967 board outline and four plated holes come from Adafruit's Eagle board file. The board is25.40 ×17.78mm. Its2.5mm holes form a20.32 ×12.70mm pattern centered on the optical package. The pedestal uses four6mm bosses with2.1mm blind pilots per board. The board's component face points outward and its sensor looks through a10 ×10mm open wall aperture. Source: https://github.com/adafruit/Adafruit-VL53L1X-PCB .

The standard display remains the55 ×55mm Waveshare ESP32-S3-Touch-LCD-1.85. It uses a55.8mm edge pocket and foam shims; no guessed board holes. The optional Waveshare4inch DSI LCD(C) remains a round720 ×720 panel in aØ126mm case with four M4 bosses on a75 ×75mm square. Its controller-board/connector envelope is measured from the vendor STEP. Verify every purchased connector and glass projection before tightening.

## Print settings and fit coupon

Print all units in millimeters at100%. The216 ×216mm base leaves only2mm per edge on a220mm bed. Use PETG,0.20mm layers, four walls for general parts, and six walls/40% gyroid for the four link halves and joint supports. The link halves print with their5mm outer plates on the bed. The head yoke prints on its rear mounting plate, rotated180° aboutX, with supports under the fork ears. The removed sensor pods, retainers, and brow bracket are not part of revision C.

Print `fit_coupon` first. It carries3.0/3.2/3.4/3.6mm bores, the16.2mm 625 pocket, and the DS3218 straight-horn slots. Tune only the affected CAD clearance; never scale the full mechanism. A625 bearing must seat squarely without force through its inner race. Clear through-holes by hand. The2.1mm base sensor and Pi pilots are for carefully started M2.5 screws. Do not force a screw or bend a PCB.

Print the light diffusers in natural/translucent PETG at0.15mm layers, three perimeters, and100% infill. Preserve the1.2mm optical skin. Do not put printed plastic, frosting, adhesive, or an untested IR window over any ToF aperture.

## Prepare metal parts

1. Make the Ø180 × 6mm mild-steel ballast from `cad/ballast_plate.dxf` or the dimensioned SVG. Drill four Ø4.4 holes at (±45,±35) and four Ø3.4 holes at (±26,−36/−8). Deburr and protect it from corrosion. Keep the center solid.
2. Confirm the 6808 bearing is 40 × 52 × 7mm and all three 625 bearings are 5 × 16 × 5mm. Test the printed pockets before assembly.
3. Inspect all four inventory DS3218MG units and straight 25T arms. Confirm the servo is the 270° position variant, not a continuous-rotation unit. Measure the metal-arm hole diameter, spacing, thickness, center-screw engagement, and clearance over the case. Dry-fit each printed radial slot before power is applied.
4. Test each M5 idler stack outside the arm. The smooth bolt section must span the bearing inner race. The washer and printed plate must not touch the rotating outer race. The captive nut must seat without splitting its boss.

## Assemble the pedestal sensors and weighted base

5. Put the steel disc on the 3mm base floor. Attach the yaw cradle at Z9 with four M3x16 bolts through the asymmetric pattern. Keep all underside hardware above the rubber-foot contact plane.
6. Identify front as +Y. The five sensor centers are front 90°, front-left 162°, rear-left 234°, rear-right 306°, and front-right 18°. Insert one M2.5x6 screw through each board hole from the PCB's inner side. Put the component face toward its wall opening and start all four screws into the matching blind pilots. Tighten in a cross pattern only until the board cannot rock. Confirm the optical package is centered and unobstructed. Connect one side STEMMA QT socket and strain-relieve the cable inside the wall.
7. Place the yaw DS3218MG with its shaft vertical, its body toward −Y, its case bottom at Z12, and its output-side case plane at Z52.5. The rear-body cradle avoids the servo's mounting ears. Fit the yaw keeper at Z52.5 with two M3x50 bolts at X±18.5/Y−27. Tighten only enough to secure the case.
8. Mount the electronics tray at Z14 on four 5mm metal spacers with four M4x25 bolts through base, ballast, spacer, and tray. Mount the Pi on its standoffs. Use the universal tray slots for the PCA9685, sensor mux, 6V regulator, and insulated distribution hardware; their exact board revisions are not assigned guessed hole coordinates. Keep the regulator ventilated. Keep the external AC adapter and any oversized high-current enclosure outside the compact base.
9. Use the round rear wall port at 270° and rectangular port at 282° for external power/USB. Add grommets and strain relief. The six lid posts sit at 42/66/126/198/258/342° between the sensor sectors and rear ports.
10. Seat the 6808 in the lid's 52.15mm pocket. The turntable's 39.75mm neck passes through its 40mm bore. Below the lid, attach the 3mm tangential horn web to the yaw servo's straight metal arm through the adjustable slots. The web stays below the lid disk and the neck carries load through the bearing. Verify free ±25° yaw before fitting six M3x12 lid screws. The turntable top and shoulder-tower datum are Z74.

## Assemble the shoulder and enclosed links

Each link has an idler half and a driven half. Local X runs from proximal pivot to distal servo shaft: 140mm for the upper link and 120mm for the forearm. The 5mm plates occupy assembled Z−31..−26 and +26..+31. Each half has 26mm-deep walls that meet at Z0. The DS3218 rear-case pocket is 41.2mm across the axis and 20.7mm across the body. The walls start at X28, leaving the proximal clevis open. A 21 × 9mm cable port remains on the inner wall. Three M3x70 bolts close each link: one at X35 and two around the distal block at Y±18.5.

The idler half has a 16.2mm bearing pocket at its proximal pivot and an M5 captive-nut boss at its distal servo axis. The driven half has the two straight-horn slots at its proximal pivot. The parent servo output-side case plane sits at local Z+20.25. Its metal arm and the 3.3mm spacer fill the gap to the driven plate inner face at Z+26. The opposite case face clears the idler boss by 0.35mm. These are digital nominal clearances and require a physical fit check.

11. Bolt the shoulder tower to the turntable with four M3x16 bolts. Fit the shoulder DS3218MG with its shaft at Z121 and its output side toward the driven link half. Install the keeper with two M3x60 bolts. Insert the first M5 captive nut into the tower idler boss before the upper link blocks access.

12. Press one625 bearing into the proximal pocket of `upper_arm_left`. Put the inner race against the shoulder idler boss with its thrust washers and M5 axle. Do not tighten the outer race. With the shoulder servo centered and unpowered, index the link at115° and attach `upper_arm_right` to the straight metal arm through one printed spacer and two measured horn fasteners.

13. Seat the elbow DS3218MG in the upper-link distal pocket with its shaft atX140. Put its output toward the driven half and cable toward the rear relief. Route its lead and the remaining head harness through the inner wall port. Insert the distal M5 captive nut. Close the two halves with threeM3x70 bolts without crushing the case.

14. Assemble the forearm the same way. Its idler625/M5 stack attaches at the elbow, and its driven half attaches to the elbow's straight horn at−70° relative angle. Seat the wrist servo atX120, route its lead, insert the last M5 captive nut, and close with threeM3x70 bolts.

15. The head yoke has one625 idler ear and one tangential straight-horn ear. Attach its bearing side to the forearm idler boss, then attach its driven side to the centered wrist horn through the last3.3mm spacer. At neutral, the broad head plate faces+Y and lies21mm forward of the wrist axis. Verify that the servo mounting tab clears the yoke notch.

16. Pass the head harness through the26 ×8mm yoke slot, over each servo case, and through both link ports. Leave a slack service loop at every joint. No moving wire may enter a horn, bearing, idler bolt, or clamshell seam.

## Assemble the standard 1.85-inch head

Head-localZ starts at the rear face and increases toward the viewer. The shell floor isZ0-3 and rim ends atZ36. There are no sensor mounts or apertures in the revision C head.

17. Attach `head_shell` to the yoke with fourM3x12 bolts on the54 ×30mm pattern. Bring only display, LED, and optional DSI wiring through the center slot.

18. Join four Adafruit1768 RGB quarter-rings into one60-pixel circle. Dry-fit the158mm OD/145mm ID board on the radius76 posts. Retain the PCB mechanically; solder bridges are electrical joints only.

19. Put the standard LCD cradle on its four headZ14 pillars. Place the55mm board in the55.8mm pocket with thin edge foam and its low-profile USB plug in the19mm relief. Fit the retainer atZ29 with fourM3x25 screws. Do not load the glass.

20. Fit the white-ring carrier on the radius45 bosses atZ32.5 with fourM3x10 screws. Attach the24-pixel RGBW ring without covering LEDs or loading solder joints.

21. Put1mm shims on the four radius60 face bosses and eight radius83 bezel bosses. Fit the outer diffuser, center baffle, and outer bezel without squeezing the optical skin. Secure the center with fourM3x12 and bezel with eightM3x10. Retain the inner diffuser with three small removable neutral-cure silicone dots.

## Assemble the optional 4-inch DSI head

This variant replaces `head_shell`, `face_center`, `white_carrier`, `inner_diffuser`, `lcd_cradle`, and `lcd_retainer` with `head_shell_dsi`, `lcd4_carrier`, and `face_ring_dsi`. It keeps the yoke, outer bezel, outer diffuser, and60-pixel halo. It has no brow bracket or head-mounted sensors and no separate white ring.

22. Attach `head_shell_dsi` to the yoke with fourM3x12 bolts. Fit the RGB halo on its radius76 posts.

23. Attach `lcd4_carrier` to the display's four M4 bosses with M4x8 screws. Do not use longer screws. Confirm the PCB, Pi standoffs, power connector, USB-C, and DSI FPC all clear the carrier window.

24. Lower the carrier onto the shell bosses at(±55,±38). Route the800mm DSI FPC and5V/GND lead through the26 ×8mm floor/yoke slots. Secure the carrier with fourM3x8 screws.

25. Put0.5mm foam on the display rim and1mm shims on the face/bezel bosses. Fit the outer diffuser, `face_ring_dsi`, and bezel. Tighten evenly without pressing the active glass. Configure and test the display before closing the base.

## Bring-up and limits

26. With every horn disconnected from the links, center one servo at a time using the bounded PCA9685 command in `software/README.md`. Mark its1500µs position. Power off before installing a horn. Set `neutral_pulse_us` and `joint_signs` from the actual indexed mechanism; the default values are not physical calibration.
27. Support the head and move each unpowered joint through the intended small range. Check the625 stacks, open clevises, servo tabs, cable loops, and straight horns. Initial commanded limits remain yaw±25° and shoulder/elbow/wrist±10° at12°/s. The supplied digital collision check covers only the indexed pose.
28. Verify the physical motor stop before armed motion. Run one joint and then one complete scene while supported. A PCA9685 check proves controller communication only. The DS3218 provides no position, current, load, or temperature feedback. Measure6V droop, regulator temperature, each accessible servo-case temperature, and joint holding behavior.
29. Recheck clamp, horn, idler, and link fasteners after the first short motion run and first thermal cycle. Stop on noise, binding, rising temperature, visible deflection, or lost PWM control. Physical fit, proof load, drop, fatigue, thermal, optical, and cable-flex tests remain first-build work.

## Load and stability calculations

The documented21.5kg-cm DS3218 stall torque at6.8V is about2.11N·m. Torque at the selected6V rail and continuous safe torque are not documented. Never treat stall torque as a holding-duty rating. A conservative folded standard-head estimate remains about0.40N·m at the shoulder; the optional DSI head estimate remains about0.48N·m. Wider link shells and actual wiring change mass, so weigh the build and repeat the calculation.

With a0.32m horizontal head moment arm, the standard0.40kg head alone produces about1.26N·m; the optional0.50kg head produces about1.57N·m, before link mass and acceleration. That is too close to the servo's6.8V stall rating for claimed continuous operation, and the project runs at6V. The restricted folded pose is deliberate. Reduce reach, add counterbalance, or change the actuator architecture before extended duty.

The1.19kg ballast alone gives a nominal1.26N·m restoring moment about the108mm base radius. This excludes dynamics, cable pulls, foot friction, and the raised center of mass. Use a commercial padded desk clamp if the measured build exceeds the validated stable envelope. This is not a lifting arm.

## Drawing and verification limits

Assembly matrices are row-major4×4 in millimeters. Right-side link halves are rotated180° about their length axis for assembly. Purchased horns, bearings, screws, wiring, and exact PCA9685 geometry are not printable substitutes. `validation/assembly_check.json` covers printable solids at the indexed pose. `cad/purchased_fit.json` covers conservative static DS3218, sensor-board, and display envelopes. Neither report certifies moving clearance, strength, electrical safety, or physical fit.

# LUMA electrical design, revision C

The Raspberry Pi 4 is the behavior and motion master. It reads five Adafruit 3967 VL53L1X STEMMA QT sensors through an I2C mux, commands four inventory Miuzei DS3218MG servos through the user-selected PCA9685 PWM board, selects face expressions, produces local speech, and drives both light rings.

The standard head uses the **Waveshare ESP32-S3-Touch-LCD-1.85, SKU 28514, without B or C suffix**. It is a 360 × 360 TFT LCD, not OLED. Its ESP32-S3 is a USB graphics peripheral; its Wi-Fi, touch, microphone, and battery functions are unused. Do not substitute a visually similar 1.85B, 1.85C, AMOLED, or bare panel. See Waveshare documentation.

```
Pi4 USB ─┬─ Waveshare ESP32-S3 1.85 LCD / face firmware
         └─ USB speaker / local espeak-ng speech
Pi I2C ──┬─ PCA9685 address0x40 — PWM channels0..3 — four DS3218MG
         └─ PCA9548/TCA9548A-compatible mux address0x70 — ports0..4 — five VL53L1X
Pi BCM10 SPI MOSI — 74AHCT125 +330Ω — 60RGB outer ring
Pi BCM18 PWM0    — 74AHCT125 +330Ω — 24RGBW inner ring (white only)
Pi BCM27        — isolated NC stop contact — GND
12V relay output — 7.5A fuse — 6V/11A regulator — four fused servo feeds
```

## Optional 4-inch DSI head

The optional head replaces the USB display with a **Waveshare 4inch DSI LCD (C)**. It is a round 720 × 720 IPS panel in a 126 mm case. The Pi renders the face directly with `display_kind: "dsi"`; there is no display coprocessor or serial heartbeat. Connect its 15-pin 1.0 mm DSI FPC and connect 5 V/GND from Pi header pins4/39 to the display's HP2.0 socket. Leave that socket's SDA/SCL contacts empty. Touch and backlight control use `I2C_bus=10` on the DSI connector, and LUMA does not read touch input.

The FPC run through the arm is about800 mm. That is longer than the vendor-supplied50mm cables and must be proven on the bench before the links close. Keep it away from the moving 6 V servo feeds. This head omits the24-pixel RGBW ring; the outer halo supplies capped white output. Vendor page.

## Power, stop, and servo distribution

Use the official Pi5.1 V/3 A USB-C supply. Use the external Mean Well GST220A12-R7B 12 V/15 A supply for the LED buck converter and the servo regulator. The enclosure contains DC only. All grounds meet at one DC star. Pi5 V and LED5 V remain separate. Wire the locking R7B connector exactly as its drawing specifies. Mean Well specification.

The four Miuzei DS3218MG units are 4.8-6.8 V PWM position servos. The vendor datasheet gives 2.2 A stall current and21.5 kg-cm stall torque at6.8 V. Four can therefore demand8.8 A at once. This is an abnormal design transient, not a permitted continuous operating point. Use the Pololu D42V110F6 fixed6 V regulator. Its nominal rating is11 A, while its actual continuous output depends on input voltage, cooling, and load. Verify6 V polarity and loaded voltage before connecting a servo. DS3218 datasheet, regulator specification.

A15 A main fuse protects the12 V trunk. A normally-open12 V relay interrupts only the servo regulator input. A latching stop's first NC contact interrupts the fused relay coil. Its second isolated NC contact grounds BCM27 in the normal state. Add the coil flyback diode with its stripe at coil positive. **The stop must remove servo power with the Pi disconnected.** Pi, display, sensing, and lights stay powered so they can report a stopped motor rail.

Feed the regulator through its own7.5 A time-delay fuse. Put a2200 µF/10 V or higher low-ESR capacitor across the6 V star. Feed each servo from that star through its own3 A time-delay fuse and18 AWG V+/GND pair. Do not send the possible8.8 A group current through a generic PCA9685 terminal block or PCB trace. At the PCA9685, connect only each servo's signal and ground; route V+ from the fused star. Confirm actual connector polarity instead of relying only on wire color.

Power PCA9685 VCC from Pi3.3 V. Connect SDA/SCL to Pi physical pins3/5, connect its ground to the DC star, and hold active-low `/OE` low. The PCA9685 address is0x40 and the sensor mux address is0x70. Software uses channels0,1,2,3 for yaw, shoulder, elbow, wrist at50 Hz. The documented DS3218 pulse range is500-2500 µs with1500 µs nominal neutral over the270-degree variant. The PWM board reports I2C communication only. It cannot report servo shaft position, load, current, or temperature. PCA9685 guide.

The Pololu D24V50F5 still converts12 V to the separate LED5 V rail through its2 A input and3 A output fuses. Software caps the outer RGB ring at20%. The outer ring allowance at that cap is about0.72 A. The inner ring uses RGB=0 and W no higher than192/255, about0.36 A. Put a1000 µF/10 V capacitor at each ring and a100 nF ceramic at the74AHCT125. Exact current and temperature require measurement.

## Five pedestal sensors and I2C allocation

All five Adafruit3967 boards mount behind open apertures in the216 mm pedestal wall. Looking down with the face aimed toward+Y, their centers are front90°, front-left162°, rear-left234°, rear-right306°, and front-right18°. Mux ports0 through4 use that same order. The two rear sectors can request a wink. The two front-side sectors can request the happy gesture. Every sector is also a mandatory obstruction input.

Each board remains at address0x29 on its own mux channel. The mux is the Adafruit5626 PCA9548, which uses the TCA9548A-compatible driver. Both the mux and PCA9685 share the upstream Pi I2C bus at3.3 V logic. Put the mux on the electronics tray in the pedestal. The five branch cables now remain in the fixed base and do not pass through yaw or the arm.

Software ranges only one emitter at a time with a33 ms short-mode timing budget. A normal five-channel sweep is about0.25-0.4 seconds. A missing sensor, invalid/no-return value, reading older than650 ms, or valid range below100 mm latches a motion hold. The five narrow horizontal cones do not cover every pinch point and are not a safety-rated perimeter.

Remove each shipping film. Keep every10 × 10 mm wall aperture open; no frosted plastic or uncharacterized IR window belongs in front of the sensor. Mount each25.40 × 17.78 mm board component-side toward its opening using fourM2.5x6 screws on its20.32 × 12.70 mm hole pattern. Tighten only until the PCB is stable. Leave side clearance for one STEMMA QT plug and add strain relief inside the base.

## Motion indexing without joint feedback

The DS3218MG has one25T output spline and no readable position channel. Yaw remains supported by the6808 bearing. Each pitch joint uses the servo's straight metal arm on the driven side and a625 bearing/M5 axle on the opposite side. The supplied metal arm's attachment holes are not dimensioned in the datasheet, so the printed interface has two radial slots. Measure each actual horn and select fasteners before applying load.

Center each disconnected, unloaded servo at1500 µs before installing its horn. The mechanical neutral pose is upper link115° from+Y horizontal, forearm45° from+Y, relative elbow−70°, and face vertical toward+Y. The shoulder axis isZ121 mm after the6 mm pedestal increase. Initial command limits remain yaw±25° and shoulder/elbow/wrist±10°. Calibrate each `neutral_pulse_us` and `joint_signs` value from the real mechanism; all1500/all+1 values are only starting values.

Hardware motion stays off unless both `calibrated=true` and `--arm` are supplied. Arming verifies fresh sensor data, display health, a closed motor stop, and PCA9685 communication, then sends the four calibrated neutral pulses. It cannot verify the physical starting angles. Support the complete arm before arming because a wrong pulse or horn index can cause immediate motion.

A behavior fault stops target updates and leaves the PCA9685 at its last commanded pulse widths. A PCA9685 I2C failure also stops new commands, but the controller or servo can retain its last output. Use the independent physical motor stop for unexpected motion. A motor-power cut releases holding torque. No software watchdog, encoder feedback, current feedback, thermal feedback, or safety-rated stop controller is claimed.

## Bench commissioning sequence

1. Keep all four horns disconnected from the links. Open the motor stop. Verify12 V input, LED5 V, Pi3.3 V logic, and the regulated6 V servo output separately. Confirm the2200 µF capacitor polarity.
2. With servo V+ still disconnected, run `i2cdetect -y 1`. Confirm PCA9685 at0x40 and the mux at0x70. Check every PCA9685 signal lead and all five mux branches against `wiring.csv`.
3. Test the five sensors with `.venv/bin/python -m luma.bench sensors --seconds 10`. Cover one pedestal aperture at a time and confirm the reported port/name order.

4. Connect one unloaded servo and one3 A fused branch. With the stop in reach, run `.venv/bin/python -m luma.bench servo --joint yaw --pulse-us 1500 --seconds 2 --confirm-unloaded`, changing `--joint` for each channel. The command disables the other three PWM channels and removes the selected PWM pulse when the bounded check ends. Mark the resulting physical neutral.
5. Power off. Fit each straight metal arm to its indexed servo. Dry-fit its two printed radial slots and the opposite625/M5 support. Confirm free rotation, bearing alignment, smooth-shank contact at the inner race, and no bolt contact with the servo case.
6. Test the standard or DSI display and both LED rings. Verify that outer RGB is at20% or less and inner W is at192/255 or less. Measure LED, regulator, Pi, and enclosure temperatures during a30-minute run.
7. Verify that the physical stop removes voltage at the6 V star with the Pi unplugged. Verify BCM27 reports stop when either NC2 wire is removed. Reset only while the head is supported.
8. Enter measured neutral pulses and joint signs in `config.json`. Set `calibrated=true` only after every unloaded joint moves in the correct direction. Run hardware mode without `--arm`, then one supported6-second scene with `--arm`.
9. Check the complete limited motion for cable clearance, collision, stability, servo/regulator temperature, and loaded6 V droop. Reduce travel or redesign the counterbalance if any joint cannot sustain the pose.

## Evidence and limits

Host behavior and PWM-conversion tests run without hardware. The CAD uses the vendor's body envelope and the Adafruit board's published mounting coordinates. Physical servo fit, horn-hole fit, PCA9685 board revision, regulator cooling, electrical continuity, holding torque, optical range, and thermal behavior remain first-build checks. Budget figures are planning allowances, not quotations.

# Point-to-point wiring schedule

Use these connection names together with the electrical chapter. Physical Raspberry Pi header numbers and BCM GPIO numbers are explicitly distinguished. Disconnect the supplies while changing wiring. Nets starting with `DSI_` apply only to the optional 4-inch DSI head and replace `FACE_USB`.

| Net | From → to | Cable / connection detail |
|---|---|---|
| 3V3_LOGIC | Raspberry Pi4: physical1 3V3 → Mux5626: VIN | STEMMAQT upstream red. All I2C pullups stay3.3V |
| GND_LOGIC | Raspberry Pi4: physical6 GND → Mux5626: GND | STEMMAQT upstream black. Connect DCSTARground also |
| I2C_SDA | Raspberry Pi4: physical3 BCM2 → Mux5626: SDA | STEMMAQT upstream blue. I2C bus1 at100kHz |
| I2C_SCL | Raspberry Pi4: physical5 BCM3 → Mux5626: SCL | STEMMAQT upstream yellow. I2C bus1 at100kHz |
| TOF_FRONT | Mux5626: port0 → VL53L1X front pedestal: STEMMAQT4pin | 4conductor JSTSH. Address0x29 isolated on channel0; wall angle90deg |
| TOF_FRONT_LEFT | Mux5626: port1 → VL53L1X front-left pedestal: STEMMAQT4pin | 4conductor JSTSH. Wall angle162deg |
| TOF_REAR_LEFT | Mux5626: port2 → VL53L1X rear-left pedestal: STEMMAQT4pin | 4conductor JSTSH. Wall angle234deg |
| TOF_REAR_RIGHT | Mux5626: port3 → VL53L1X rear-right pedestal: STEMMAQT4pin | 4conductor JSTSH. Wall angle306deg |
| TOF_FRONT_RIGHT | Mux5626: port4 → VL53L1X front-right pedestal: STEMMAQT4pin | 4conductor JSTSH. Wall angle18deg; leave every optical aperture clear |
| PWM_VCC | Raspberry Pi4: physical1 3V3 → PCA9685: VCC | 22AWG. Powers controller logic only; never powers a servo |
| PWM_GROUND | DCstar: GND → PCA9685: GND | 18AWG. Common reference to Pi sensor bus and all servo signal returns |
| PWM_SDA | Raspberry Pi4: physical3 BCM2 → PCA9685: SDA | 26AWG. Shared upstream I2C bus; address0x40 |
| PWM_SCL | Raspberry Pi4: physical5 BCM3 → PCA9685: SCL | 26AWG. Shared upstream I2C bus;50Hz PWM configured in software |
| PWM_ENABLE | DCstar: GND → PCA9685: /OE | 26AWG. Active-low output enable held low |
| RGB_DATA3V3 | Raspberry Pi4: physical19 BCM10 MOSI → 74AHCT125: DIP2 1A | 26AWG. Do not connect SPI clock to ring |
| RGB_DATA5V | 74AHCT125: DIP3 1Y → RGB ring quadrant1: DIN via330ohm | 26AWG. Resistor at first ring input |
| RGB_CHAIN1 | RGB quadrant1: DOUT → RGB quadrant2: DIN | solder jumper. Connect common5V andGND at all quadrants |
| RGB_CHAIN2 | RGB quadrant2: DOUT → RGB quadrant3: DIN | solder jumper. Mechanical backing supports solder joints |

| Net | From → to | Cable / connection detail |
|---|---|---|
| RGB_CHAIN3 | RGB quadrant3: DOUT → RGB quadrant4: DIN | solder jumper. Do not connect lastDOUT back to firstDIN |
| WHITE_DATA3V3 | Raspberry Pi4: physical12 BCM18 PWM0 → 74AHCT125: DIP5 2A | 26AWG. Disable onboard analog audio |
| WHITE_DATA5V | 74AHCT125: DIP6 2Y → 24RGBW inner ring: DIN via330ohm | 26AWG. SeparateGRBW data stream |
| SHIFT_ENABLE | DCstar: GND → 74AHCT125: DIP1 /1OE andDIP4 /2OE | 26AWG. Active outputs; tie both low |
| SHIFT_UNUSED | 5VLED: +5V → 74AHCT125: DIP10 /3OE andDIP13 /4OE | 26AWG. Unused outputs disabled |
| SHIFT_UNUSED_INPUT | DCstar: GND → 74AHCT125: DIP9 3A andDIP12 4A | 26AWG. UnusedDIP8 andDIP11 outputs unconnected |
| SHIFT_SUPPLY | 5VLED: +5V → 74AHCT125: DIP14 VCC | 24AWG. 100nF betweenDIP14 andDIP7 |
| SHIFT_GROUND | DCstar: GND → 74AHCT125: DIP7 GND | 24AWG. Shared reference toPi andboth rings |
| PI_GROUND | Raspberry Pi4: physical9 GND → DCstar: GND | 22AWG. Common ground only; never tie5VLED toPi5V |
| PI_POWER | Pi official supply: USB-C5.1V3A → Raspberry Pi4: USB-Cpower | factory cable. Independent from motor12V |
| FACE_USB | Raspberry Pi4: USB-Aport1 → DisplaySKU28514: USB-C | data cable. Do not add a second5V power feed |
| AUDIO_USB | Raspberry Pi4: USB-Aport3 → USB speaker3369: USB | captive cable. LimitUSB total current toPi specification |
| DC_INPUT | MeanWellGST220A12: R7B+12V → Main15Afuse: input | 16AWG. Use correctparallel contacts from supply drawing |
| DC_MAIN | Main15Afuse: output → DC12Vdistribution: +12V | 16AWG. Supply remains outside printed enclosure |
| DC_RETURN | MeanWellGST220A12: R7Breturn → DCstar: GND | 16AWG. All high-current returns star here |
| STOP_COIL_FEED | DC12Vdistribution: +12V → 0.5Afuse: input | 22AWG. Coil branch |
| STOP_NC1 | 0.5Afuse: output → StopNC1: COM | 22AWG. NoPiGPIO in this branch |
| STOP_COIL_PLUS | StopNC1: NC → Relay: coilpositive | 22AWG. Diodecathode stripe here |
| STOP_COIL_MINUS | Relay: coilnegative → DCstar: GND | 22AWG. Diodeanode here |
| MOTOR_RELAY_IN | DC12Vdistribution: +12V → Relay: COM highcurrent | 16AWG. NOcontact>=15A12VDC |
| MOTOR_RELAY_OUT | Relay: NO highcurrent → Servo regulator7.5Afuse: input | 16AWG. Stop removes regulator input and all motor V+ independent of software |
| SERVO_REG_INPUT | Servo regulator7.5Afuse: output → PololuD42V110F6: VIN | 16AWG. 12V nominal input; mount with airflow |

| Net | From → to | Cable / connection detail |
|---|---|---|
| SERVO_6V | PololuD42V110F6: VOUT6V → Servo4way fusedstar: input | 16AWG. Verify6V and polarity before connecting any servo |
| SERVO_BULK | Servo6Vstar: +6V/GND → 2200uF10Vcapacitor: +/- | short18AWG. Observe polarity and place near star distribution |
| SERVO_POWER1 | Servo3Afuse1: output → YawDS3218MG: V+ | 18AWG. Separate6V feed; do not carry power through PCA9685 traces |
| SERVO_POWER2 | Servo3Afuse2: output → ShoulderDS3218MG: V+ | 18AWG. Separate6V feed |
| SERVO_POWER3 | Servo3Afuse3: output → ElbowDS3218MG: V+ | 18AWG. Separate6V feed |
| SERVO_POWER4 | Servo3Afuse4: output → WristDS3218MG: V+ | 18AWG. Separate6V feed |
| SERVO_GROUNDS | AllfourDS3218MG: GND → DCstar: GND | 18AWG. One power return per servo; common with PCA9685 logic ground |
| SERVO_PWM1 | PCA9685: channel0 signal → YawDS3218MG: PWM | 26AWG. 500-2500us range;1500us nominal neutral; signal and ground only at controller |
| SERVO_PWM2 | PCA9685: channel1 signal → ShoulderDS3218MG: PWM | 26AWG. Confirm direction before fitting loaded horn |
| SERVO_PWM3 | PCA9685: channel2 signal → ElbowDS3218MG: PWM | 26AWG. Confirm direction before fitting loaded horn |
| SERVO_PWM4 | PCA9685: channel3 signal → WristDS3218MG: PWM | 26AWG. Confirm direction before fitting loaded horn |
| STOP_SENSE | Raspberry Pi4: physical13 BCM27 → StopNC2: COM | 26AWG. External10kohm toPi3.3V |
| STOP_SENSE_RETURN | StopNC2: NC → Raspberry Pi4: physical14 GND | 26AWG. Opencontact orbrokenwire readsstop |
| BUCK_INPUT | DC12Vdistribution: +12V via2Afuse → PololuD24V50F5: VIN | 22AWG. ENleftunconnected |
| BUCK_GND | PololuD24V50F5: GND → DCstar: GND | 22AWG. Shortreturn |
| LED_FUSED | PololuD24V50F5: VOUT5V → 3Afuse: input | 22AWG. Verify5VbeforeconnectingLEDs |
| LED_RGB_POWER | 3Afuse: output5V → Outer60RGB: 5V atoppositequadrants | 22AWG. 1000uF10Vcap at ring |
| LED_WHITE_POWER | 3Afuse: output5V → Inner24RGBW: 5V | 22AWG. 1000uF10Vcap at ring |
| LED_RETURNS | Bothrings: GND → DCstar: GND | 22AWG. KeepLEDreturnseparatefromI2Creturnuntilstar |
| DSI_DATA | Raspberry Pi4: DSI connector → 4inch DSI LCD (C): 15-pin DSI FPC | 15-pin1.0mm FFC800mm type A. Optional head only;replaces FACE_USB;route through both arm cable ports with a service loop at each joint |
| DSI_POWER | Raspberry Pi4: physical4 5V → 4inch DSI LCD (C): HP2.0 4-pin 5V | 22AWG. Optional head only;display draws from the Pi supply in place of the USB display |

| Net | From → to | Cable / connection detail |
|---|---|---|
| DSI_GROUND | Raspberry Pi4: physical39 GND → 4inch DSI LCD (C): HP2.0 4-pin GND | 22AWG. Optional head only;SDA and SCL contacts of this plug stay empty;touch and backlight use the DSI connector I2C bus10 |

# LUMA Pi4 controller and face firmware

The project includes a deterministic simulator and Raspberry Pi adapters for five pedestal ranging sensors, four Miuzei DS3218MG servos through a PCA9685, two independent LED rings, USB face display and local spoken Hi. The simulation needs Python3.11 or later.

```
cd software
python -m luma.app --scene wink
python -m luma.app --scene hi
python -m luma.app --scene happy
python -m unittest discover -s tests -v
```

Simulation prints timestamped pose/light/face events as JSON. It advances deterministically without sleeping. `--seconds` sets duration. Scenes last6seconds, then return to idle. The three promotional scenes remain within the initial CAD travel envelope; they are repeatable face/motion scripts and do not require a cloud service.

## Raspberry Pi installation

Use Raspberry Pi OS Bookworm64bit on Pi4. The following setup commands affect only a Pi used for this project; do not execute them on the development Windows computer.

```
sudo apt update
sudo apt install python3-venv python3-pip python3-dev i2c-tools espeak-ng alsa-utils
sudo raspi-config
```

In Interface Options enable I2C and SPI. In `/boot/firmware/config.txt`, confirm `dtparam=i2c_arm=on`, `dtparam=i2c_arm_baudrate=100000`, and `dtparam=spi=on`. Set `dtparam=audio=off` because BCM18's PWM peripheral drives theRGBW ring and analog audio conflicts. USB audio is separate. Reboot after these settings.

```
cd software
python3 -m venv .venv
.venv/bin/pip install -e '.[hardware]'
ls -l /dev/serial/by-id/
aplay -l
```

Enter the actual display `/dev/serial/by-id/` path in `config.json`. The PCA9685 is I2C address64 (`0x40`), with yaw/shoulder/elbow/wrist on channels0/1/2/3. Select the USB speaker as the ALSA output or set `audio_device` to the identifier from `aplay -L`. The standard display and speaker share Pi4 USB power; use a powered hub if measurement requires it. Motors and LEDs never take power from Pi USB or GPIO.

Before connecting motor power, test the sensor chain and display independently. These commands do not instantiate the PCA9685 or LED drivers and do not require calibrated joints:

```
.venv/bin/python -m luma.bench sensors --seconds 10
.venv/bin/python -m luma.bench display --port /dev/serial/by-id/YOUR_DISPLAY_DEVICE --scene all
--seconds 10
```

The sensor check reports millimeters, age and mux channel in front/front-left/rear-left/rear-right/front-right order. Present a matte target at each pedestal aperture. `near` means a valid value below100mm: communication can pass, but armed motion would fault. The display check cycles idle/wink/Hi/happy and checks firmware replies. Inspect the pixels yourself. Exit status0 is pass,1 fail and130 interrupted. I2C normally needs membership in the `i2c` group; USB serial normally needs `dialout`.

Center only one disconnected, unloaded servo at a time. Keep its straight horn off the arm and keep the physical stop in reach. The command is capped at10seconds, disables the other three PCA9685 channels, and removes the selected PWM pulse when it exits:

```
.venv/bin/python -m luma.bench servo --joint yaw --pulse-us 1500 --seconds 2 --confirm-unloaded
```

Repeat with `shoulder`, `elbow` and `wrist`. Power off before installing each horn. The DS3218 datasheet gives500-2500µs over270° with1500µs nominal neutral. Record the actual mechanical neutral in `neutral_pulse_us`; do not use this command on a loaded joint.

The PWM-backed NeoPixel library normally requires root on Pi4, so the full hardware command uses the virtual environment interpreter with sudo. Leave `calibrated=false` while wiring/testing. Hardware mode without `--arm` requires PCA9685 communication but leaves all four PWM outputs disabled; the separate6V motor rail can remain off.

```
sudo .venv/bin/python -m luma.app --hardware --scene hi --seconds 6
```

Complete the commissioning procedure in `../electronics/architecture.md`, including unloaded neutral indexing, polarity/sign checks, physical stop tests and five valid pedestal ranges. Only then set `calibrated=true`. Support the arm in its indexed pose before every armed start; software cannot read its physical angles:

```
sudo .venv/bin/python -m luma.app --hardware --arm --scene happy --seconds 6
```

Any fault latches for that process. It stops new targets and leaves the PCA9685 at the last pulse widths. Inspect the reason, support the arm, remove the cause and restart explicitly. PCA9685 health proves I2C communication only, not servo movement or position. The physical motor stop removes6V holding power. There is no unattended boot or auto-arm service.

## Display firmware

`display_firmware/` targets **ESP32-S3-Touch-LCD-1.85 SKU28514 only**. It uses USB serial115200, an ST77916 QSPI controller on CS21/SCK40/D046/D145/D242/D341, and a TCA9554 at0x20 over SDA11/SCL10 for LCD resetEXIO2/P1. Backlight is GPIO5. This pin mapping is from the Waveshare board documentation and schematic. Firmware reads panel register0x04 and chooses one of the exact two vendor initialization tables: ID00-7F-7F-7F or00-02-7F-7F. Unknown IDs produce an error and do not advertise READY. Register command/data/delay values were converted from the vendor's April2025 demo to Arduino_GFX operations. A successful compile does not prove physical display operation; verify the purchased panel on the bench.

Install PlatformIO and build/flash the connected board:

```
python -m pip install platformio
cd display_firmware
python -m platformio run
python -m platformio run --target upload --upload-port YOUR_SERIAL_PORT
python -m platformio device monitor --port YOUR_SERIAL_PORT --baud 115200
```

Commands are newline-delimited ASCII. `PING` replies `PONG`. `FACE idle`, `FACE wink`, `FACE hi`, `FACE happy`, and `FACE fault` reply `OK <name>`. Startup announces `READY LUMA1`. Invalid or overlong commands reply `ERR`. After1.5seconds without a host command/heartbeat the display shows WAITING. The host sends heartbeat every250ms and latches a motion hold if replies stop. No touch/audio/SD/battery functions are enabled on the display board; speech is generated on the Pi and played by the USB speaker.

## Optional 4-inch DSI head (Pi-rendered face)

Instead of the ESP32 face board, LUMA can use a Waveshare 4inch DSI LCD (C) — a round720x720 panel wired directly to the Pi's DSI port. There is no coprocessor for this head: the Pi itself renders the face every frame with `luma/face.py`, which reproduces `paint()` from `display_firmware/src/main.cpp` in Python (same colors, same shapes, same scene timing) scaled from the firmware's360x360 canvas up to the panel's720x720 resolution.

Select this head by setting `"display_kind": "dsi"` in `config.json` (leave `display_port` alone; it is ignored for this kind). Install the extra dependency alongside the existing hardware group:

```
.venv/bin/pip install -e '.[hardware,dsi]'
```

Add these lines to `/boot/firmware/config.txt` for the Waveshare4inch DSI LCD(C):

```
dtoverlay=WS_xinchDSI_Screen,SCREEN_type=10,I2C_bus=10
dtoverlay=WS_xinchDSI_Touch,I2C_bus=10
```

Both overlays are the vendor's documented setup; LUMA's software never reads touch events — the lamp has no touch-driven behavior. If the picture is upside down for the mounted orientation (connectors at the bottom), rotate it with the vendor's documented `display_lcd_rotate`/Wayland rotation setting rather than in LUMA.

Running outside a desktop session (the normal case for an unattended lamp) needs pygame's KMS/DRM backend. `sudo` drops environment variables, so pass the driver through `env`:

```
sudo env SDL_VIDEODRIVER=kmsdrm .venv/bin/python -m luma.app --hardware --scene hi --seconds 6
```

Bench-check the head on its own, the same way as the USB face board:

```
.venv/bin/python -m luma.bench display --kind dsi --scene all --seconds 10
```

`--port` is not required (and is ignored) when `--kind dsi`; it stays required for the default `--kind usb_serial`. There is no serial heartbeat to a coprocessor for this head, so `Hardware/bench.display`'s notion of "heartbeat" becomes "a frame was drawn within the last1.5seconds" — `DsiDisplay.update()` returns that boolean the same way `Display.update()` returns the USB PONG heartbeat, so the rest of the control/fault-latching logic in `luma/control.py` is unaffected by which head is attached.

## Source/verification notes

`luma/servo_pwm.py` uses the maintained Adafruit CircuitPython PCA9685 library at50Hz. It converts calibrated angles through the DS3218's documented500-2500µs/270° range, starts with all four outputs disabled, and checks controller registers over I2C. The bounded bench path enables one unloaded channel only. Tests cover the duty conversion, channel isolation, config validation, scene limits, five renamed sensor inputs and fault behavior. Hardware paths have not been exercised on a physical Pi/arm.

The pinned firmware toolchain is self-contained in `platformio.ini`; downloaded SDK/toolchain caches are development artifacts and do not need redistribution. Python hardware dependency versions are bounded for installation flexibility, not a physical acceptance-tested lockfile. Record actual installed versions with `pip freeze` when commissioning the Pi.

# First-build verification and maintenance

## Verify before enclosing

Power and exercise the display, rings and each pedestal sensor on the bench. Confirm that the large ring shows the intended colors, the small ring uses only its white channel during lamp operation, and every wall direction is associated with its labeled multiplexer port.

Connect one unloaded DS3218MG at a time to its fused6V feed and PCA9685 channel. Use the bounded one-channel command in the software guide to center it at1500µs, then remove power before attaching the horn. Save each measured neutral pulse and direction with the assembled unit. PWM servos have no readable shaft position.

## Mechanical acceptance record

| Check | Acceptance and record |
| --- | --- |
| Printed interfaces | Actual purchased boards and horns seat without forced deflection; record component revision. |
| Fasteners | Required washers/nuts present, rotating surfaces free and wires not under screw heads. |
| Bearings and joints | Free motion when unpowered; no binding, visible cracks or loose axial play. |
| Cable routing | Slack remains throughout allowed travel; no pinched insulation or tight connector bends. |
| Stability | Base secured as specified; verify measured mass and center of gravity at the maximum allowed reach. |
| Head mass | Weigh the completed head and replace assumed mass in the torque calculation. |
| Horns | Metal horns secured with the servo's correct center screw; no printed spline substitutes. |
| Pitch idlers | Three625 bearings square in their pockets; M5 smooth shanks load only inner races; captive nuts seated; no axial bind or play. |
| Pedestal sensors | Five boards on four screws each; component face outward; apertures open; one strain-relieved STEMMA QT cable per board. |
| Enclosed links | DS3218 case seated in the41.2 ×20.7mm pocket; halves meet without crushing it; harness has slack; horn, idler and servo tab clear the commissioned travel. |
| Optional DSI head | Carrier pads bear on all four display bosses; face ring bears on the foam rim only, never on the glass; FFC bend radius above 10 mm at every turn; picture stable at full backlight for 30 minutes with the arm closed. |

## Electrical and motion acceptance

Start with the arm supported and a single joint enabled at reduced speed. Move a few degrees about the calibrated neutral position, verify the sign of rotation, then return to neutral. Repeat for each joint before a coordinated expression. A wrong sign or zero offset must be corrected before enlarging the motion range.

Confirm that the hardware motor-power disconnect removes the regulated6V rail independently of software. Support the head when testing a power cut. Verify the software fault response for a missing sensor, invalid range, lost PCA9685 communication and lost display connection. A live PCA9685 does not prove that a servo moved.

Run the normal white-light and RGB settings with the head assembled. Record temperatures at5,15 and30 minutes at the LED support, screen,6V regulator and most heavily loaded servo. The DS3218 gives no telemetry, so measure accessible surfaces directly. Use45°C on a printed enclosure surface as a conservative investigation threshold for the first PETG build, not as a product certification.

Verify front, front-left, rear-left, rear-right and front-right with a matte target at several distances. Record ranges and reject invalid readings rather than treating them as open space. Check for wall-edge reflections. These five horizontal cones do not cover every direction or arm pinch point.

## Verification worksheet

| Measurement | First-build result |
|---|---|
| Build date / builder | ______________________________ |
| Printer and filament | ______________________________ |
| Component revisions | ______________________________ |
| Completed head mass | ______________________________ |
| Base / ballast mass | ______________________________ |
| Motor supply at motion peak | ______________________________ |
| Regulated servo rail / PCA9685 check | ______________________________ |
| LED supply at normal brightness | ______________________________ |
| Pi undervoltage check | ______________________________ |
| Maximum observed enclosure / servo / regulator temperature | ______________________________ |
| Mechanical travel limits and neutral pulse widths | ______________________________ |
| Sensor directions and valid range test | ______________________________ |
| Hardware power-cut test | ______________________________ |

## Care and service

Inspect the horn screws, bearing mounts and cable strain relief after the first hour of motion and again after the first week. Stop using a printed part if a layer crack, whitening around a fastener or increasing joint play appears. Correct the cause before replacing the part.

Clean the diffuser with a soft damp cloth. Keep solvents away from printed surfaces and the display. Dust near a time-of-flight aperture can change measurements; remove it without pushing debris onto the optical package.

For service, shut down the Pi, disconnect the low-voltage supplies and support the arm before opening a joint or loosening the head. Store the final calibration, BOM revision and any dimensional changes with the project so the next printed replacement matches the machine.

# Finished design and promotional films

The promotional films show the LUMA digital prototype built from the supplied CAD geometry. The lighting, camera exposure, screen graphics and sound are presentation choices; the films do not establish the physical lamp's brightness, motor sound or production finish.

Finished LUMA digital prototype, rendered from the project geometry.

## The three films

| Film | Included file | Action |
|---|---|---|
| The wink | `media/videos/01_wink.mp4` | A brief one-eye wink with a coordinated head gesture and colored halo. |
| Hi, there | `media/videos/02_hi.mp4` | A friendly face greeting, arm gesture and an audible “Hi!” |
| Happy to see you | `media/videos/03_happy.mp4` | Cheerful eyes and a small happy movement, with warm halo color. |

Open `START_HERE.html` in a browser after extracting the archive to play all three clips. The videos are also individual files for editing or presentation use. The project includes editable animation sources under `media/sources/`.


Frames from the three included promotional animations.

## Assembly visualization

Exploded view of the prototype. Explode distances expose components and do not prescribe insertion travel.

Optional 4-inch DSI head, rendered from `cad/assembly_dsi.json` with the same enclosed arm and base.

The engineering sheets at the back of the manual are the dimensional references. The rendered images explain appearance and assembly relationships; use the numbered assembly sequence for fasteners, wiring and the optical stack.

# Digital release checks

The included validation files record what was checked on the development computer. They should travel with the source and manual when the project is revised. Passing these checks does not replace the first-build measurements in the preceding chapter.

## Executed host software checks

The controller, bench diagnostic, PWM and face-renderer suite passed38 tests. It exercises all three expressions, angle/slew limits, the five pedestal channel names and order, stale/invalid/near ranges, latched faults, stop polarity, lost display or PCA9685 communication, and the one-shot greeting.

Additional tests verify the500-2500µs/270° conversion, a known PCA9685 duty value, one-channel commissioning with the other channels disabled, neutral calibration gating, and white-channel limits. Face tests render the optional DSI output off-screen and check expression/fault pixels. The simulator and tests activate no physical hardware.

## Meshes and assembly

`cad/validation.json` records exporter mesh properties. `validation/assembly_check.json` covers27 printable types and reports zero printed-part overlaps above0.5mm³ in the standard24-instance and DSI21-instance assemblies. `cad/purchased_fit.json` reports no intersection for four DS3218MG case/tab envelopes, five Adafruit3967 pedestal-board envelopes, and either display envelope.

The intersection check is limited to the indexed pose. It excludes real horns/splines,625 bearings, screws and moving cables. It cannot prove the supplied straight-horn hole pattern, physical sensor alignment, or a collision-free moving envelope. The4-inch DSI connector position remains estimated from a vendor photograph.

## PDF and media

`validation/final_report.json` records the final PDF page/bookmark count, text bounds and full decoding of each promotional MP4. The verification script also generates a contact sheet of the manual for visual review. Video metadata and frame counts verify the encoded files; the animation is a visualization of the digital design. The optional head is rendered as a still only; the films show the standard head.

## Display firmware build status

The firmware source includes the exact-board pin mapping and the vendor's two panel initialization variants with attribution. **Its cross-compilation was not completed in this environment:** the required PlatformIO compiler download did not finish, and an alternate download was truncated. No firmware binary or successful cross-compile is claimed. Run the supplied PlatformIO build on a connected development computer before flashing, and test the purchased display on the bench. The optional DSI head does not use this firmware.

## Physical work still required

Actual Raspberry Pi GPIO/I2C/USB operation, PCA9685 output,6V regulator cooling, horn/idler fit, servo holding ability, electrical continuity, stop behavior, sensor optics, light diffusion, stability and enclosed temperatures have not been tested on a physical unit. The DS3218 servos expose no position/current/temperature telemetry. The800mm optional DSI cable is also untested. Record all first-build results on the worksheet.

# Component source register

Vendor information checked for this digital release. Order by exact part number and verify the revision supplied. Prices in the BOM are budgeting references, not quotations.

**S01** · Autonomous Lamp — appearance reference
Inspiration for an articulated desk companion; no commercial-lamp geometry reused.

**S02** · Raspberry Pi 4 Model B
Master computer and USB-C supply selection.

**S03** · Waveshare ESP32-S3-Touch-LCD-1.85 documentation
Exact integrated LCD board, display resolution, interface and vendor firmware reference.

**S04** · Adafruit VL53L1X STEMMA QT — product 3967
Five sensor boards, nominal dimensions and default I2C address.

**S05** · Adafruit VL53L1X guide
Sensor software and wiring reference.

**S06** · Adafruit RGBW 24-pixel ring — product 2862
Inner white-light ring, RGBW format and mechanical envelope.

**S07** · Adafruit 60-pixel RGB ring quarter — product 1768
Four supported arcs form the outer RGB halo.

**S08** · Adafruit NeoPixel rings guide
Ring diameter reference.

**S09** · Miuzei DS3218 6V 20kg servo datasheet
Inventory servo body dimensions, voltage, current, torque, PWM pulse range and 270-degree option.

**S10** · Adafruit PCA9685 16-channel PWM/servo driver guide
User-selected PCA9685 control topology, separate VCC/V+ power and Python driver.

**S11** · Pololu D42V110F6 6V 11A regulator
Regulated 6V servo rail, current capability, dimensions and mounting holes.

**S12** · Adafruit PCA9548 STEMMA QT multiplexer — product 5626
Independent channels for five sensors with identical addresses.

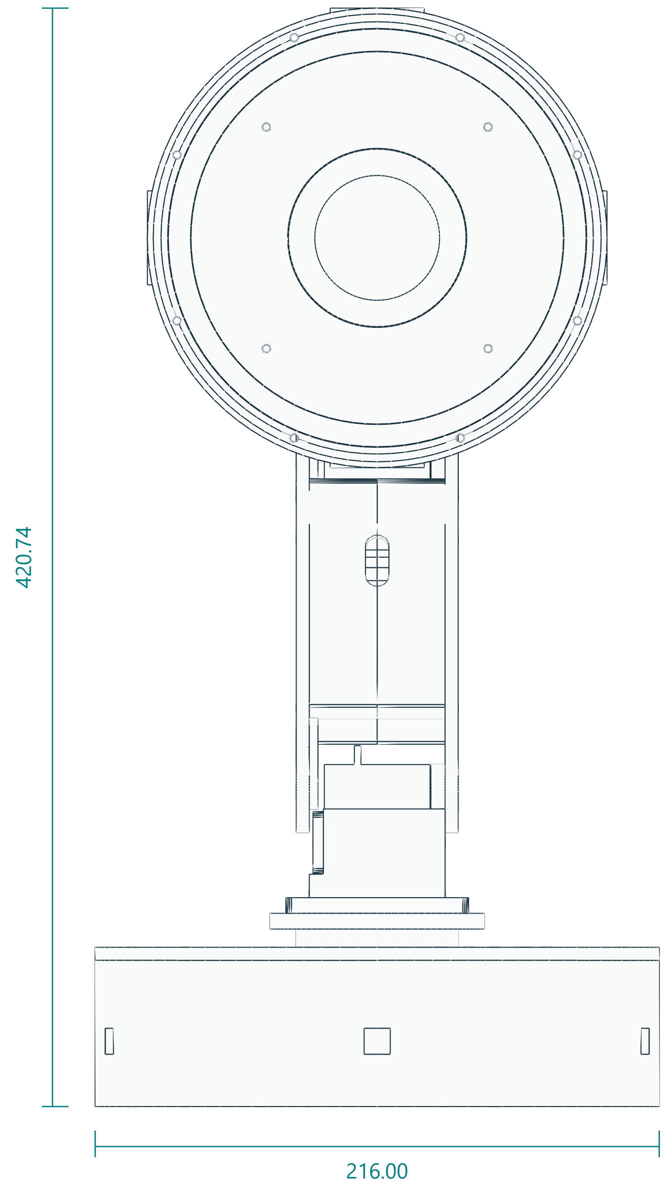
**S13** · Waveshare 4inch DSI LCD (C) — round 720×720 DSI display
Optional head display: outline, case thickness, M4 boss pattern and PCB envelope from the vendor drawing and STEP model; config.txt overlays from the vendor wiki.

**S14** · Adafruit VL53L1X PCB source
Exact25.40×17.78mm board outline and four2.5mm holes on20.32×12.70mm centers.

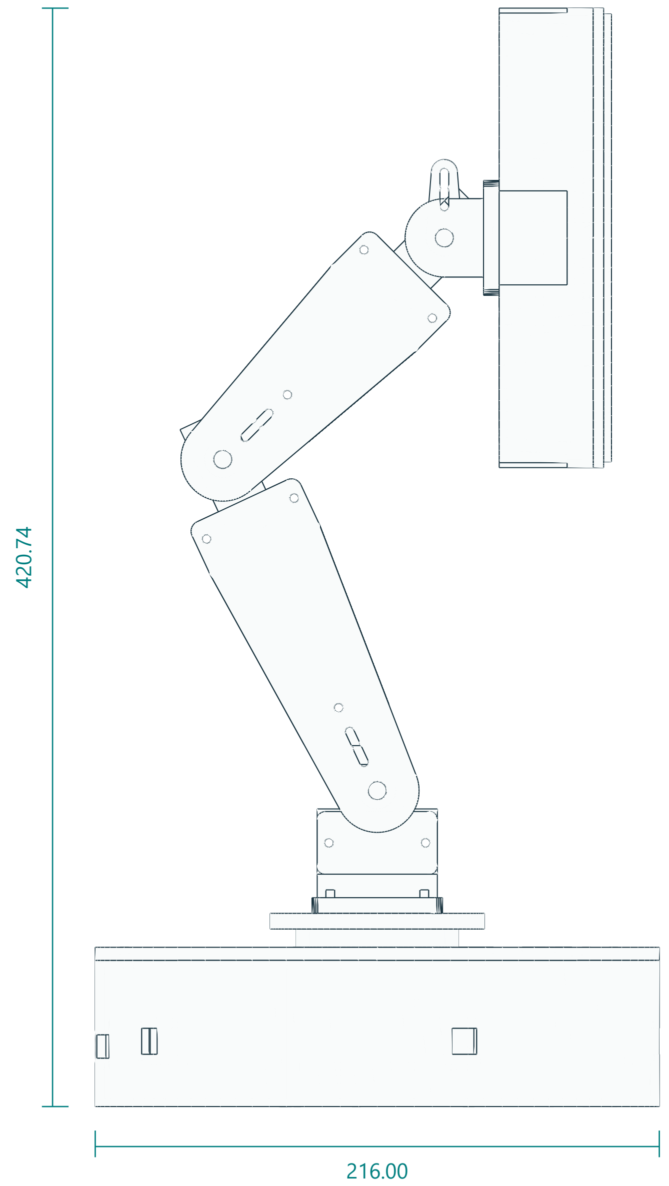
# Finished assembly / indexed neutral pose

Assembled geometry from the same STL files supplied for printing. Purchased parts are simplified envelopes.

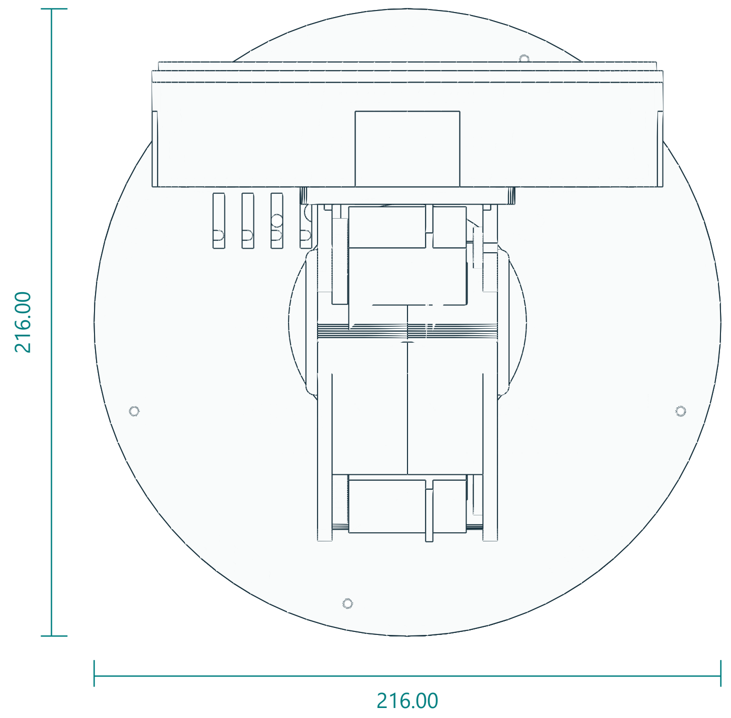
## FRONT / LCD FACING VIEW




## RIGHT SIDE / NEUTRAL POSE




## TOP / FOOTPRINT




Four axes: yaw about Z; shoulder, elbow and wrist about local X. Datum is the base underside at Z=0.
Nominal linkage pitches: shoulder–elbow 140 mm; elbow–wrist 120 mm. The illustrated height is a pose dimension, not a motion envelope.
Neutral: shoulder link 115° above horizontal; forearm 45° above horizontal; face directed toward +Y. Follow indexed calibration in the manual.
Five Adafruit3967 boards mount inside the pedestal wall. Four DS3218MG servos use driven straight horns and opposite625 supports at pitch joints.
Do not infer screw lengths from the illustration. Fasteners and service loops are specified in the assembly chapter.

**UNITS mm · THIRD-ANGLE PART VIEWS · NUMERICAL DIMENSIONS CONTROL**

Views fitted to sheet. Do not measure this PDF. Digital prototype; check physical fit before batch printing.

**G01**

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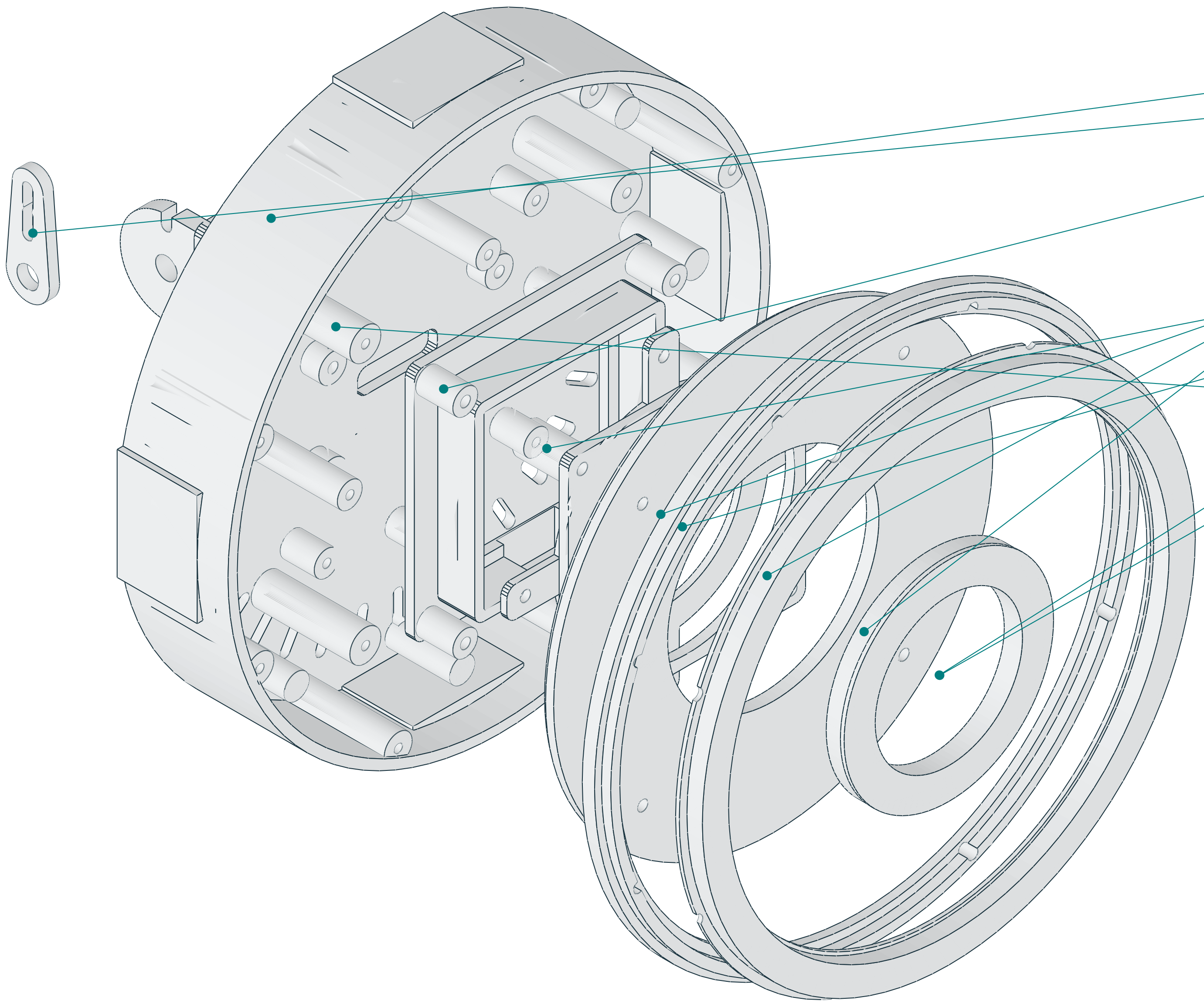
# Head / layered assembly study

Exploded positions expose the optical parts; displacements are illustrative and are not insertion paths.

### EXPLODED HEAD / MODEL-DERIVED VIEW



### LAYER REFERENCES

01 Head tilt fork
02 head driven horn spacer
03 Head Shell
04 Outer Bezel
05 Face Center
06 White Carrier
07 Lcd Cradle
08 Lcd Retainer
09 Outer Diffuser
10 Inner Diffuser
11 Purchased LCD envelope

Central LCD board is behind the inner white ring. Keep all optical components at their documented axial heights.
Use natural PETG for both light diffusers and opaque PETG for the optical baffle. Leave LCD glass clear; all ToF apertures are in the pedestal.
The four outer ring PCB quarters require continuous mechanical support. Solder joints are electrical connections only.
Refer to the head assembly chapter for screw order, spacers, wire relief and board retention. Purchased LEDs are not printable parts.

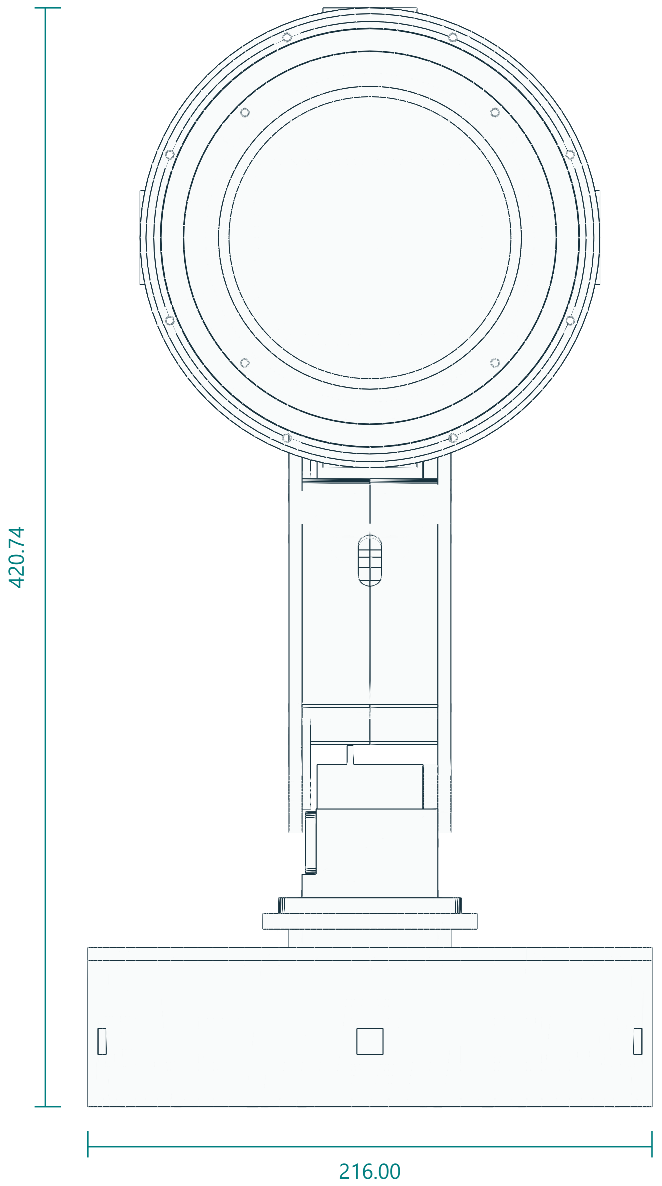
**UNITS mm · THIRD-ANGLE PART VIEWS · NUMERICAL DIMENSIONS CONTROL**

Views fitted to sheet. Do not measure this PDF. Digital prototype; check physical fit before batch printing.

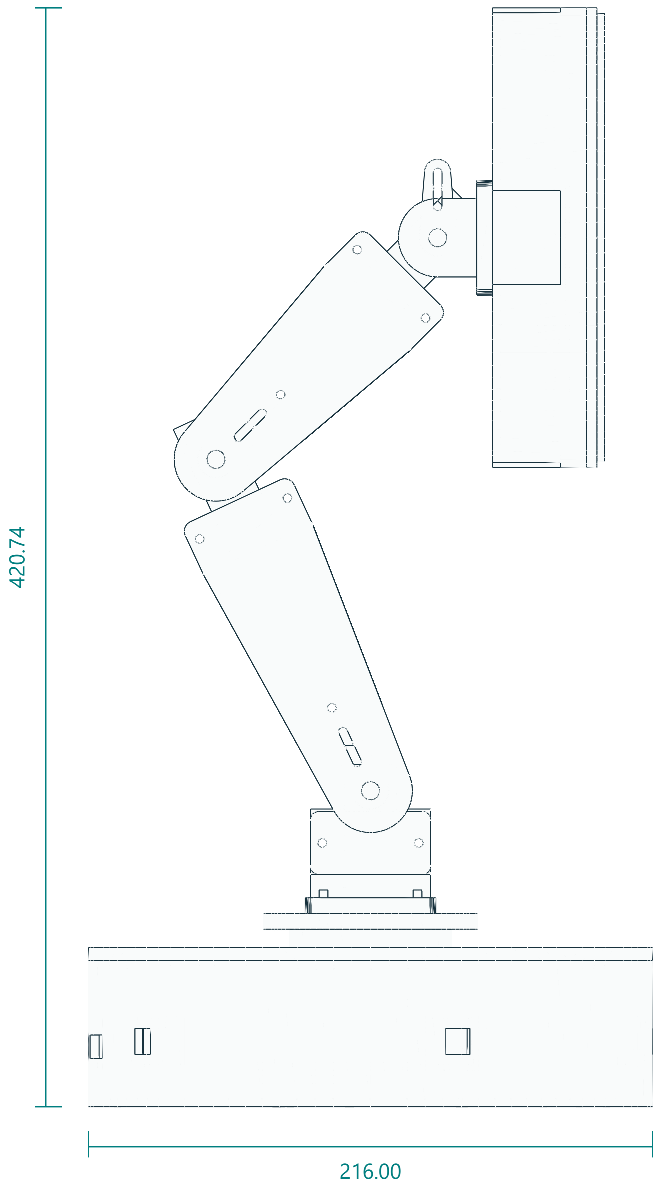
# Optional 4-inch DSI head / indexed neutral pose

Same sensor-equipped pedestal, arm links and fork as G01. Head shell, carrier and face ring are the optional parts; the display envelope is measured from the vendor STEP.

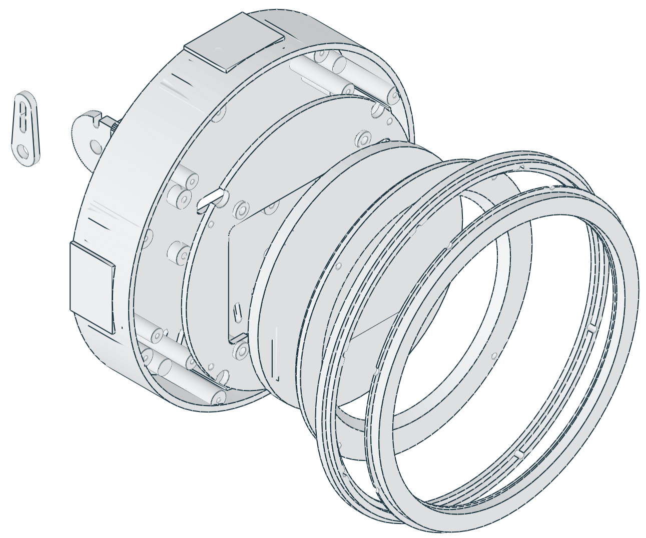
### FRONT / LCD FACING VIEW




### RIGHT SIDE / NEUTRAL POSE




### EXPLODED DSI HEAD



Waveshare 4inch DSI LCD (C): Ø126 case, 6 mm thick, four M4 bosses on a 75 × 75 pattern. Rim front at head Z35.5; carrier plate Z20.5–23.5 with pads to the boss ends at Z25.5.
Face ring opening Ø108 chamfered to Ø116 leaves a 3 mm black rim around the Ø101.5 active area. Outer60-RGB halo, bezel, diffuser and fork are shared with the standard head.
No inner white ring on this head: the halo provides capped white output. Revision C has no head-mounted sensors or brow bracket.
Cables: 15-pin DSI FFC plus 5 V/GND leave through the 26 × 8 slot in the shell floor and fork plate, then run inside the enclosed links through their wall ports.

**UNITS mm · THIRD-ANGLE PART VIEWS · NUMERICAL DIMENSIONS CONTROL**

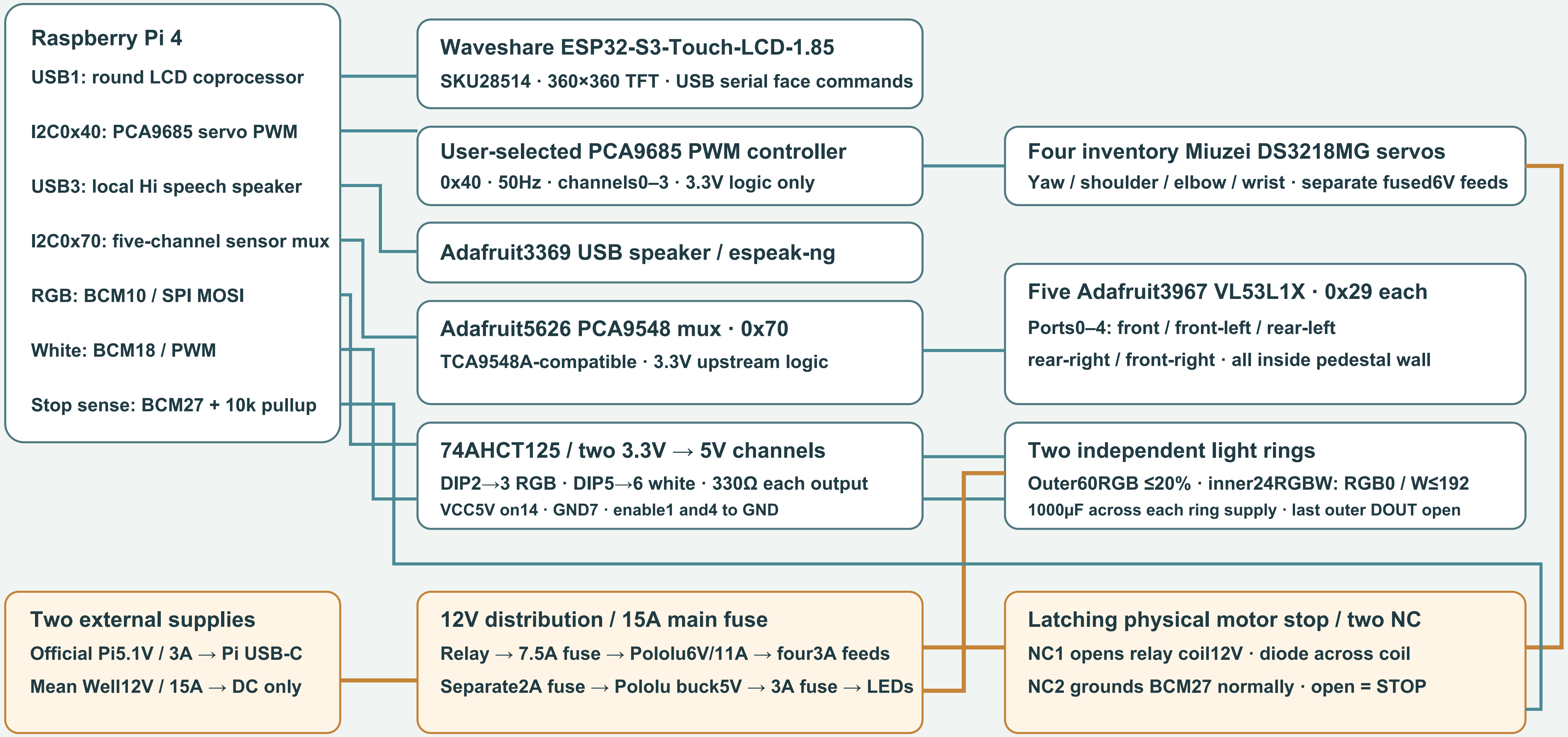
Views fitted to sheet. Do not measure this PDF. Digital prototype; check physical fit before batch printing.

# Electrical connections / system wiring

Follow the point-to-point wiring schedule for connector pins, polarity and conductor ratings.

## LUMA / Electrical connections

**Revision C · Raspberry Pi 4 master · four PWM servos · five pedestal-wall ranging sensors**




**Read wiring.csv for exact connections, fuses, DIP pins and cable notes.**

All DC grounds meet at one star point. Never join Pi5V and LED5V. Do not carry servo power through PCA9685 traces.

Motor stop cuts6V power: support the arm. PWM confirms no shaft position; five narrow ToF cones are not a safety perimeter.

Power wires shown in amber; signal routes in blue. Schematic overview: routing lengths and connector positions are not to scale.

UNITS mm · THIRD-ANGLE PART VIEWS · NUMERICAL DIMENSIONS CONTROL

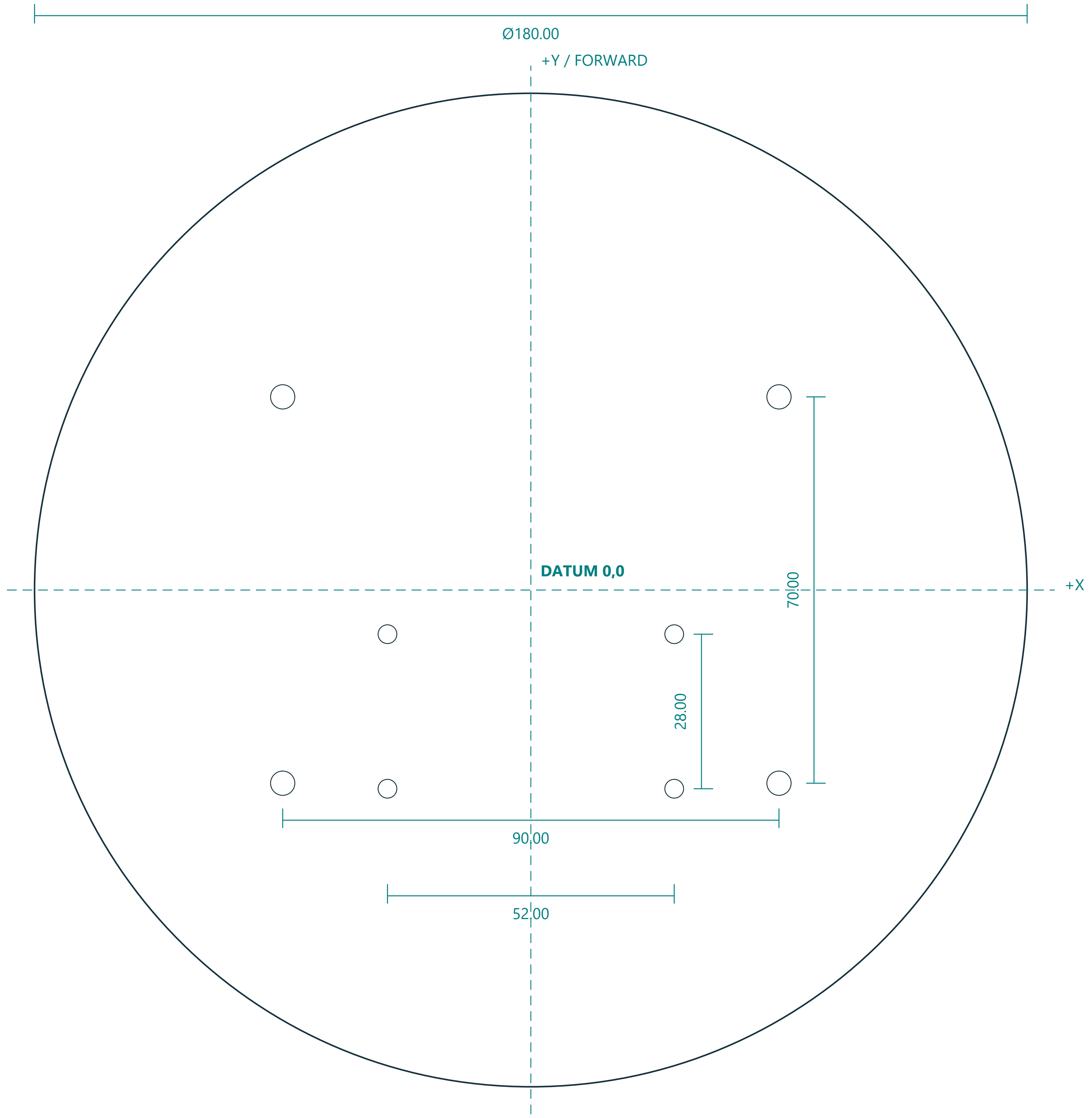
Views fitted to sheet. Do not measure this PDF. Digital prototype; check physical fit before batch printing.

**E01**

REV C | 07 SEP 2026 | A3

# Steel ballast plate / fabrication drawing

Purchased/custom-fabricated metal part M01. This sheet is not a printable-plastic substitute.




## MATERIAL + INTERFACES

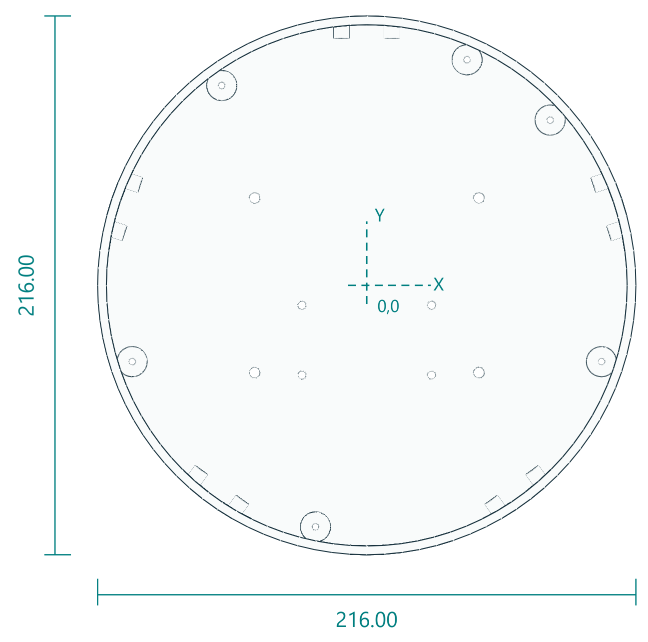
Material: mild steel; finished thickness6.00mm.
Outer diameter180.00mm; deburr all edges and holes.
Datum origin is the disc center. All hole axes normal to plate.
4 × Ø4.40 through: X=±45.00, Y=±35.00.
4 × Ø3.40 through: X=±26.00, Y=-36.00 and-8.00.
Do not add a center hole. Rear yaw bolt pattern points-Y.
Nominal mass before drilling:1.19kg at steel density7.85g/cm$^3$.
Fit to printed base floor before final coating. Inspect burrs and screw-head clearance.
Suggested workshop positional allowance±0.20mm; confirm actual printed mating pattern before drilling.

## SIDE VIEW / THICKNESS

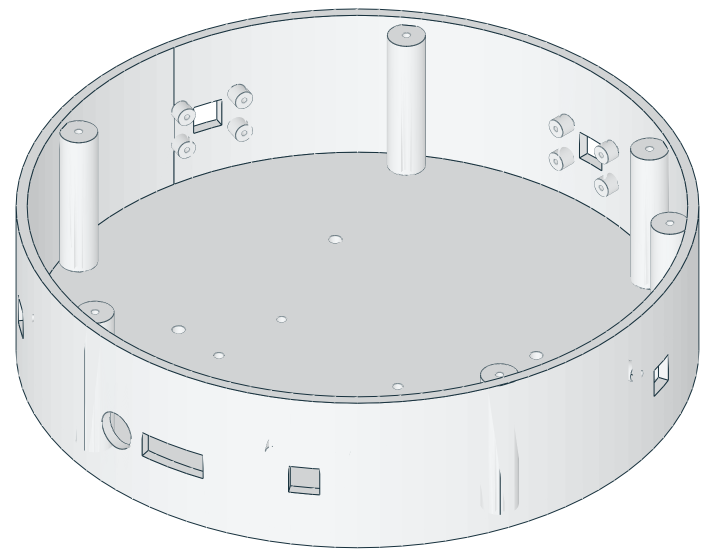
# Weighted base enclosure with five internal ToF mounts

base_tub.stl | PRINT QUANTITY 1 | PETG

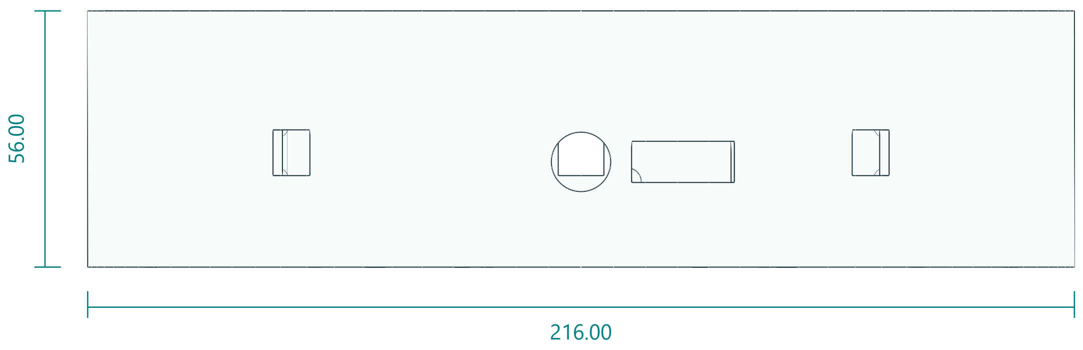
### TOP / XY



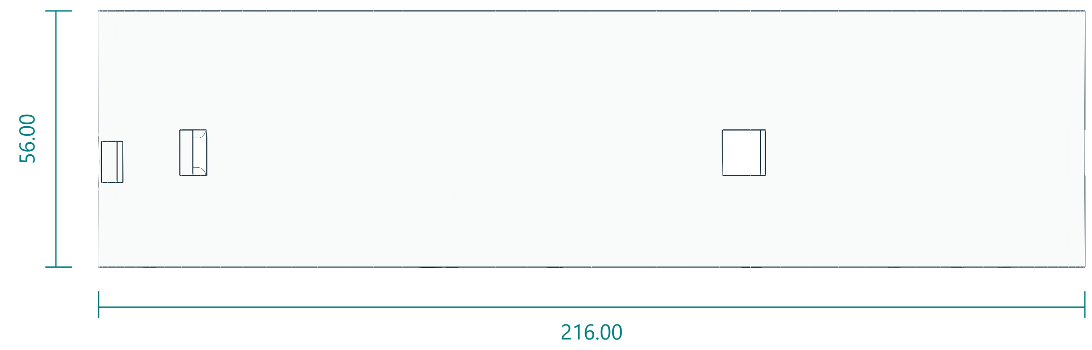

### ISOMETRIC / REFERENCE



### FRONT / XZ




### RIGHT / YZ

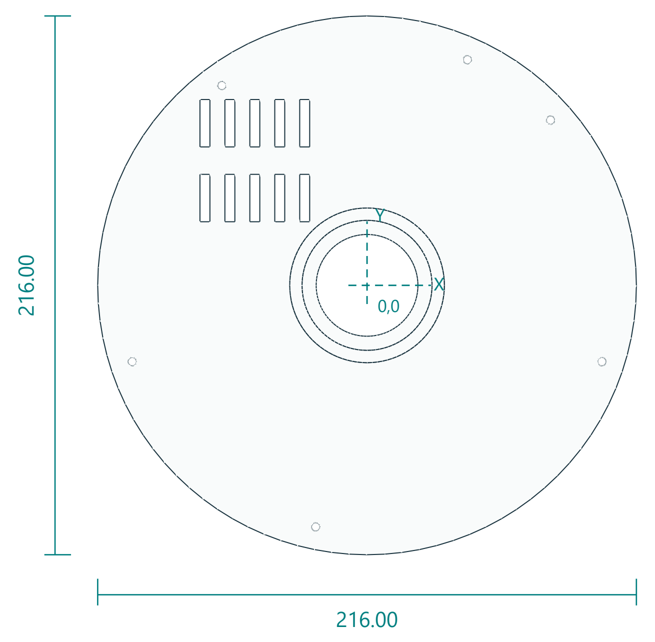



Interfaces: 216 diameter; 56 high; 3 floor; 3.5 wall; five sensor apertures and 20 internal bosses.
Fasteners / retention: 6 M3x12 lid screws; 4 M4 ballast retainers; 20 M2.5x6 sensor screws.
Print orientation: floor down. Layers 0.2 mm; 4 walls; 25% infill baseline.
Feature locations: M4 clearance4.4 at (+/-45,+/-35). Yaw bolts3.4 at (+/-26,-36) and (+/-26,-8). Six lid pilots2.8 at radius99, angles42/66/126/198/258/342deg. Five 10x10 wall apertures at18/90/162/234/306deg, Z25. Each board has four2.1 blind pilots on20.32x12.70 at tangent radius99. Rear ports occupy270/282deg.
General fit: inspect and deburr holes; check the fit coupon and actual hardware. No unlisted machining tolerances are implied.

# Vented base lid and yaw bearing seat

base_lid.stl | PRINT QUANTITY 1 | PETG

### TOP / XY

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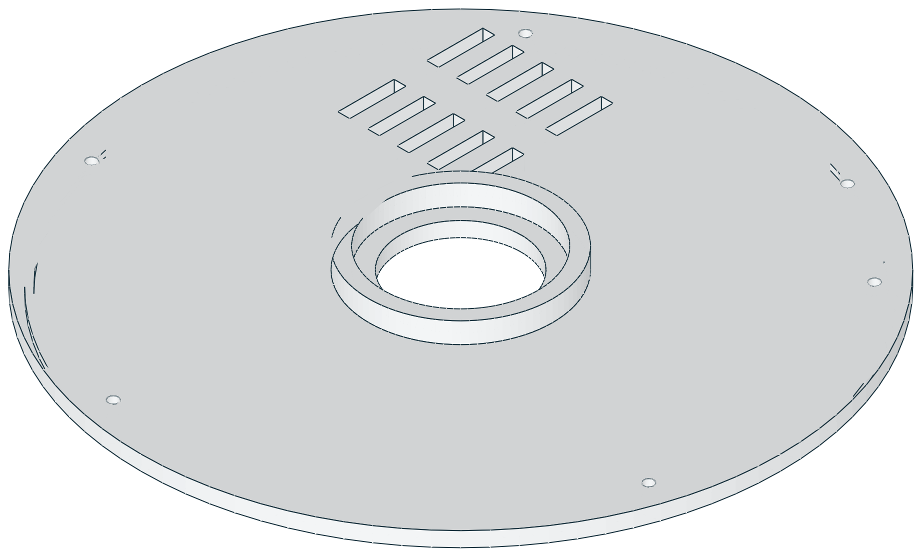
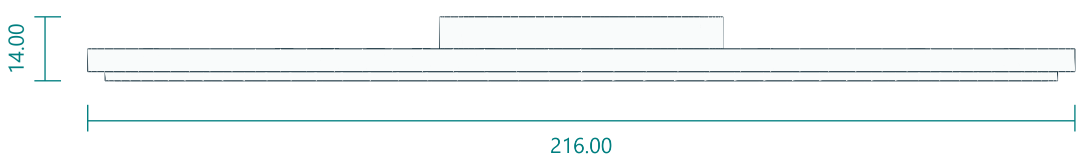


### FRONT / XZ




### RIGHT / YZ


Interfaces: 216 diameter; bearing pocket52.15 x7; 40.8 cable/hub bore.
Fasteners / retention: 6 M3x12.
Print orientation: flat annulus down. Layers 0.2 mm; 4 walls; 25% infill baseline.
Feature locations: Six3.4 holes radius99 at42/66/126/198/258/342deg. Bearing52.15 bore by7 depth. STL bottom datum is2mm below principal lid underside.
General fit: inspect and deburr holes; check the fit coupon and actual hardware. No unlisted machining tolerances are implied.

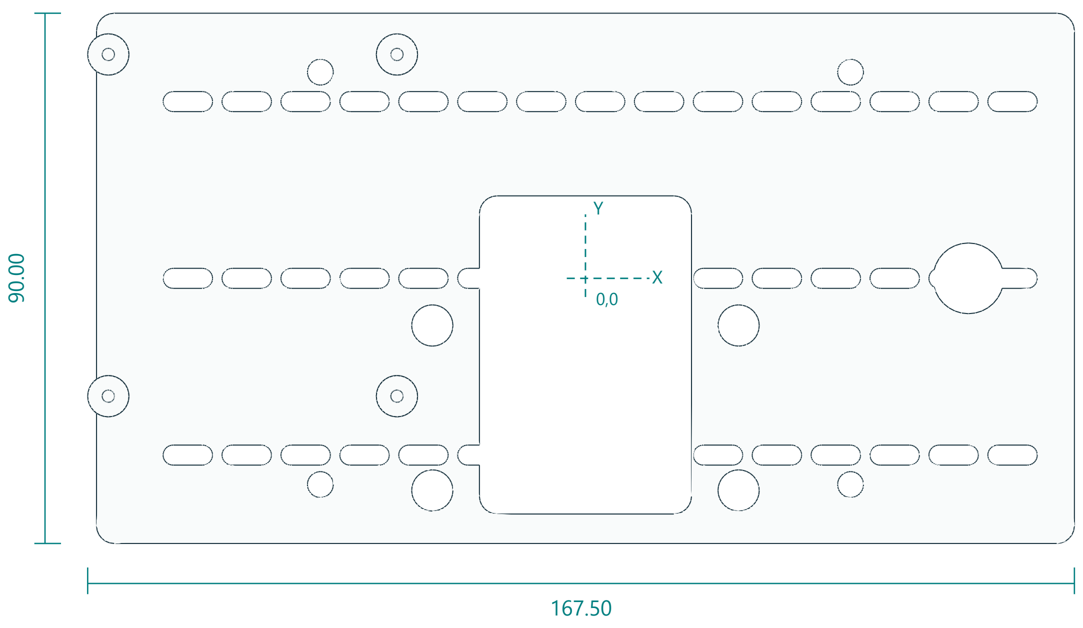
**UNITS mm · THIRD-ANGLE PART VIEWS · NUMERICAL DIMENSIONS CONTROL**

Views fitted to sheet. Do not measure this PDF. Digital prototype; check physical fit before batch printing.

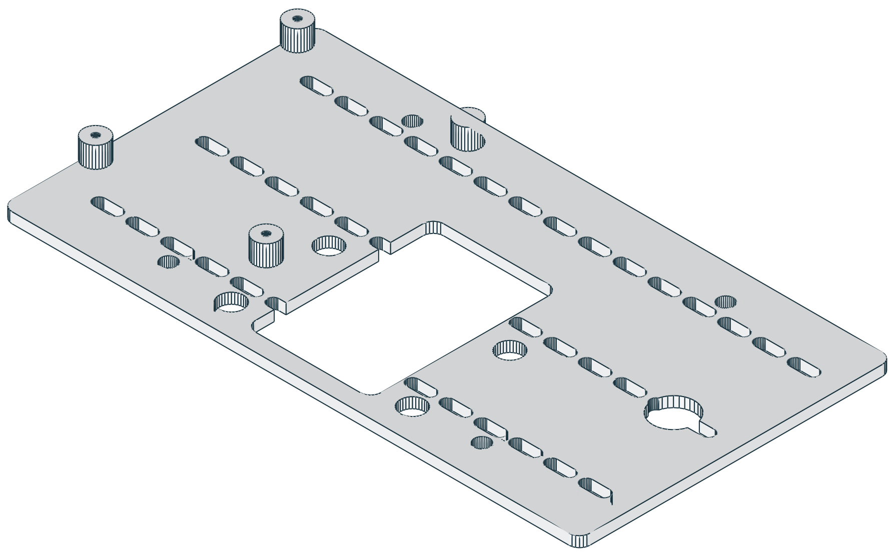
# Universal electronics tray with Pi4 standoffs

electronics_tray.stl | PRINT QUANTITY 1 | PETG

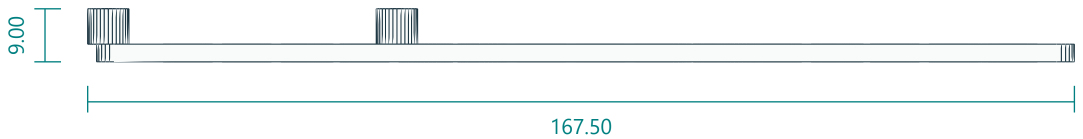
TOP / XY



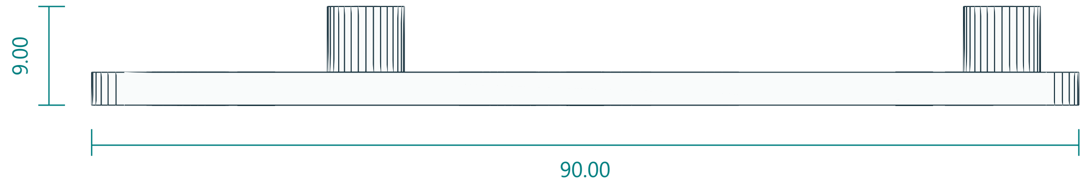

ISOMETRIC / REFERENCE



FRONT / XZ




RIGHT / YZ




Interfaces: 166 x90; Pi4 hole pitch49 x58; 2.1mm M2.5 tapping pilots.
Fasteners / retention: 4 M2.5x6 Pi; 4 M4x25 with metal spacers.
Print orientation: plate down. Layers 0.2 mm; 4 walls; 25% infill baseline.
Feature locations: M4 holes (+/-45,+/-35). Pi M2.5 pilots2.1 at X=-81,-32 and Y=-20,38. Central rounded36x54 clearance centered(0,-13). Four7mm tool apertures at yaw bolts.
General fit: inspect and deburr holes; check the fit coupon and actual hardware. No unlisted machining tolerances are implied.

**UNITS mm · THIRD-ANGLE PART VIEWS · NUMERICAL DIMENSIONS CONTROL**

Views fitted to sheet. Do not measure this PDF. Digital prototype; check physical fit before batch printing.

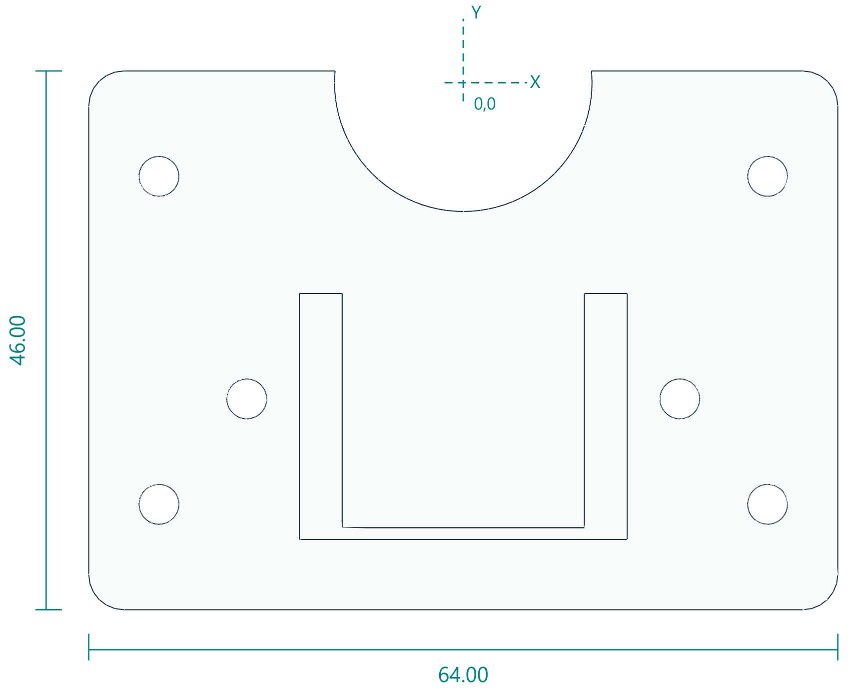
**P03**

REV C | 07 SEP 2026 | A3

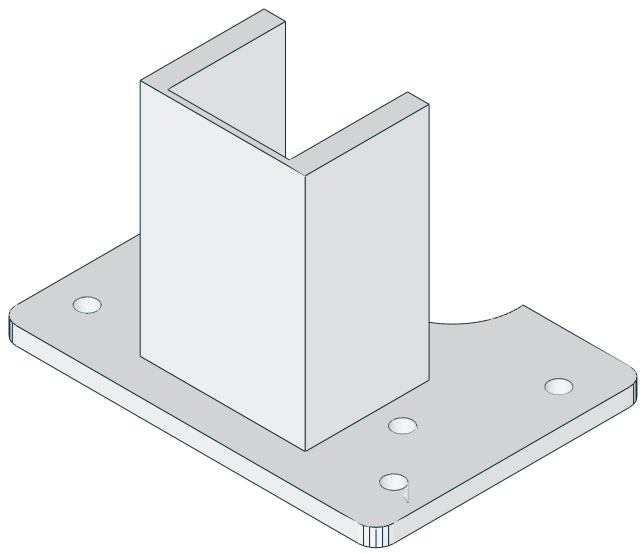
# DS3218MG yaw servo rear-body cradle

yaw_mount.stl | PRINT QUANTITY 1 | PETG

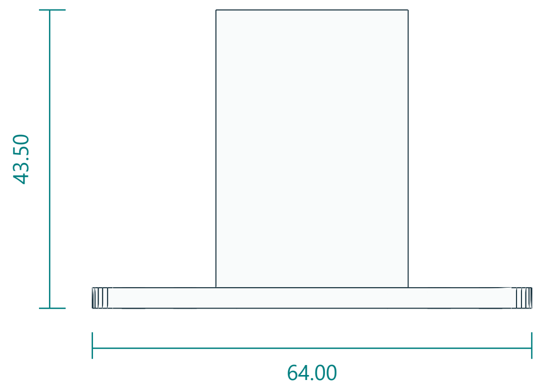
## TOP / XY



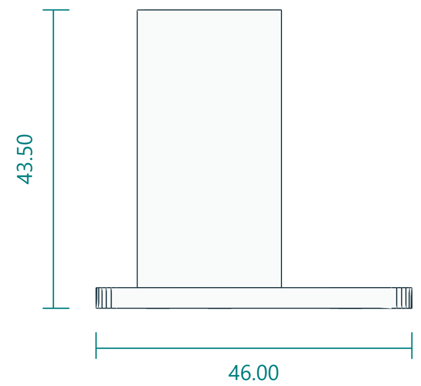

## ISOMETRIC / REFERENCE



## FRONT / XZ




## RIGHT / YZ




Interfaces: 64 x46; pocket20.7; 40.5-high rear body clamp.
Fasteners / retention: 4 M3x12 mount; 2 M3x50 clamp.
Print orientation: base down. Layers 0.2 mm; 4 walls; 25% infill baseline.
Feature locations: Four3.4 floor holes (+/-26,-36/-8). Keeper bolts at (+/-18.5,-27). DS3218 shaft atXY(0,0), output case plane Z43.5 in part coordinates.
General fit: inspect and deburr holes; check the fit coupon and actual hardware. No unlisted machining tolerances are implied.

**UNITS mm · THIRD-ANGLE PART VIEWS · NUMERICAL DIMENSIONS CONTROL**

Views fitted to sheet. Do not measure this PDF. Digital prototype; check physical fit before batch printing.

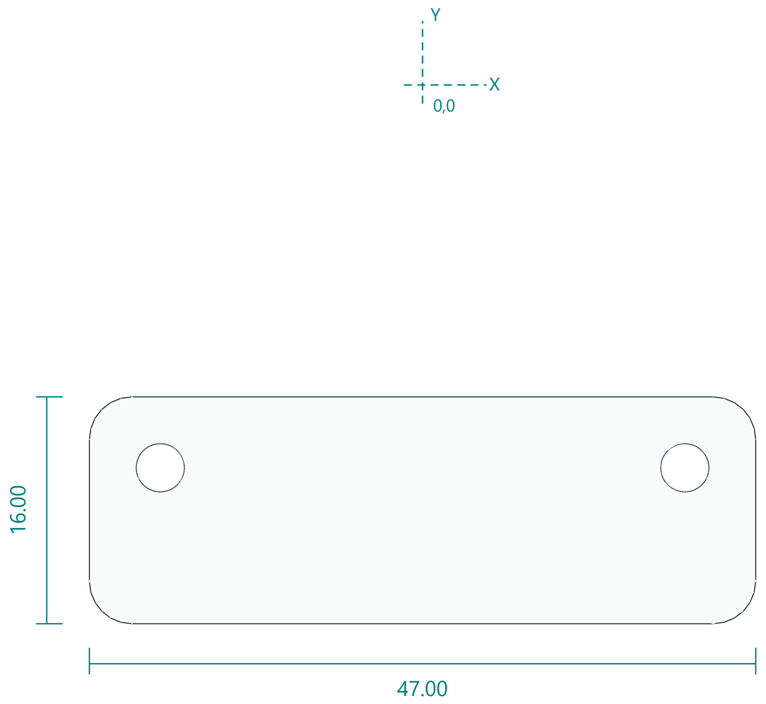
**P04**

REV C | 07 SEP 2026 | A3

# Yaw clamp keeper

yaw_cap.stl | PRINT QUANTITY 1 | PETG

### TOP / XY

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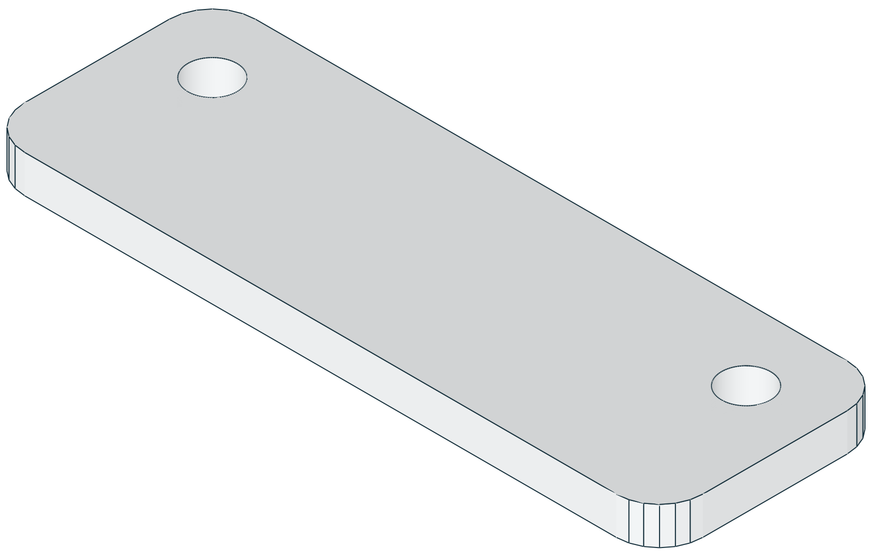
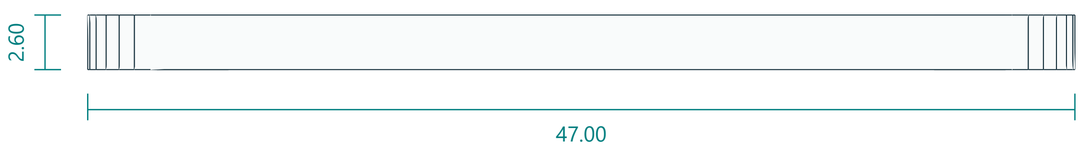
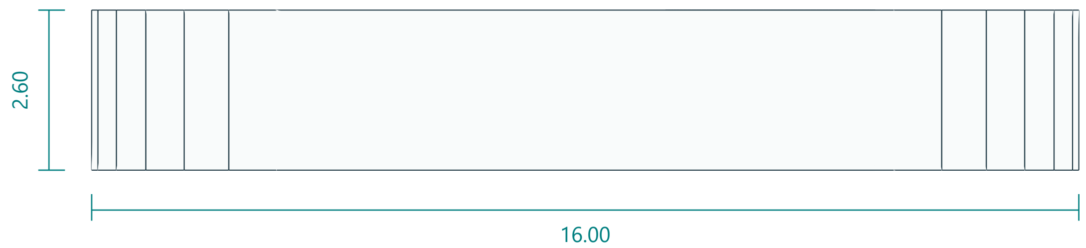


### FRONT / XZ




### RIGHT / YZ




Interfaces: 47 x16 x2.6; clamp hole pitch37.
Fasteners / retention: 2 M3x50.
Print orientation: flat down. Layers 0.2 mm; 4 walls; 25% infill baseline.
Feature locations: Two3.4 holes (+/-18.5,-27). Keeper center(0,-30); thickness2.6.
General fit: inspect and deburr holes; check the fit coupon and actual hardware. No unlisted machining tolerances are implied.

**UNITS mm · THIRD-ANGLE PART VIEWS · NUMERICAL DIMENSIONS CONTROL**

Views fitted to sheet. Do not measure this PDF. Digital prototype; check physical fit before batch printing.

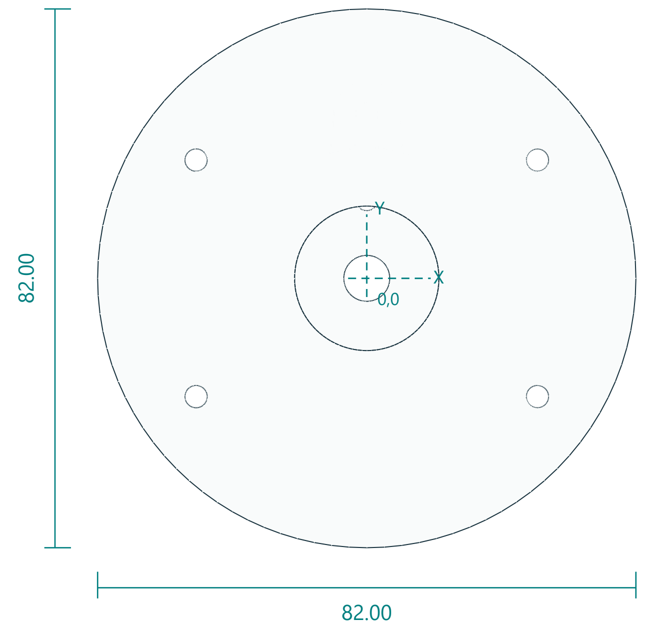
**P05**

REV C | 07 SEP 2026 | A3

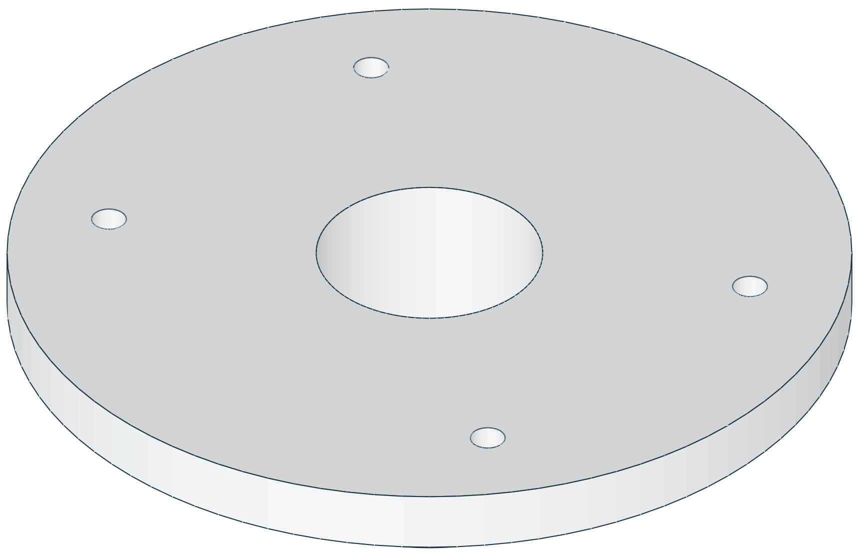
# Bearing-supported yaw turntable for straight servo arm

turntable.stl | PRINT QUANTITY 1 | PETG

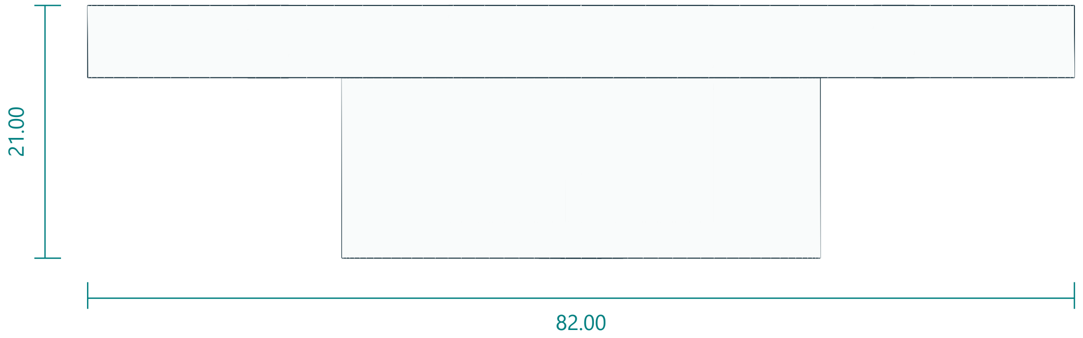
### TOP / XY



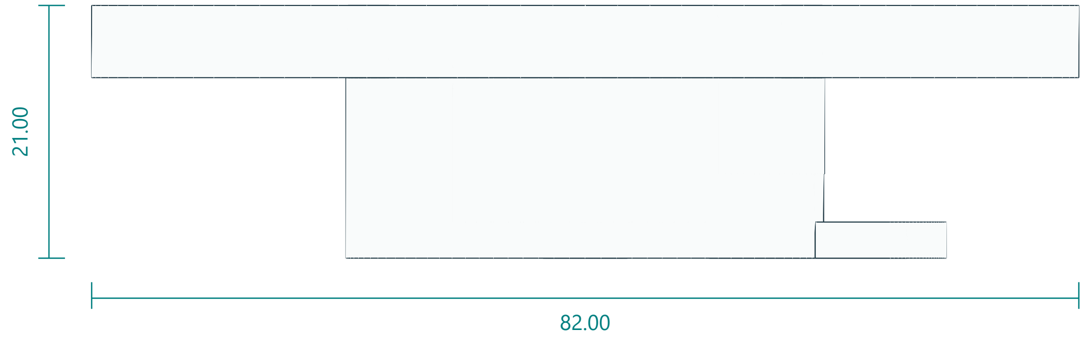

### ISOMETRIC / REFERENCE



### FRONT / XZ




### RIGHT / YZ




Interfaces: 82 diameter; neck39.75 x14.2; two adjustable radial horn slots; tower52 x36.
Fasteners / retention: 2 horn-arm bolts; 4 M3x16 tower.
Print orientation: neck down. Layers 0.2 mm; 4 walls; 25% infill baseline.
Feature locations: Straight-horn center clearance7 with radial3.4 slots12-18 and20-25 in3mm web. Tower4x3.4 at(+/-26,+/-18). Top central tool aperture22.
General fit: inspect and deburr holes; check the fit coupon and actual hardware. No unlisted machining tolerances are implied.

**UNITS mm · THIRD-ANGLE PART VIEWS · NUMERICAL DIMENSIONS CONTROL**

Views fitted to sheet. Do not measure this PDF. Digital prototype; check physical fit before batch printing.

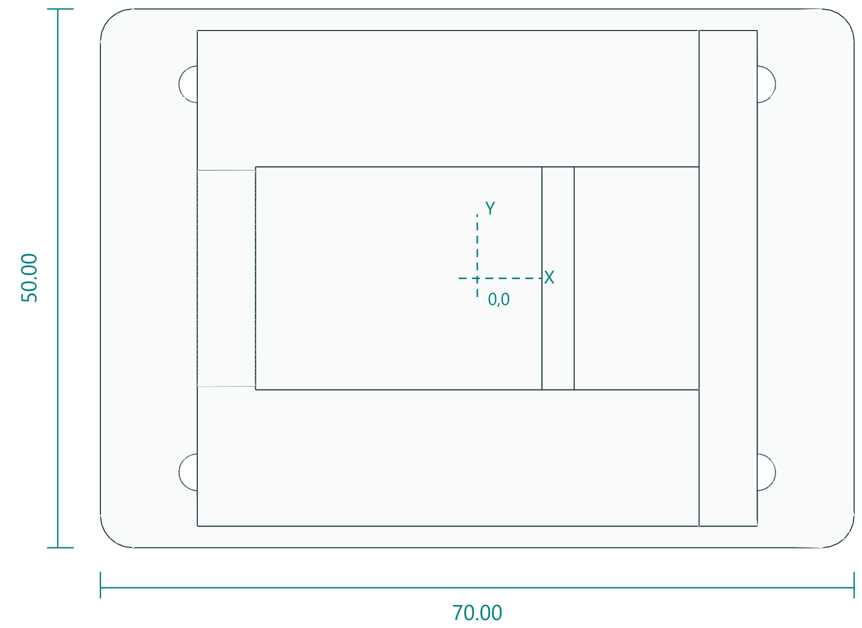
**P06**

REV C | 07 SEP 2026 | A3

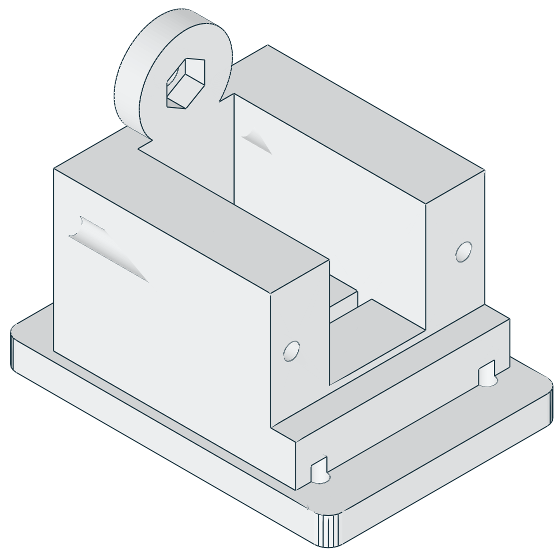
# DS3218MG shoulder body tower with idler support

shoulder_tower.stl | PRINT QUANTITY 1 | PETG

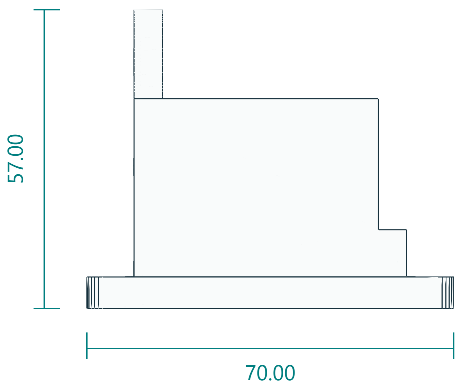
### TOP / XY



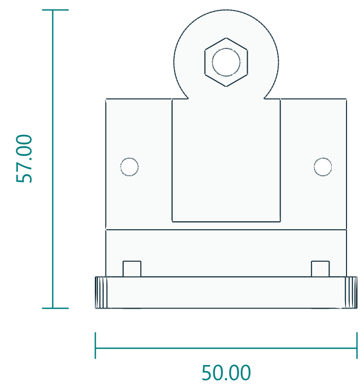

### ISOMETRIC / REFERENCE



### FRONT / XZ




### RIGHT / YZ




Interfaces: 70 x50; pitch axis47 above foot; body pocket41.2 x20.7; M5 idler nut trap.
Fasteners / retention: 4 M3x16; 2 M3x50 clamp; 1 M5 idler bolt and 625 bearing.
Print orientation: foot down. Layers 0.2 mm; 4 walls; 25% infill baseline.
Feature locations: Foot4x3.4 at(+/-26,+/-18). Clamp holes alongX atY+/-18.5,Z29. Pitch axis47 abovefoot. Left idler M5 clearance and captive nut in20mm boss.
General fit: inspect and deburr holes; check the fit coupon and actual hardware. No unlisted machining tolerances are implied.

**UNITS mm · THIRD-ANGLE PART VIEWS · NUMERICAL DIMENSIONS CONTROL**

Views fitted to sheet. Do not measure this PDF. Digital prototype; check physical fit before batch printing.

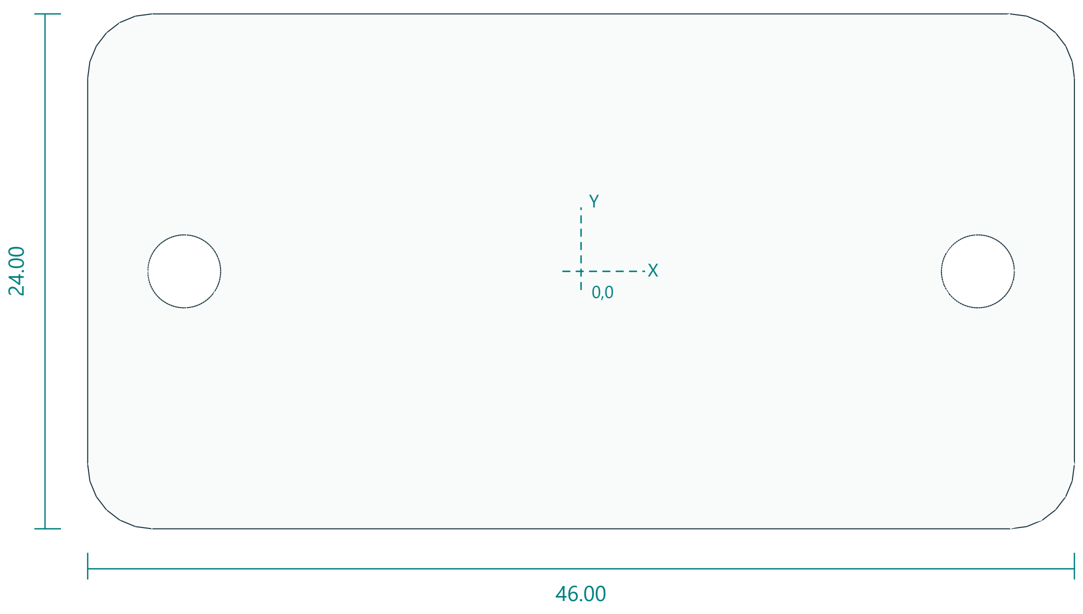
**P07**

REV C | 07 SEP 2026 | A3

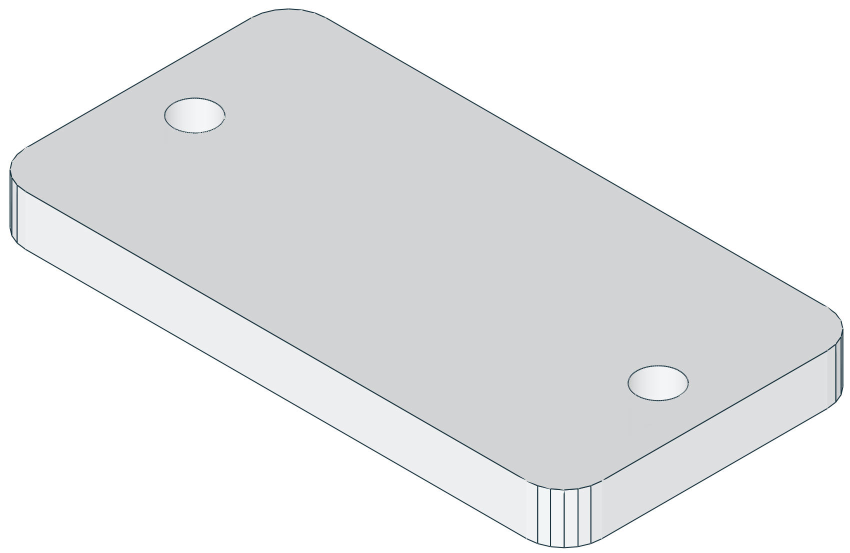
# Shoulder clamp keeper

shoulder_cap.stl | PRINT QUANTITY 1 | PETG

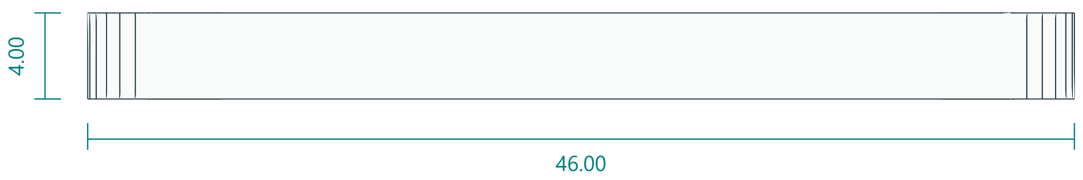
**TOP / XY**



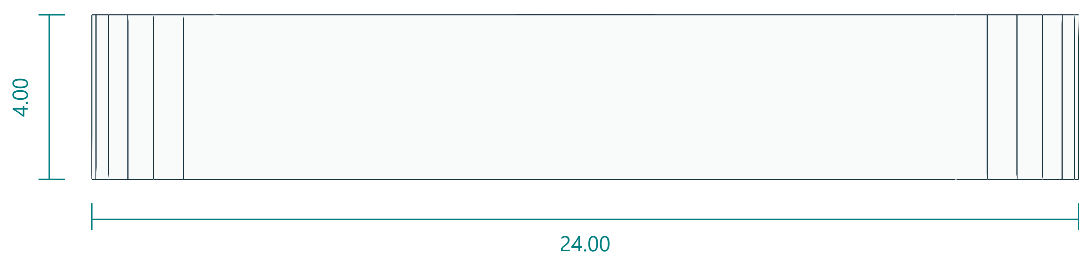

**ISOMETRIC / REFERENCE**



**FRONT / XZ**




**RIGHT / YZ**




Interfaces: 46 x28 x4; clamp pitch37.
Fasteners / retention: 2 M3x50.
Print orientation: flat down. Layers 0.2 mm; 4 walls; 25% infill baseline.
Feature locations: Two3.4 holes (+/-18.5,0). Printedface46x28.
General fit: inspect and deburr holes; check the fit coupon and actual hardware. No unlisted machining tolerances are implied.

**UNITS mm · THIRD-ANGLE PART VIEWS · NUMERICAL DIMENSIONS CONTROL**

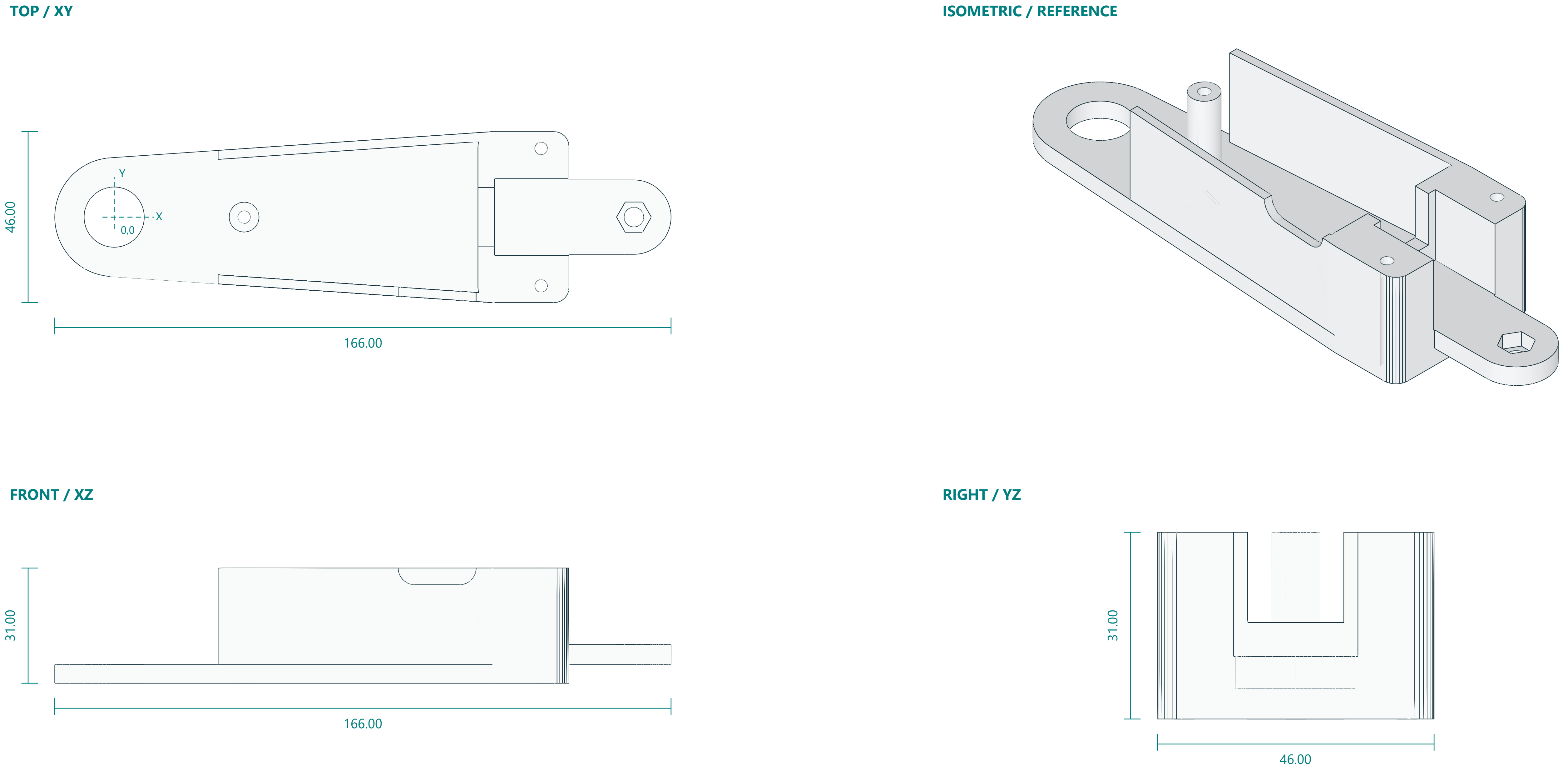
Views fitted to sheet. Do not measure this PDF. Digital prototype; check physical fit before batch printing.

**P08**

REV C | 07 SEP 2026 | A3

# Upper arm enclosed link, idler half

upper_arm_left.stl | PRINT QUANTITY 1 | PETG




Interfaces: 140 pivot pitch; 62 assembled width; 5 plate; 2.4 walls26 deep; 625 bearing pocket; distal M5 idler nut trap.
Fasteners / retention: 1 M5 idler bolt; 3 M3x70 through bolts shared with right half.
Print orientation: plate down. Layers 0.2 mm; 6 walls; 40% infill baseline.
Feature locations: Proximal 625 pocket16.2. Through bolt3.4 at(35,0); block bolts at(115,+/-18.5). DS3218 pocket41.2x20.7; walls fromX28; distal M5 nut trap atX140; cable port21x9 centredX87.
General fit: inspect and deburr holes; check the fit coupon and actual hardware. No unlisted machining tolerances are implied.

**UNITS mm · THIRD-ANGLE PART VIEWS · NUMERICAL DIMENSIONS CONTROL**

Views fitted to sheet. Do not measure this PDF. Digital prototype; check physical fit before batch printing.

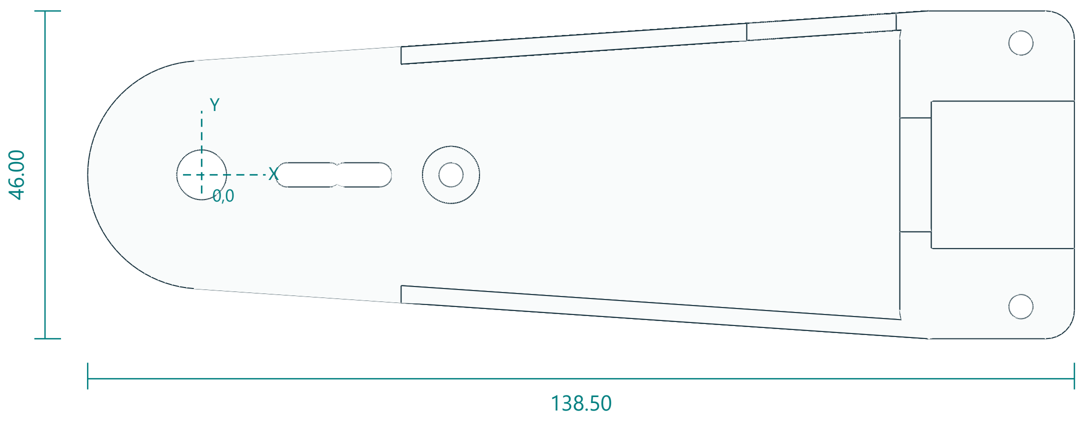
**P09**

REV C | 07 SEP 2026 | A3

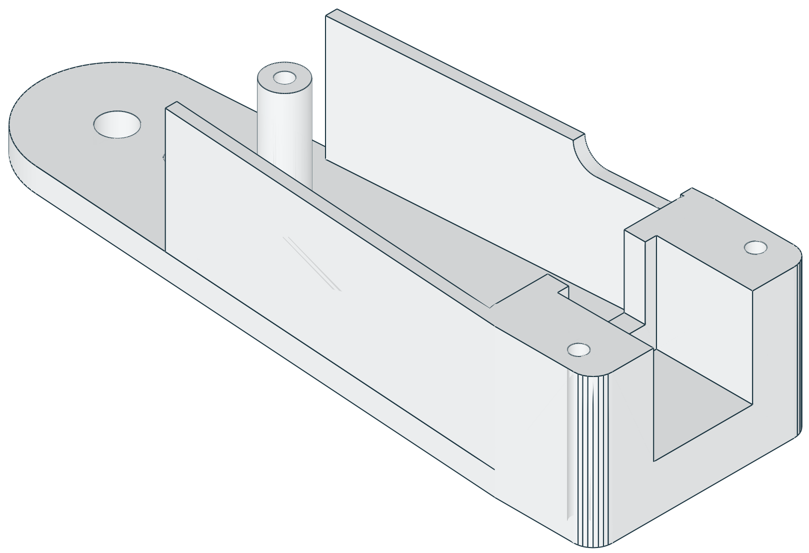
# Upper arm enclosed link, driven half

upper_arm_right.stl | PRINT QUANTITY 1 | PETG

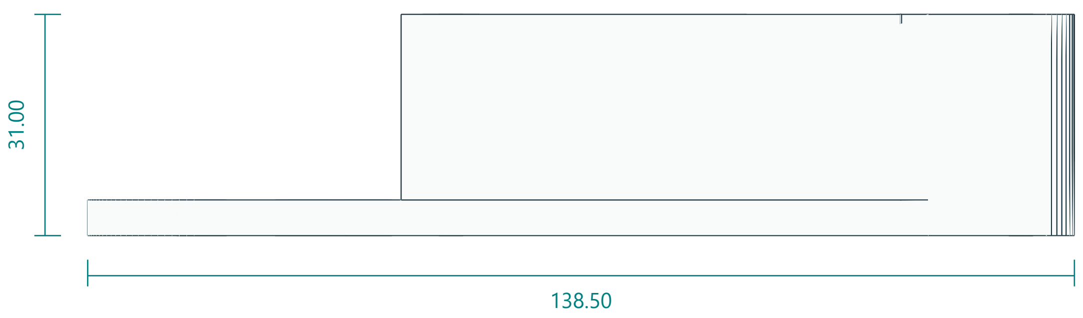
### TOP / XY



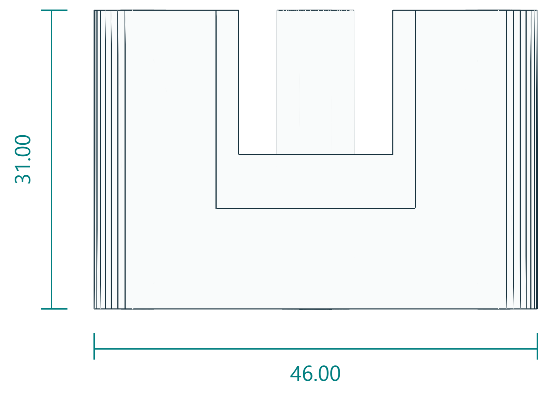

### ISOMETRIC / REFERENCE



### FRONT / XZ




### RIGHT / YZ

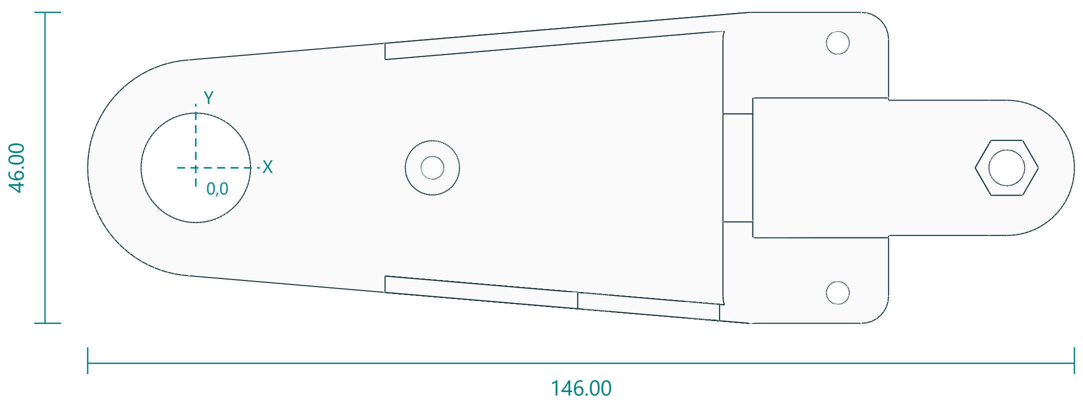



Interfaces: 140 pivot pitch; 62 assembled width; two proximal horn-arm slots; cable port on inner elbow wall.
Fasteners / retention: 2 horn-arm bolts; shared M3x70.
Print orientation: plate down. Layers 0.2 mm; 6 walls; 40% infill baseline.
Feature locations: Driven side has center7 and radial3.4 slots12-18 and20-25. Same clamp-bolt pattern as left; cable port remains on the inner elbow wall after assembly.
General fit: inspect and deburr holes; check the fit coupon and actual hardware. No unlisted machining tolerances are implied.

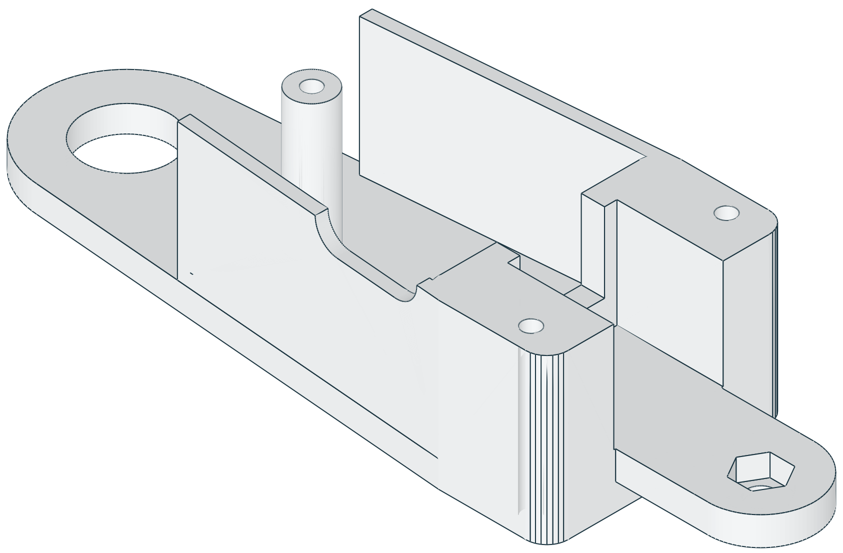
# Forearm enclosed link, idler half

forearm_left.stl | PRINT QUANTITY 1 | PETG

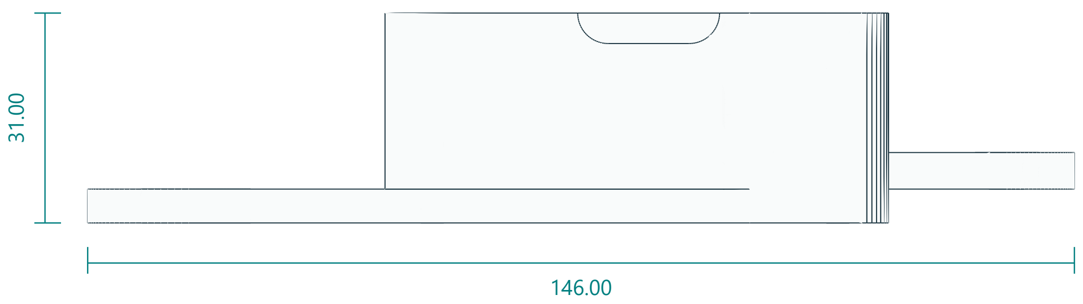
## TOP / XY



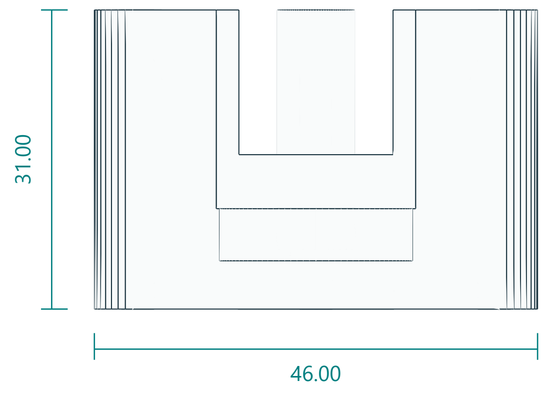

## ISOMETRIC / REFERENCE



## FRONT / XZ




## RIGHT / YZ

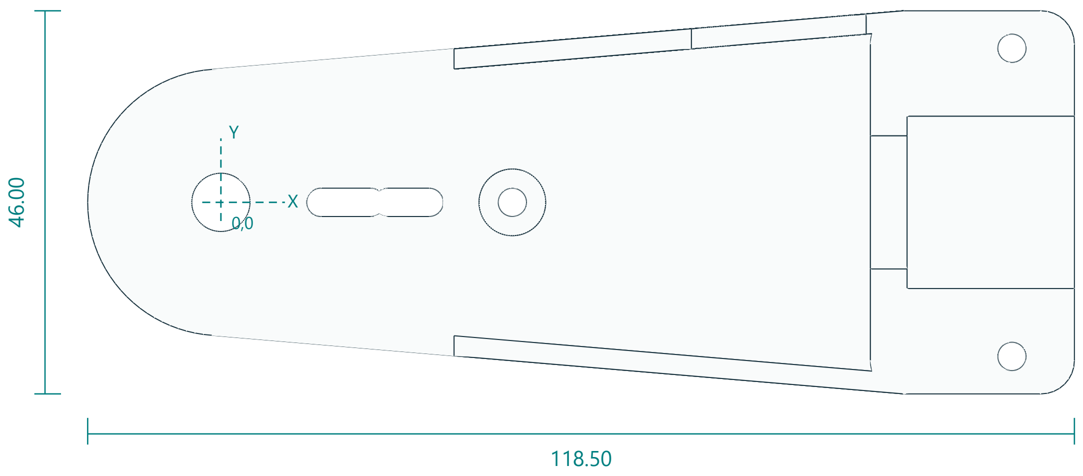



Interfaces: 120 pivot pitch; 62 assembled width; 625 bearing pocket; distal M5 idler nut trap.
Fasteners / retention: 1 M5 idler bolt; 3 M3x70 through bolts shared with right half.
Print orientation: plate down. Layers 0.2 mm; 6 walls; 40% infill baseline.
Feature locations: Proximal 625 pocket16.2. Through bolt3.4 at(35,0); block bolts at(95,+/-18.5). DS3218 pocket41.2x20.7; distal M5 nut trap atX120; cable port21x9 centredX67.
General fit: inspect and deburr holes; check the fit coupon and actual hardware. No unlisted machining tolerances are implied.

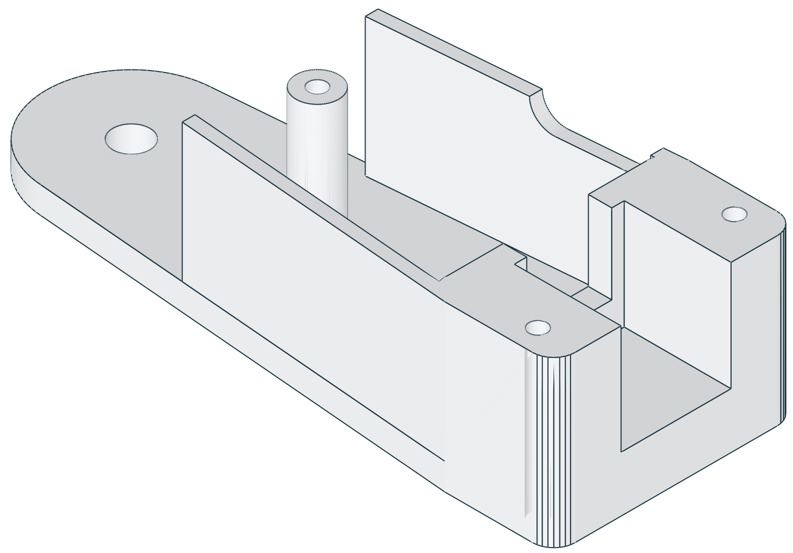
# Forearm enclosed link, driven half

forearm_right.stl | PRINT QUANTITY 1 | PETG

## TOP / XY

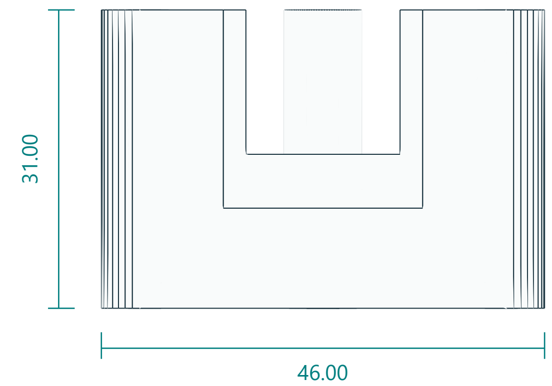



## ISOMETRIC / REFERENCE



## FRONT / XZ


## RIGHT / YZ




Interfaces: 120 pivot pitch; 62 assembled width; two proximal horn-arm slots; cable port on inner wrist wall.
Fasteners / retention: 2 horn-arm bolts; shared M3x70.
Print orientation: plate down. Layers 0.2 mm; 6 walls; 40% infill baseline.
Feature locations: Driven side has center7 and radial3.4 slots12-18 and20-25. Same clamp-bolt pattern as left; cable port remains on the inner wrist wall after assembly.
General fit: inspect and deburr holes; check the fit coupon and actual hardware. No unlisted machining tolerances are implied.

**UNITS mm · THIRD-ANGLE PART VIEWS · NUMERICAL DIMENSIONS CONTROL**

Views fitted to sheet. Do not measure this PDF. Digital prototype; check physical fit before batch printing.

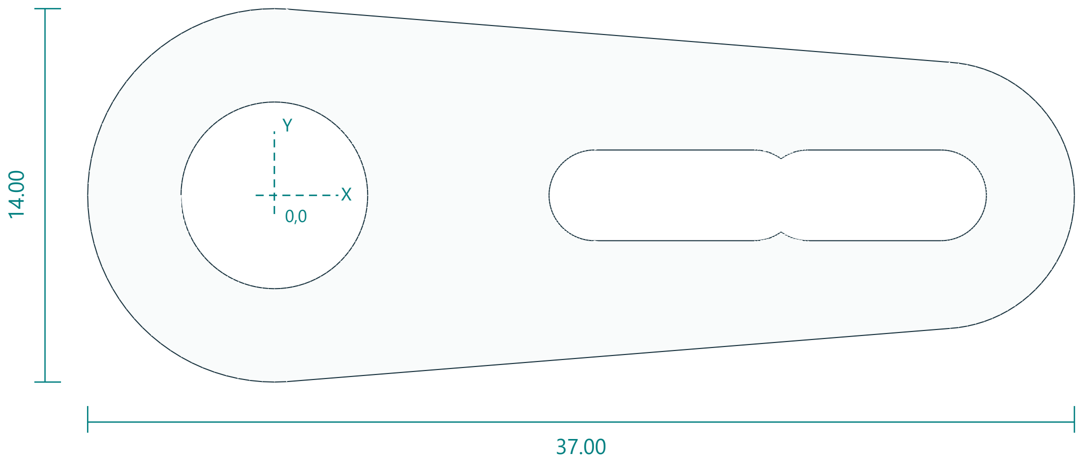
**P12**

REV C | 07 SEP 2026 | A3

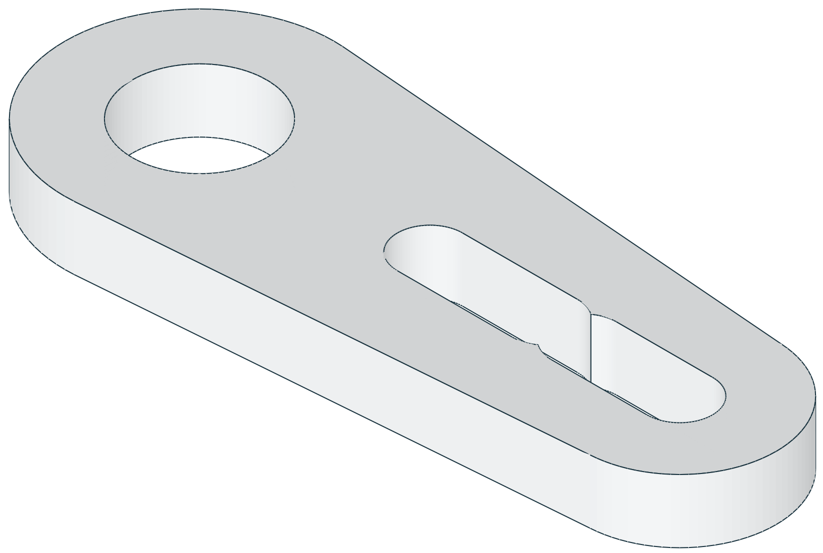
# DS3218 straight-horn to driven-link spacer

horn_spacer.stl | PRINT QUANTITY 3 | PETG

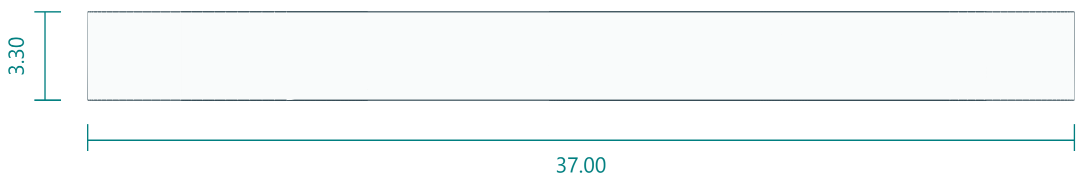
### TOP / XY



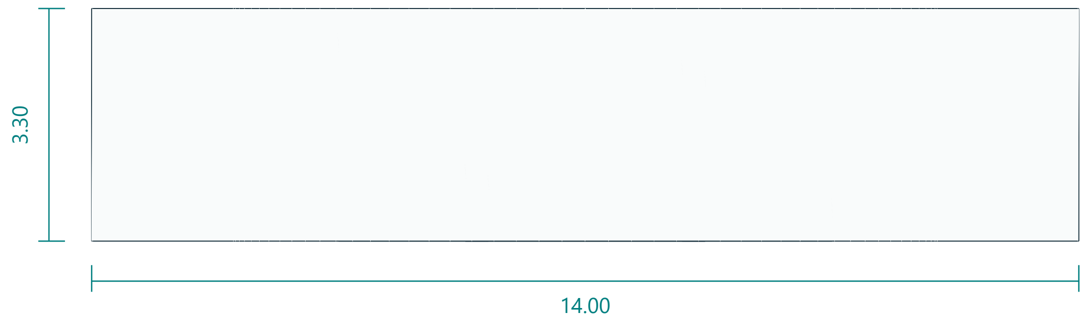

### ISOMETRIC / REFERENCE



### FRONT / XZ




### RIGHT / YZ




Interfaces: 30 x14 x3.3; center7; adjustable radial slots12-18 and20-25.
Fasteners / retention: 2 horn-arm bolts each.
Print orientation: flat down. Layers 0.2 mm; 4 walls; 25% infill baseline.
Feature locations: Center7; radial3.4 slots12-18 and20-25; thickness3.3; one driven-side spacer per pitch joint.
General fit: inspect and deburr holes; check the fit coupon and actual hardware. No unlisted machining tolerances are implied.

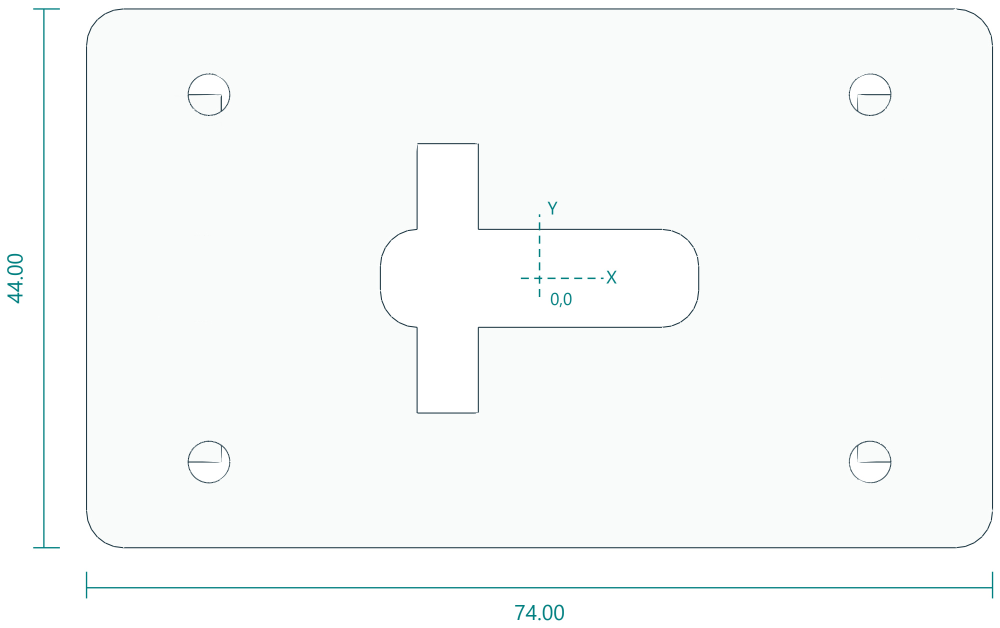
**UNITS mm · THIRD-ANGLE PART VIEWS · NUMERICAL DIMENSIONS CONTROL**

Views fitted to sheet. Do not measure this PDF. Digital prototype; check physical fit before batch printing.

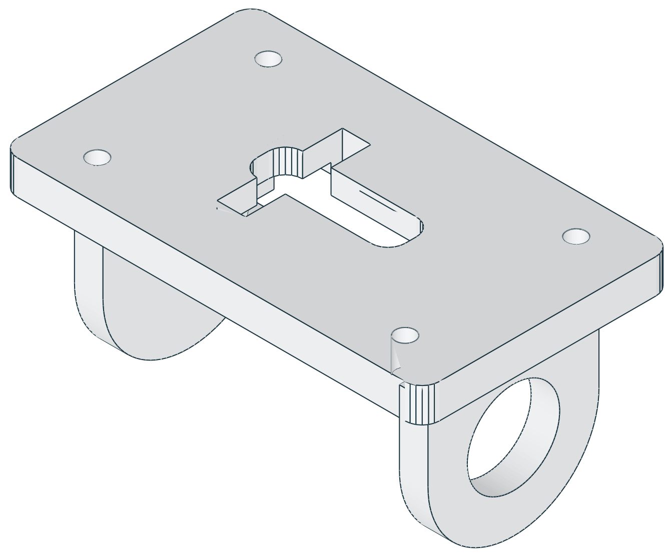
# Head tilt fork with driven and 625-idler ears

head_yoke.stl | PRINT QUANTITY 1 | PETG

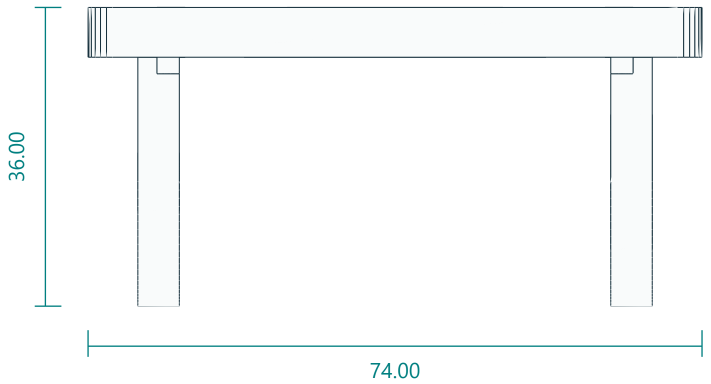
### TOP / XY



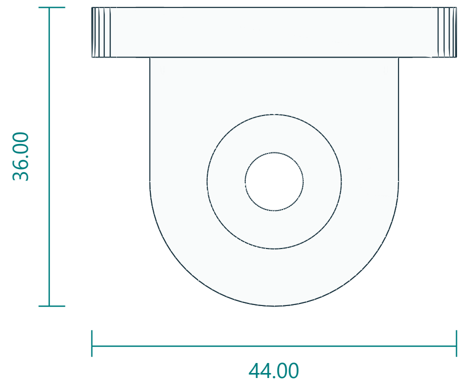

### ISOMETRIC / REFERENCE



### FRONT / XZ




### RIGHT / YZ

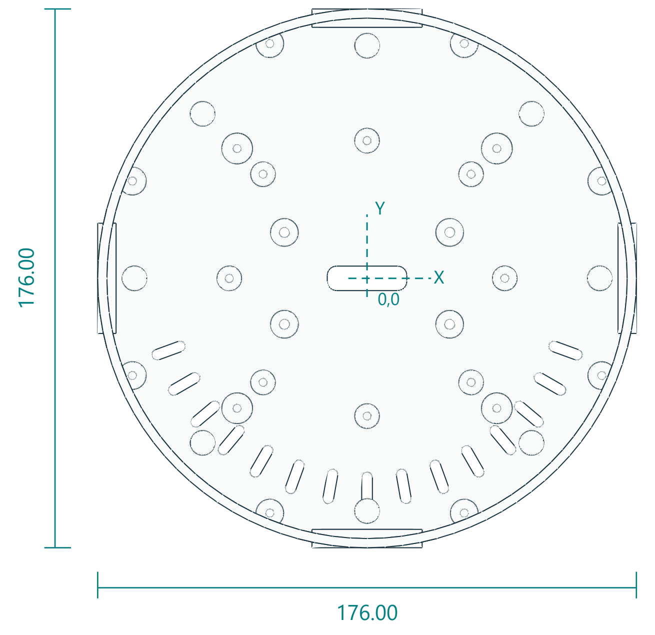



Interfaces: 62 fork inside span; axis21 behind head rear; rear holes54 x30; 26 x8 centre cable slot.
Fasteners / retention: 2 horn-arm bolts; 1 M5 idler bolt; 1 625 bearing; 4 M3x12 rear screws.
Print orientation: rear mounting plate on bed; rotate180X. Layers 0.2 mm; 4 walls; 25% infill baseline.
Feature locations: Back bolts3.4 XY(+/-27,+/-15); attachmentplaneZ36. AxisX atY0,Z15; one625 pocket16.2 and one tangential straight-horn interface; fork inside width52. Centre cable slot26x8.
General fit: inspect and deburr holes; check the fit coupon and actual hardware. No unlisted machining tolerances are implied.

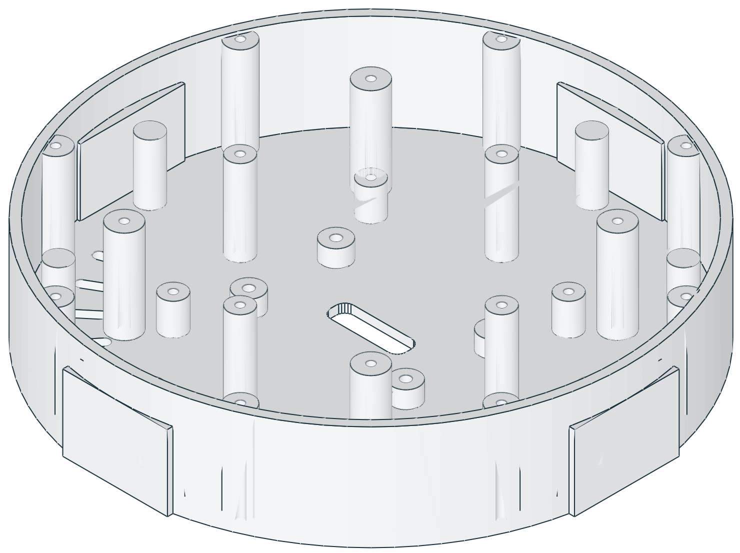
# Optical head rear enclosure

head_shell.stl | PRINT QUANTITY 1 | PETG

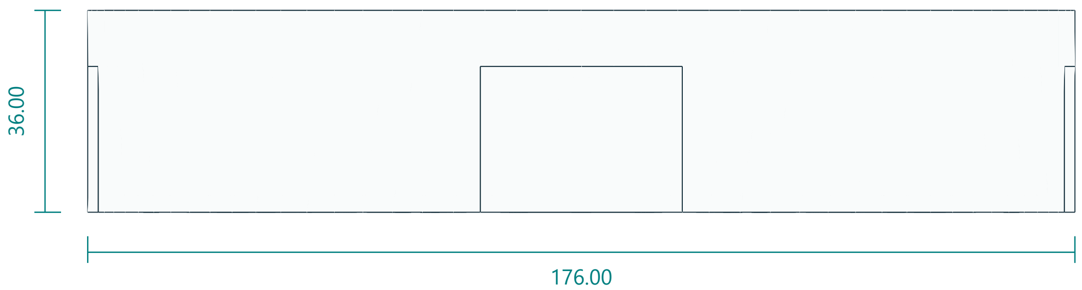
### TOP / XY



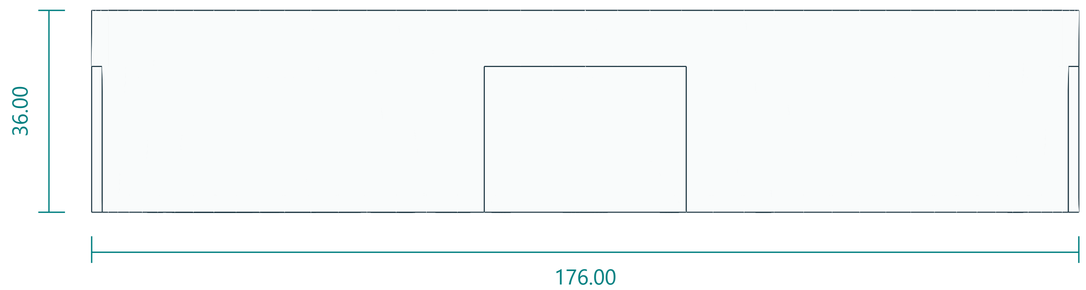

### ISOMETRIC / REFERENCE



### FRONT / XZ




### RIGHT / YZ




Interfaces: 176 diameter x36; rear thickness3; white carrier bosses radius45 top32.5; 26 x8 centre cable slot.
Fasteners / retention: 8 M3x10 bezel; 4 M3x12 yoke.
Print orientation: rear disc down. Layers 0.2 mm; 4 walls; 25% infill baseline.
Feature locations: Rear bolts(+/-27,+/-15); LCDpilots(+/-34,+/-34),top14. Whitecarrierpilotsradius45 cardinal,top32.5. Facebossradius60 angles45+90n,top35. Bezelradius83 angles22.5+45n,top35. Centre cable slot26x8. No sensor penetrations.
General fit: inspect and deburr holes; check the fit coupon and actual hardware. No unlisted machining tolerances are implied.

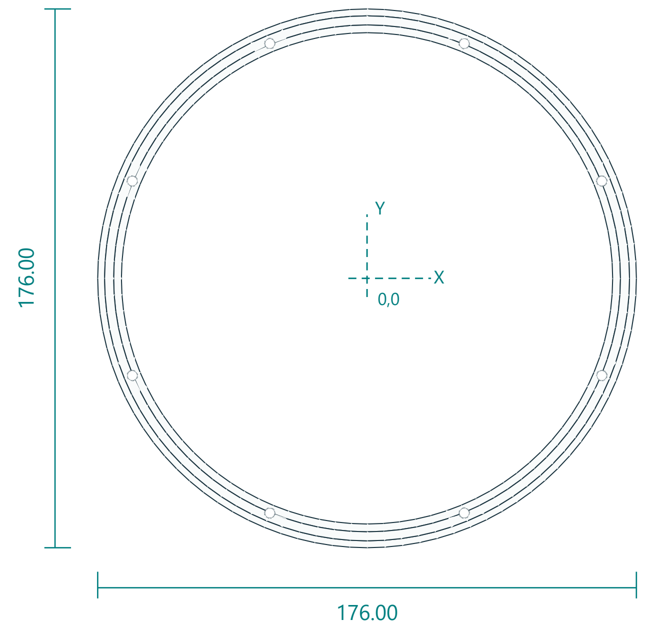
**UNITS mm · THIRD-ANGLE PART VIEWS · NUMERICAL DIMENSIONS CONTROL**

Views fitted to sheet. Do not measure this PDF. Digital prototype; check physical fit before batch printing.

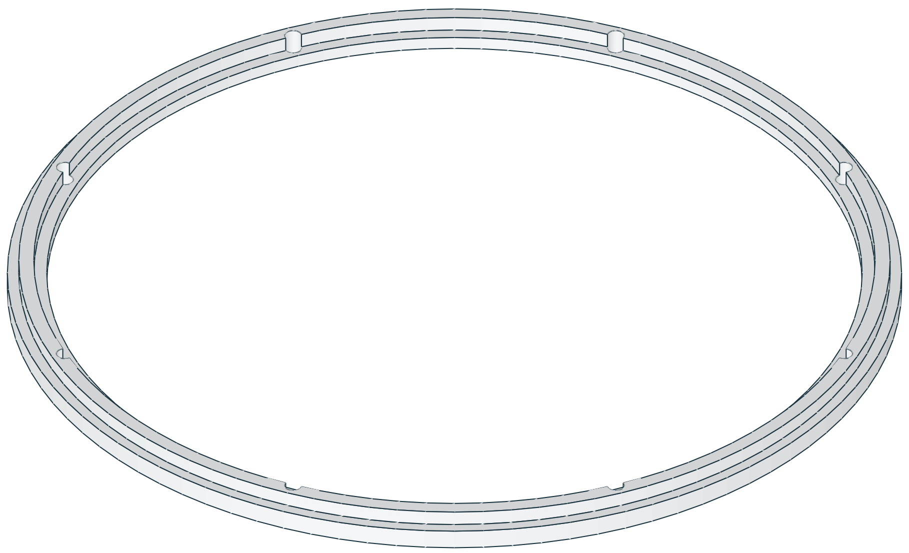
# Outer RGB diffuser retaining bezel

outer_bezel.stl | PRINT QUANTITY 1 | PETG

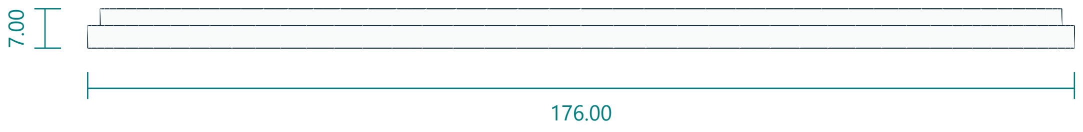
### TOP / XY



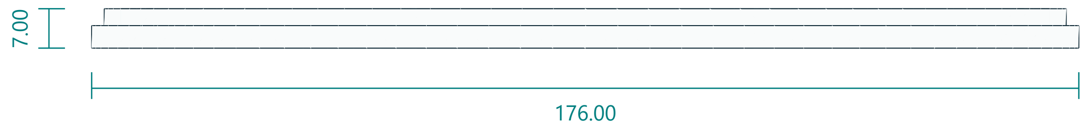

### ISOMETRIC / REFERENCE



### FRONT / XZ




### RIGHT / YZ




Interfaces: 176 diameter; window160.5; 8 holes radius83.
Fasteners / retention: 8 M3x10.
Print orientation: front lip down. Layers 0.2 mm; 4 walls; 25% infill baseline.
Feature locations: Eight3.4 holes radius83 angles22.5+45n. Rearflangerecessdiameter165.5 by1.4 deep; facewindow160.5.
General fit: inspect and deburr holes; check the fit coupon and actual hardware. No unlisted machining tolerances are implied.

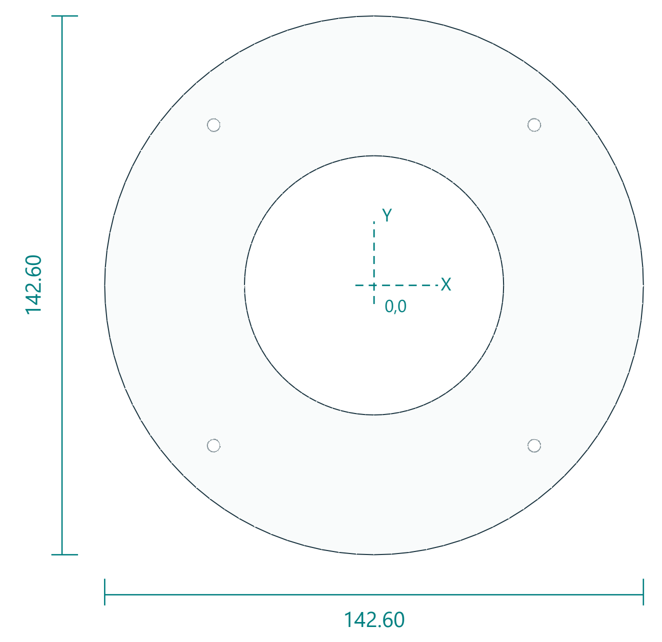
**UNITS mm · THIRD-ANGLE PART VIEWS · NUMERICAL DIMENSIONS CONTROL**

Views fitted to sheet. Do not measure this PDF. Digital prototype; check physical fit before batch printing.

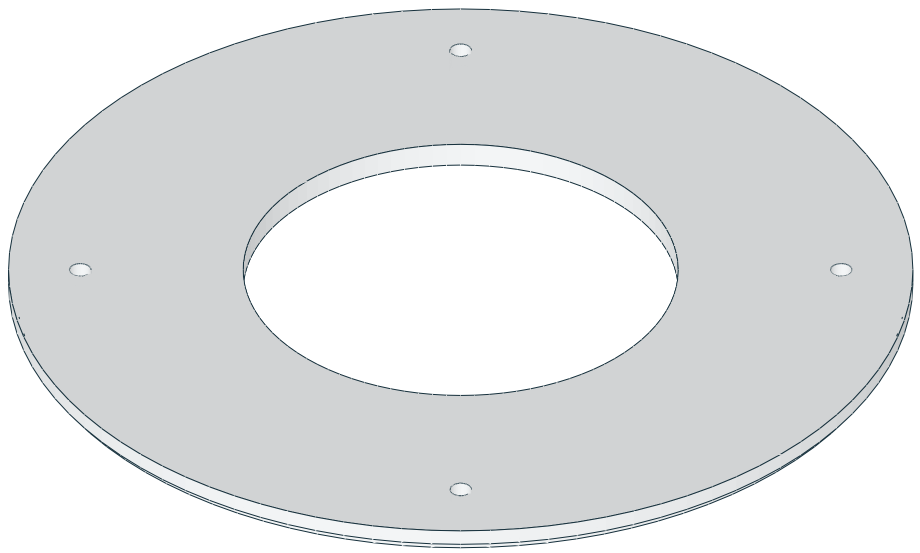
# Central optical baffle plate

face_center.stl | PRINT QUANTITY 1 | black PETG

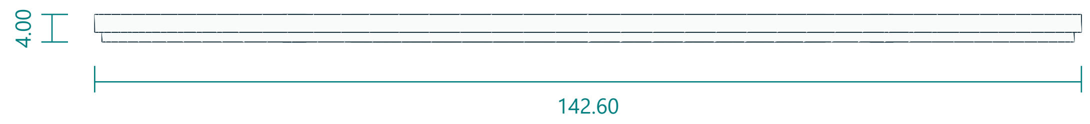
### TOP / XY




### ISOMETRIC / REFERENCE



### FRONT / XZ



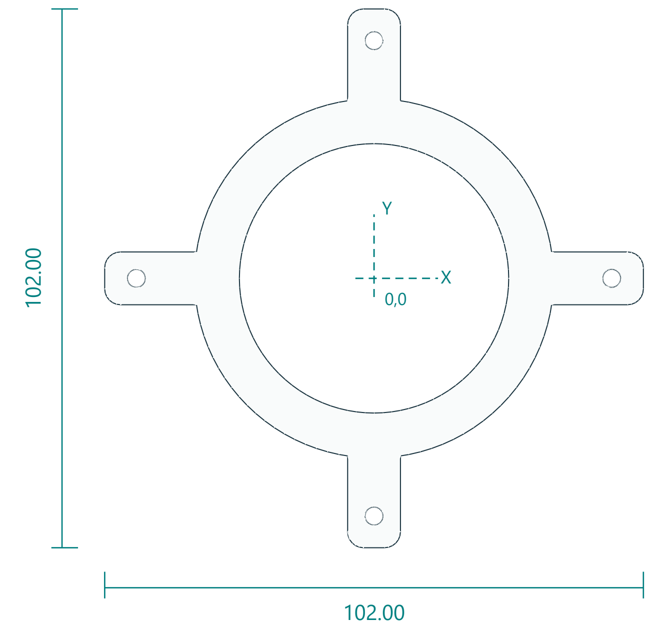

### RIGHT / YZ


Interfaces: 142.6 diameter x4; center68.6; screws radius60.
Fasteners / retention: 4 M3x12 into shell bosses.
Print orientation: flat front down. Layers 0.2 mm; 4 walls; 25% infill baseline.
Feature locations: Four3.4 holesradius60 angles45+90n. No sensor penetrations.
General fit: inspect and deburr holes; check the fit coupon and actual hardware. No unlisted machining tolerances are implied.

# White ring carrier and mounting arms

white_carrier.stl | PRINT QUANTITY 1 | PETG

### TOP / XY

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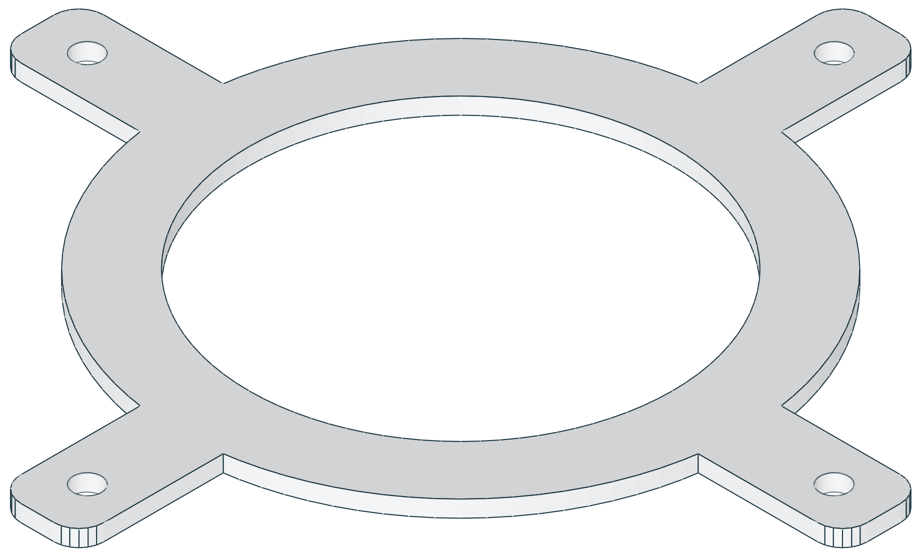
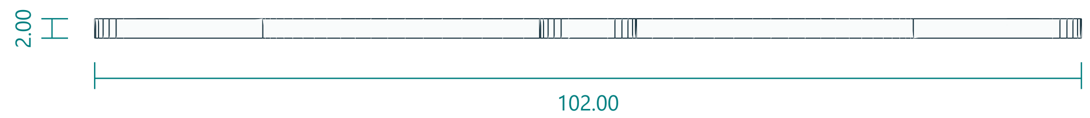
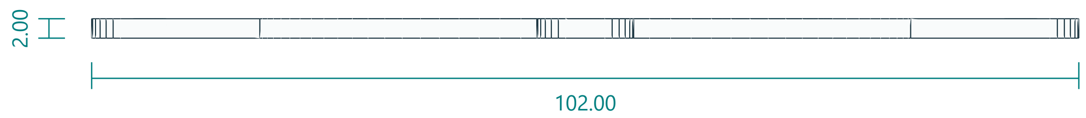


### FRONT / XZ




### RIGHT / YZ

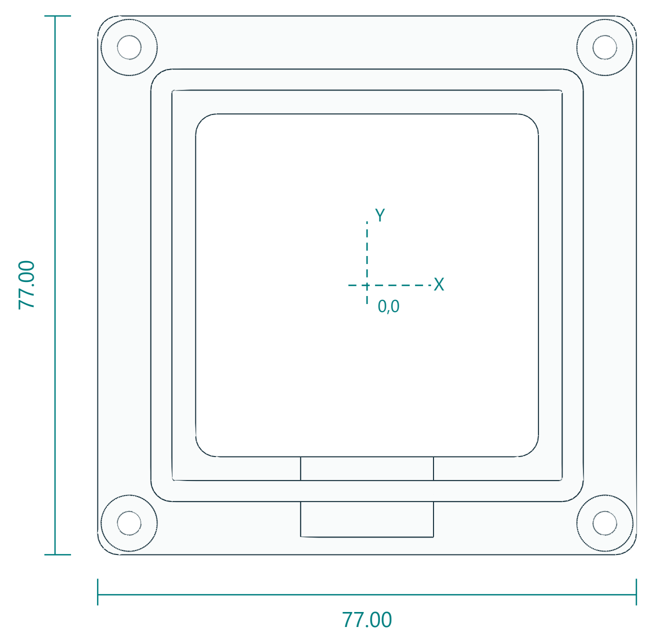



Interfaces: 68 OD51 ID; 4 cardinal mounting holes radius45; 2 thick.
Fasteners / retention: 4 M3x10.
Print orientation: flat down. Layers 0.2 mm; 4 walls; 25% infill baseline.
Feature locations: Four3.4 holes at(+/-45,0),(0,+/-45). Allarmsandring2mmthick.
General fit: inspect and deburr holes; check the fit coupon and actual hardware. No unlisted machining tolerances are implied.

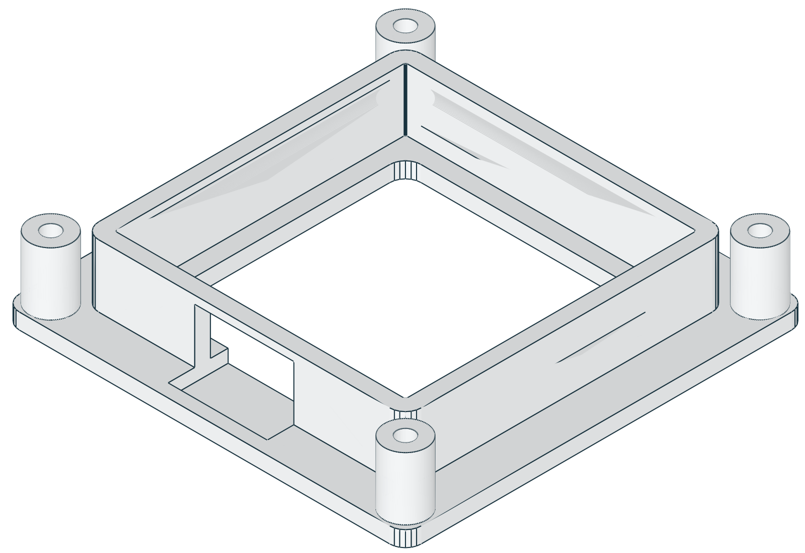
# 55 mm LCD edge cradle

lcd_cradle.stl | PRINT QUANTITY 1 | PETG

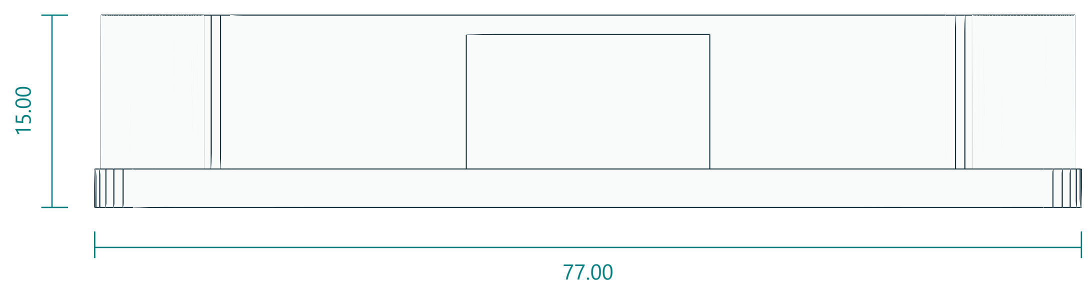
### TOP / XY



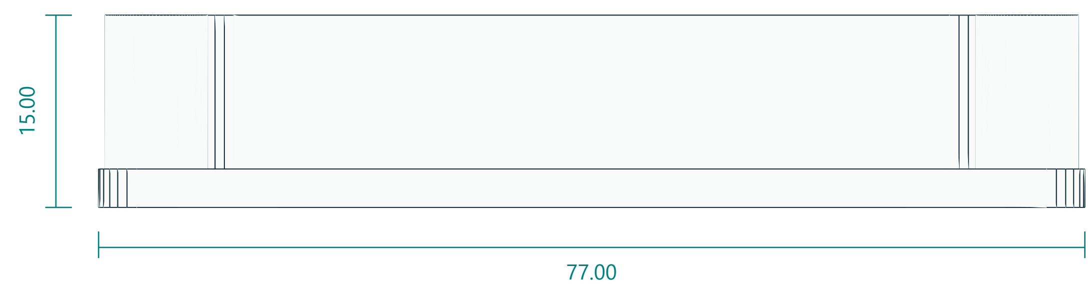

### ISOMETRIC / REFERENCE



### FRONT / XZ




### RIGHT / YZ

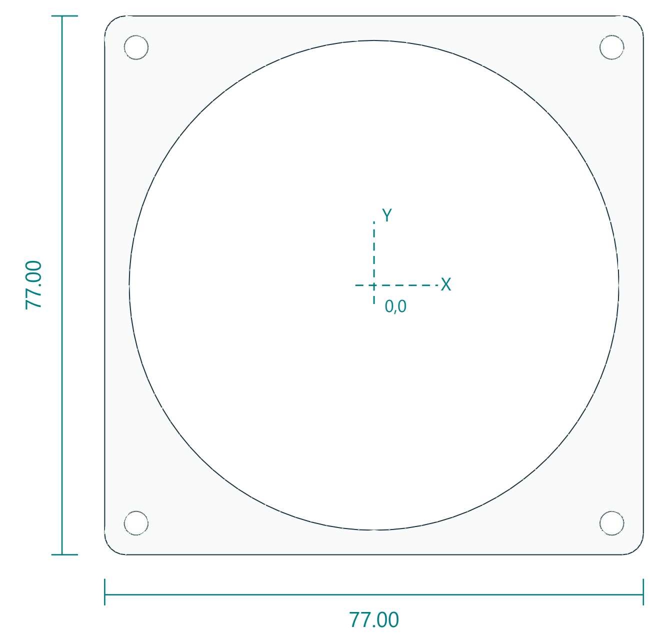



Interfaces: 55.8 square pocket; 15 depth; USB19 x13 relief.
Fasteners / retention: 4 M3x25; thin closed cell foam shims.
Print orientation: flat base down. Layers 0.2 mm; 4 walls; 25% infill baseline.
Feature locations: Four3.4 holes(+/-34,+/-34). Inner55.8square cavity. Rear49squareaccess. USBopening19wide at-Y.
General fit: inspect and deburr holes; check the fit coupon and actual hardware. No unlisted machining tolerances are implied.

# LCD PCB corner retaining plate

lcd_retainer.stl | PRINT QUANTITY 1 | black PETG

TOP / XY

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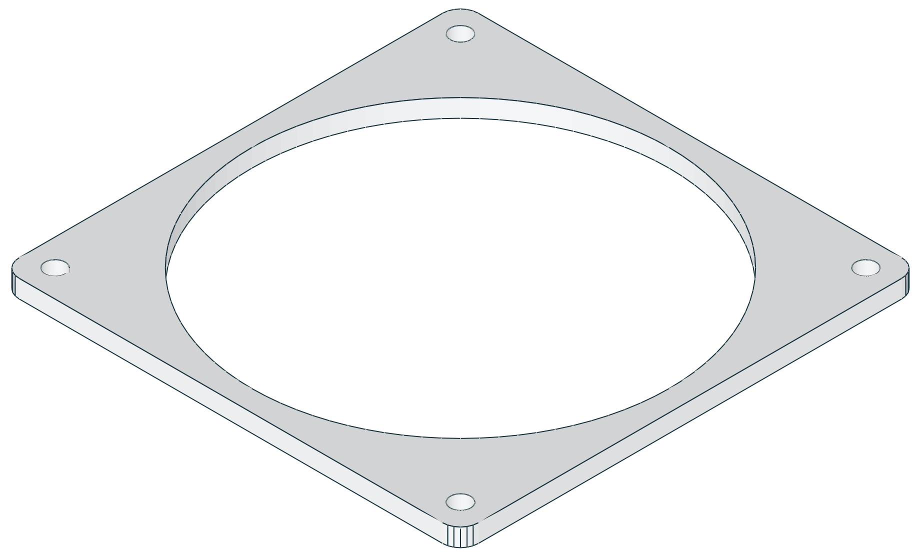
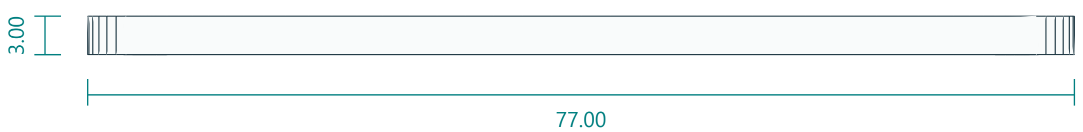
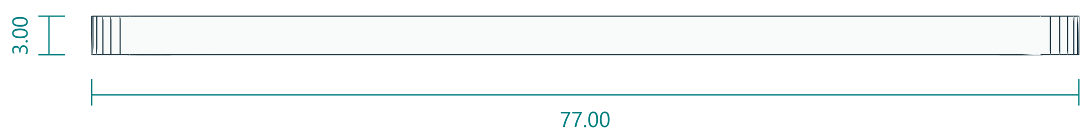


FRONT / XZ




RIGHT / YZ

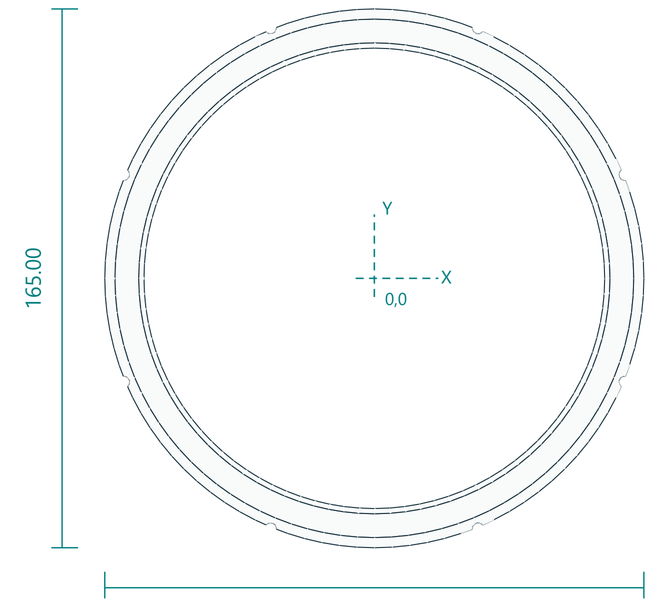



Interfaces: 77 square x3; opening70; mounting pitch68.
Fasteners / retention: 4 M3x25 shared.
Print orientation: flat down. Layers 0.2 mm; 4 walls; 25% infill baseline.
Feature locations: Four3.4 holes(+/-34,+/-34). Diameter70centeropening retainsPCBcorners, not roundglass.
General fit: inspect and deburr holes; check the fit coupon and actual hardware. No unlisted machining tolerances are implied.

# Frosted outer RGB annular diffuser

outer_diffuser.stl | PRINT QUANTITY 1 | natural/translucent PETG

### TOP / XY

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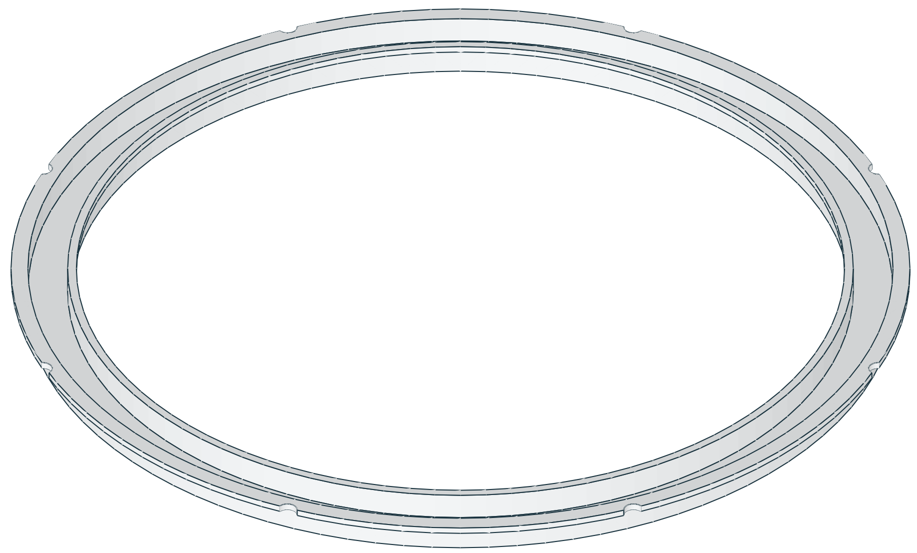


### FRONT / XZ


### RIGHT / YZ


Interfaces: 160 face OD143 ID; 165 flange; 1.2 face; 6.2 depth.
Fasteners / retention: captured by outer bezel/center plate.
Print orientation: smooth optical face on bed. Layers 0.15 mm; 3 walls; 100% infill baseline.
Feature locations: FaceOD160 ID143. RearflangesOD165/ID158.8 andOD144.2/ID141. M3edgeclearancenotchesradius83.
General fit: inspect and deburr holes; check the fit coupon and actual hardware. No unlisted machining tolerances are implied.

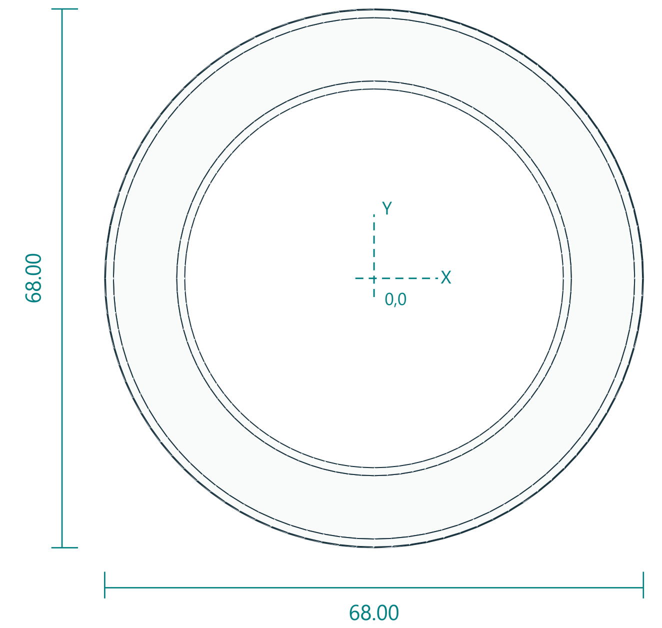
**UNITS mm · THIRD-ANGLE PART VIEWS · NUMERICAL DIMENSIONS CONTROL**

Views fitted to sheet. Do not measure this PDF. Digital prototype; check physical fit before batch printing.

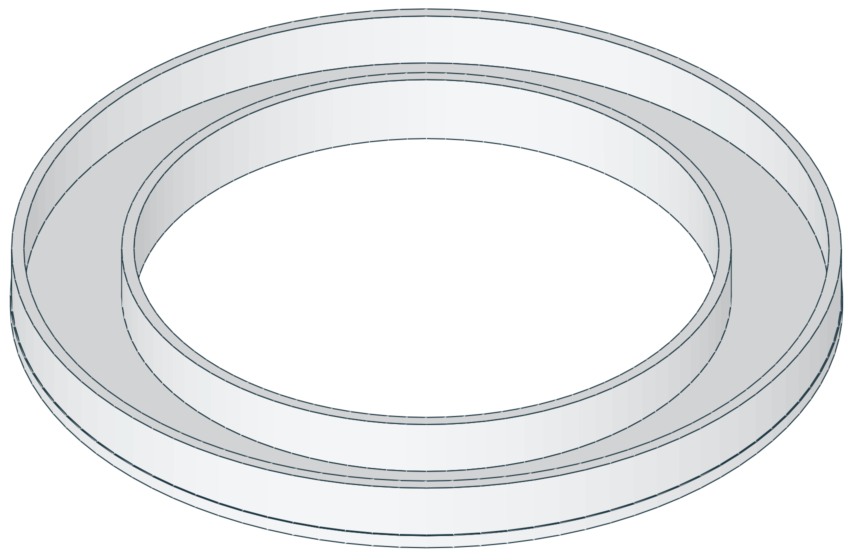
# Frosted inner RGBW light diffuser

inner_diffuser.stl | PRINT QUANTITY 1 | natural/translucent PETG

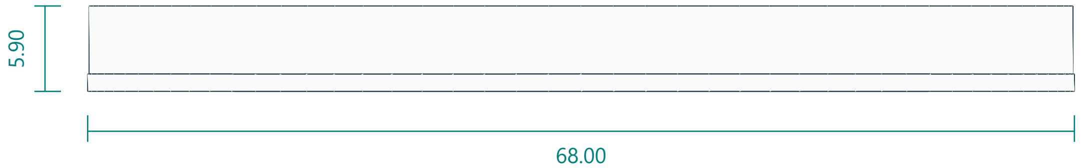
### TOP / XY



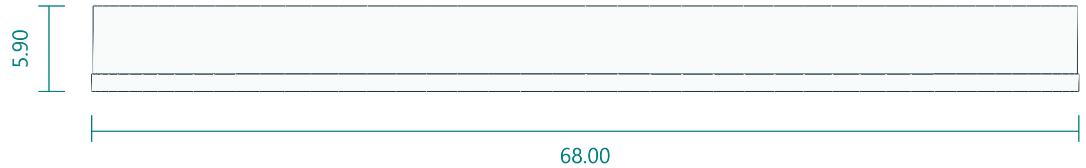

### ISOMETRIC / REFERENCE



### FRONT / XZ




### RIGHT / YZ

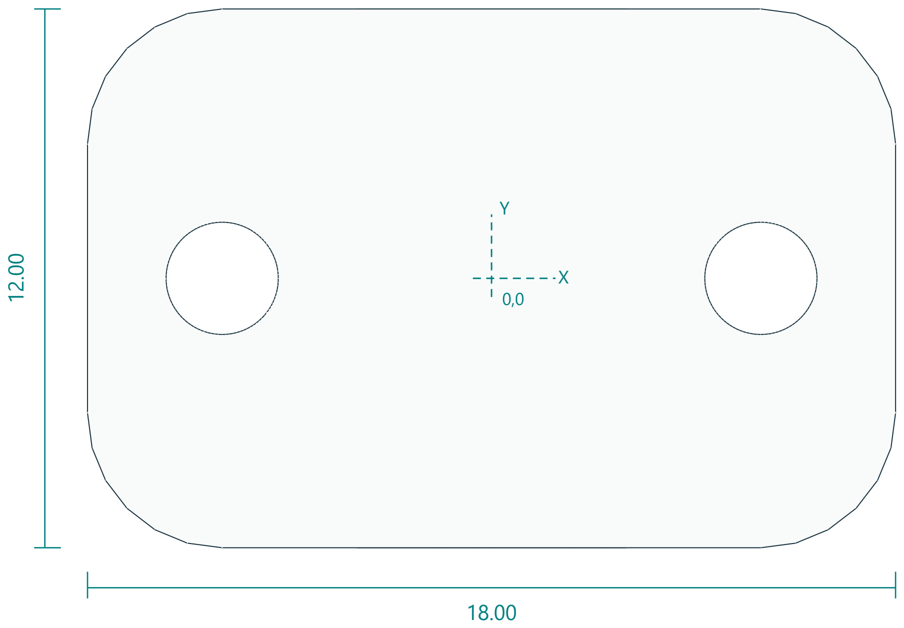



Interfaces: 68 OD47.8 ID; 1.2 face; 5.9 depth.
Fasteners / retention: 3 tiny neutral cure silicone retention dots.
Print orientation: smooth optical face on bed. Layers 0.15 mm; 3 walls; 100% infill baseline.
Feature locations: OD68 ID47.8; skin1.2; depth5.9. No sensorwindowmaterial.
General fit: inspect and deburr holes; check the fit coupon and actual hardware. No unlisted machining tolerances are implied.

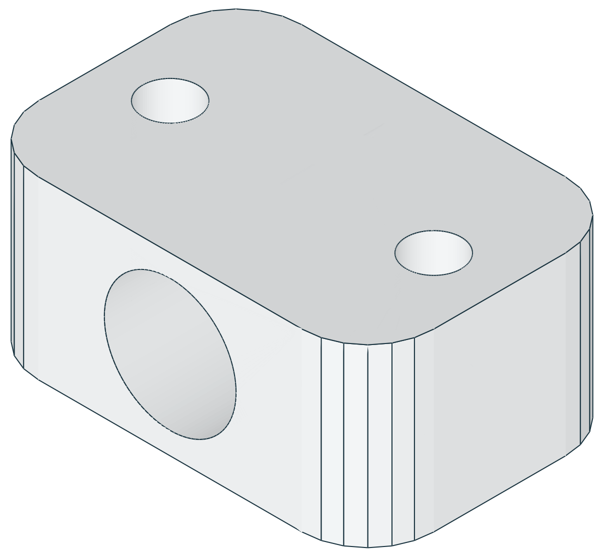
# Harness saddle clip

cable_clip.stl | PRINT QUANTITY 6 | PETG

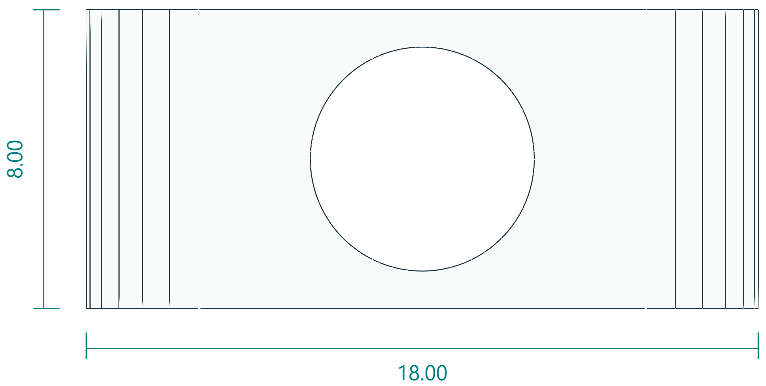
### TOP / XY



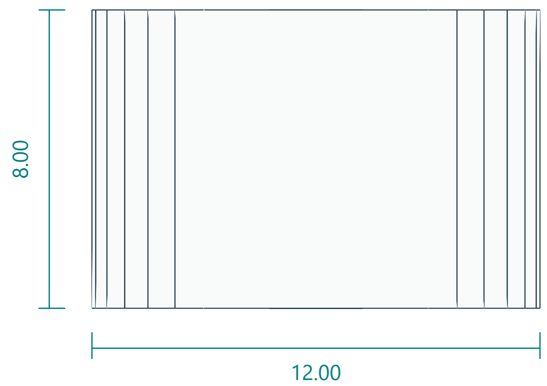

### ISOMETRIC / REFERENCE



### FRONT / XZ




### RIGHT / YZ




Interfaces: 18 x12 x8; cable6; screws12 pitch.
Fasteners / retention: 2 M2.5x12 each optional.
Print orientation: flat down. Layers 0.2 mm; 4 walls; 25% infill baseline.
Feature locations: Two2.5holes(+/-6,0); cablebore6 alongY centeredZ4.
General fit: inspect and deburr holes; check the fit coupon and actual hardware. No unlisted machining tolerances are implied.

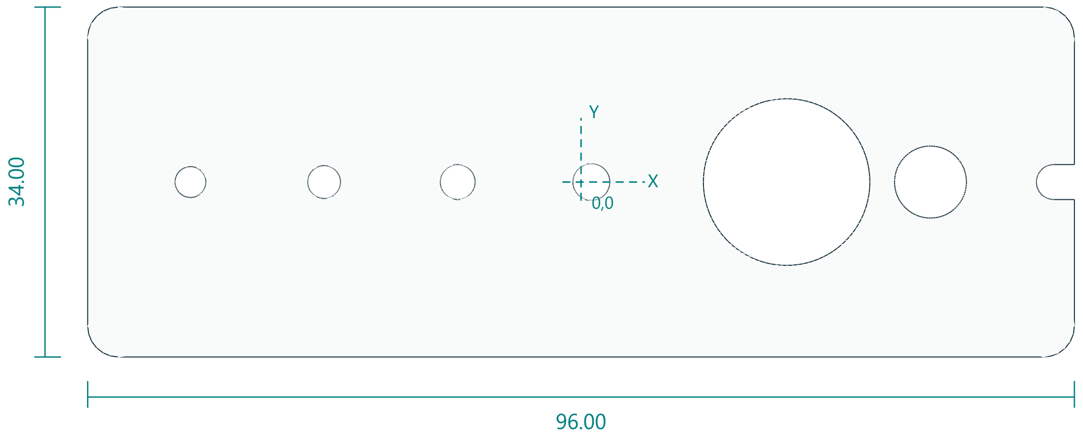
**UNITS mm · THIRD-ANGLE PART VIEWS · NUMERICAL DIMENSIONS CONTROL**

Views fitted to sheet. Do not measure this PDF. Digital prototype; check physical fit before batch printing.

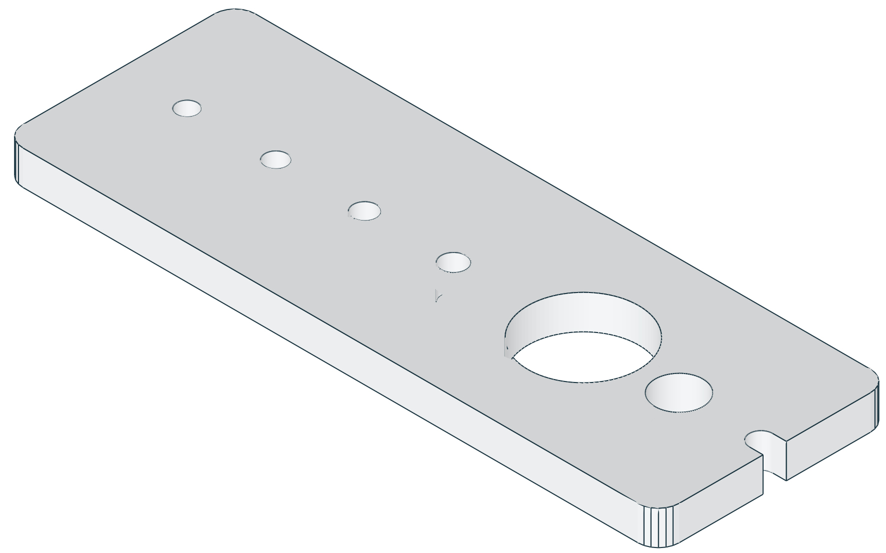
# Fastener, 625 bearing and horn-slot clearance coupon

fit_coupon.stl | PRINT QUANTITY 1 | PETG

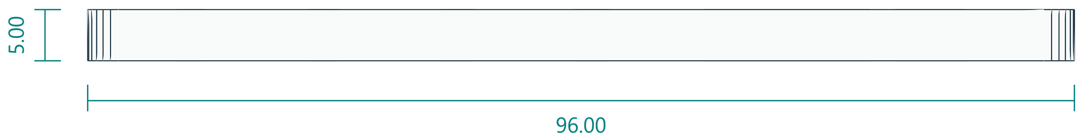
### TOP / XY



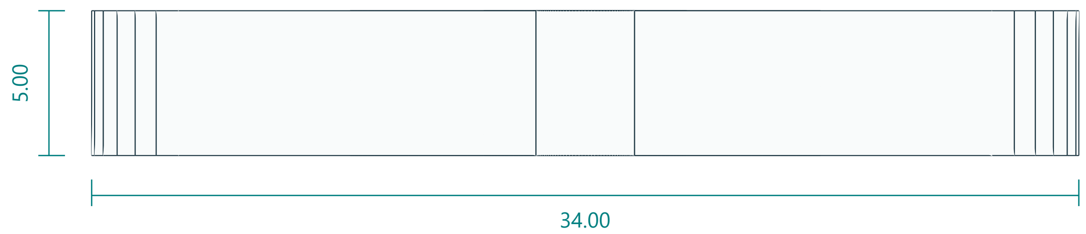

### ISOMETRIC / REFERENCE



### FRONT / XZ




### RIGHT / YZ




Interfaces: 3.0 /3.2 /3.4 /3.6 bores; 16.2 bearing pocket; DS3218 horn slots.
Fasteners / retention: none.
Print orientation: flat down. Layers 0.2 mm; 4 walls; 25% infill baseline.
Feature locations: Bolt bores3.0,3.2,3.4,3.6; 625 bearing pocket16.2; straight-horn center7 and radial slots12-18 and20-25.
General fit: inspect and deburr holes; check the fit coupon and actual hardware. No unlisted machining tolerances are implied.

**UNITS mm · THIRD-ANGLE PART VIEWS · NUMERICAL DIMENSIONS CONTROL**

Views fitted to sheet. Do not measure this PDF. Digital prototype; check physical fit before batch printing.

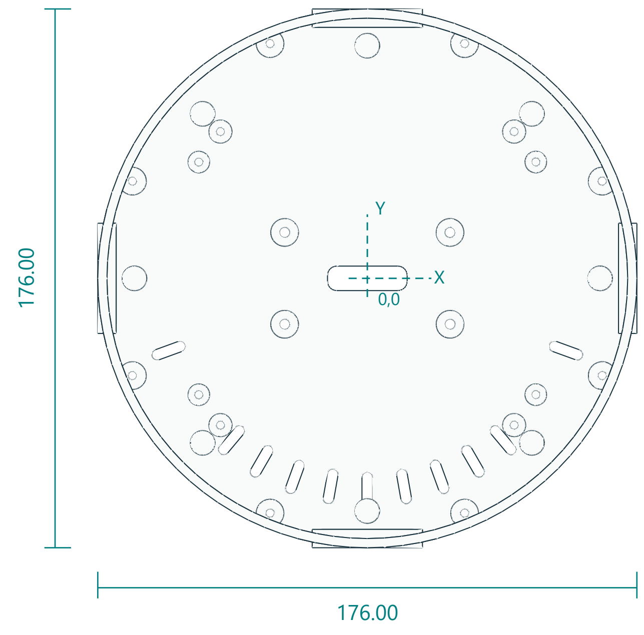
**P24**

REV C | 07 SEP 2026 | A3

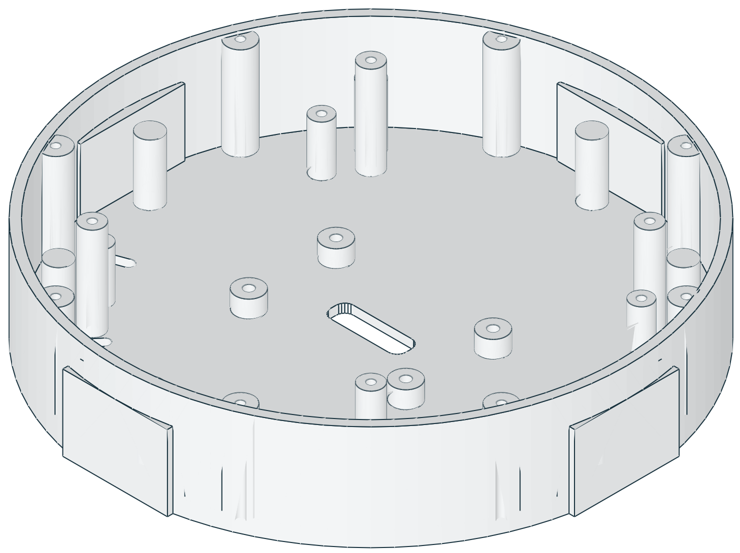
# Optional DSI head rear enclosure

head_shell_dsi.stl | PRINT QUANTITY 1 | PETG | OPTIONAL 4-INCH DSI HEAD ONLY

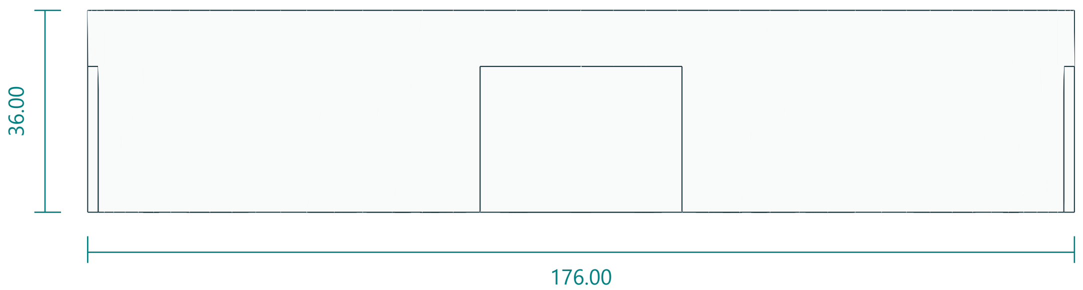
TOP / XY



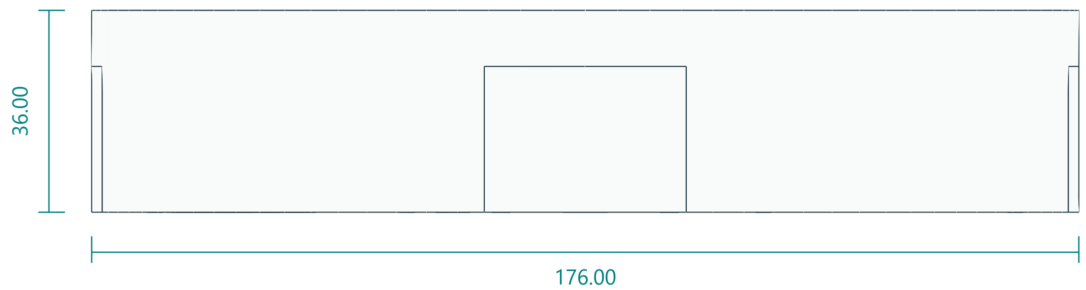

ISOMETRIC / REFERENCE



FRONT / XZ




RIGHT / YZ

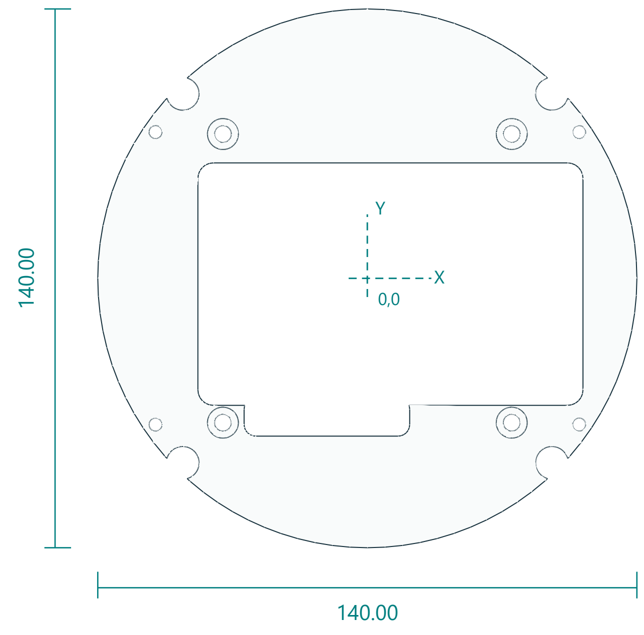



Interfaces: 176 diameter x36; carrier bosses (+/-55,+/-38) top20.5; face ring bosses radius67.8 top35.
Fasteners / retention: 8 M3x10 bezel; 4 M3x12 yoke; 4 M3x8 carrier; 4 M3x12 face ring.
Print orientation: rear disc down. Layers 0.2 mm; 4 walls; 25% infill baseline.
Feature locations: Rear bolts(+/-27,+/-15). Carrier pilots2.8 at(+/-55,+/-38) top20.5. Face ring pilots2.8 radius67.8 angles45+90n top35. Bezel radius83 angles22.5+45n top35. Centre cable slot26x8. No sensor penetrations.
General fit: inspect and deburr holes; check the fit coupon and actual hardware. No unlisted machining tolerances are implied.

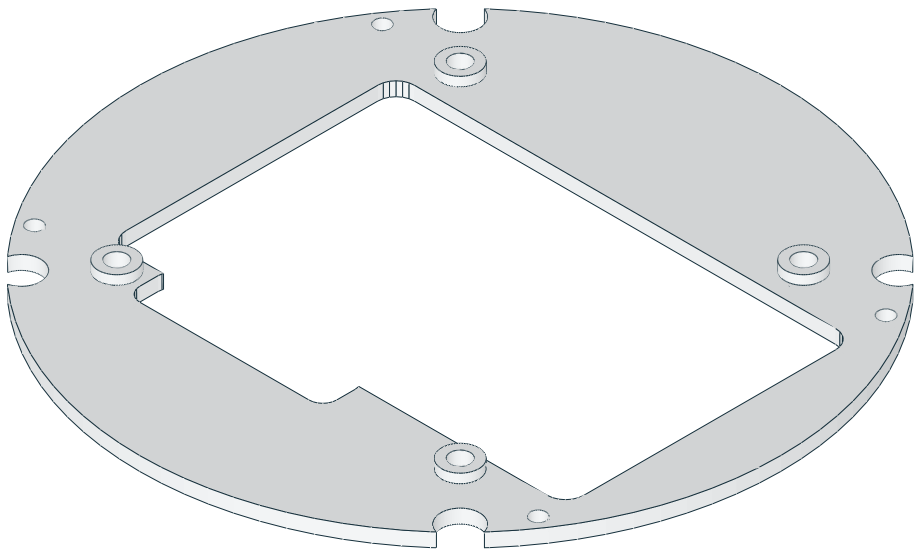
# Optional 4 inch LCD carrier plate

lcd4_carrier.stl | PRINT QUANTITY 1 | PETG | OPTIONAL 4-INCH DSI HEAD ONLY

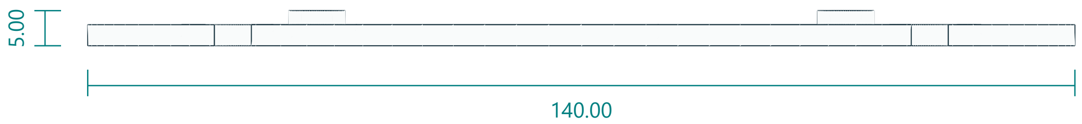
### TOP / XY



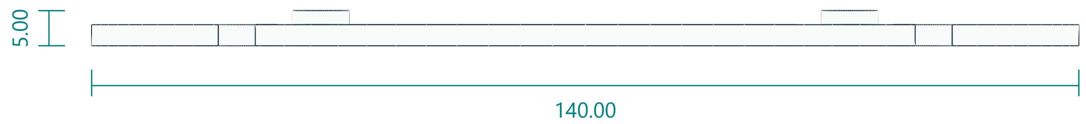

### ISOMETRIC / REFERENCE



### FRONT / XZ




### RIGHT / YZ




Interfaces: 140 diameter x3; boss pads to5; LCD M4 pattern75 x75; PCB window100 x63 with connector notch.
Fasteners / retention: 4 M4x8 into LCD case bosses; 4 M3x8 into shell bosses.
Print orientation: flat down. Layers 0.2 mm; 4 walls; 25% infill baseline.
Feature locations: M4 clearance4.4 at(+/-37.5,+/-37.5) on5mm pads. Shell screws3.4 at(+/-55,+/-38). Face ring boss clearance8.5 radius67.8. PCB window X-44..56 Y-33..30 plus connector notch X-32..11 Y-41..-31.
General fit: inspect and deburr holes; check the fit coupon and actual hardware. No unlisted machining tolerances are implied.

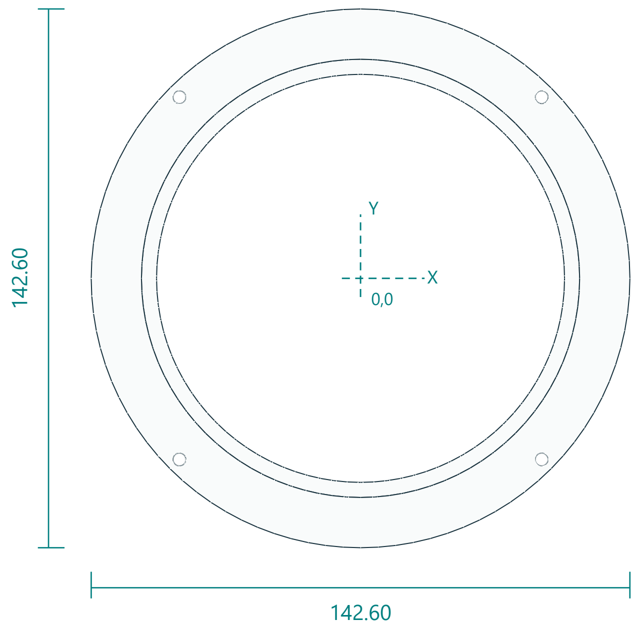
**UNITS mm · THIRD-ANGLE PART VIEWS · NUMERICAL DIMENSIONS CONTROL**

Views fitted to sheet. Do not measure this PDF. Digital prototype; check physical fit before batch printing.

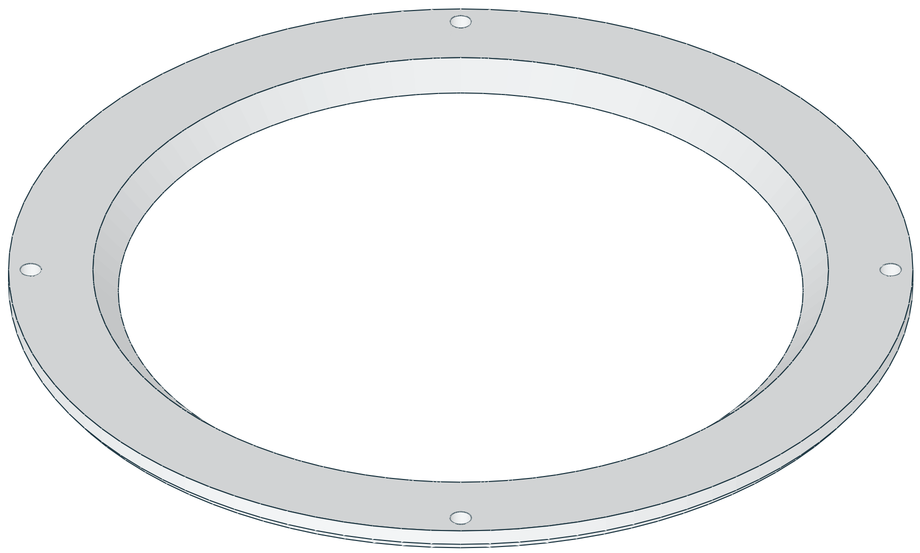
# Optional DSI face ring

face_ring_dsi.stl | PRINT QUANTITY 1 | black PETG | OPTIONAL 4-INCH DSI HEAD ONLY

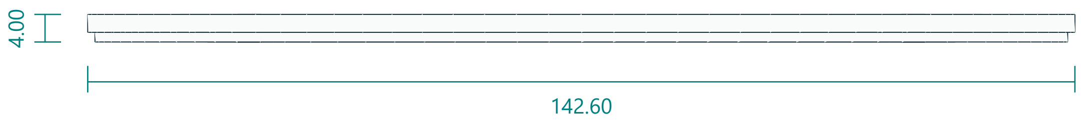
**TOP / XY**




**ISOMETRIC / REFERENCE**



**FRONT / XZ**




**RIGHT / YZ**


Interfaces: 142.6 diameter x4; opening108 chamfered to116; screws radius67.8.
Fasteners / retention: 4 M3x12 into shell bosses.
Print orientation: flat front down. Layers 0.2 mm; 4 walls; 25% infill baseline.
Feature locations: Four3.4 holes radius67.8 angles45+90n. Rear opening108 chamfered to116 at the front. Rear flange recess145/140.5 by1.4.
General fit: inspect and deburr holes; check the fit coupon and actual hardware. No unlisted machining tolerances are implied.

**UNITS mm · THIRD-ANGLE PART VIEWS · NUMERICAL DIMENSIONS CONTROL**

Views fitted to sheet. Do not measure this PDF. Digital prototype; check physical fit before batch printing.